The two poles of Grothendieck's localizer axioms

A proof audit of reversible excision and fibration stability

AI-produced by GPT-6 Astra (OpenAI, ChatGPT Pro mode)

10 November 2026

Conventions and evidentiary status

We fix universes $\mathcal{U} \in \mathcal{V}$. Categories are $\mathcal{U}$-small, and classes and localizations are formed in $\mathcal{V}$ when needed. We write $BA = |NA|$, e for the terminal category, $\varnothing$ for the empty category, and $\mathcal{W}_\infty$ for the functors inducing weak homotopy equivalences on nerves. We put $\mathbf{Hot} = \mathbf{Cat}[\mathcal{W}_\infty^{-1}]$. A category A is $\mathcal{W}$-aspherical if $A \to e$ belongs to $\mathcal{W}$.

We use the prompt's modern relative-Theorem-A convention for a fundamental localizer: identities belong to $\mathcal{W}$; two out of three holds; if $i : A \to B$ and $r : B \to A$ satisfy $ri = \mathrm{id}_A$ and $ir \in \mathcal{W}$, then $i, r \in \mathcal{W}$; every category with a terminal object is $\mathcal{W}$-aspherical; and, for a commutative triangle $A \xrightarrow{u} B \to C$, if each induced $A/c \to B/c$ belongs to $\mathcal{W}$, then $u \in \mathcal{W}$. In the actual Chapter XVI this last axiom is Loc (3 bis), while Loc (3) is only the absolute Theorem A, corresponding to $C = B$ and the identity structure functor of B. The prompt conflates these labels. We retain the modern definition for the mathematical leads and explicitly state stronger hypotheses when auditing deductions under the weaker manuscript definition. Full arrow-retract closure and strong saturation are theorems for the modern definition, not extra assumptions silently inserted in historical arguments.

For $u : A \to B$, the objects of A/b are $(a, u(a) \to b)$; the objects of $b\backslash A$ are $(a, b \to u(a))$. A final functor in the usual covariant convention has nonempty connected $b\backslash A$. A sieve is a full subcategory closed under sources of arrows; a cosieve is closed under targets. The manuscript normally uses presheaf coefficients. We state variance at each comparison where it changes a conclusion.

Every displayed claim is assigned one of the statuses Proved, Refuted, Known, or Open. Known means that the cited result has been checked in the indicated source; it does not mean that every surrounding inference in that source is accepted. Open means unresolved in this audit, unless a claim explicitly identifies an open problem in the literature. We claim no priority. The attached companion, CATHOT, and TOTASPH texts are drafts to be checked, not authorities.

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A The axioms, their scope, and their implications

A.1. The historical and modern definitions differ. [Known]

In XVI, manuscript pp. 1–3, Loc (1) requires isomorphisms, two out of three, and the implication

$$gf, fg \in \mathcal{W} \quad \Longrightarrow \quad f, g \in \mathcal{W}$$

for $f : X \to Y$ and $g : Y \to X$. Loc (1 bis) adds closure under retracts in the arrow category. Loc (1 ter) says that $f \in \mathcal{W}$ whenever its image in $\mathbf{Cat}[\mathcal{W}^{-1}]$ is invertible. Loc (2) concerns categories with a terminal object. The literal Loc (3) is the absolute Theorem A statement: $u : A \to B$ belongs to $\mathcal{W}$ if every A/b is $\mathcal{W}$-aspherical. The relative Theorem A in our conventions is printed as Loc (3 bis), not Loc (3) (Grothendieck, 1991c, ms. 1–3).

The distinction is explicit in the manuscript; its footnote 1 even proposes exchanging the two labels. Our use of "fundamental localizer" follows the definition requested in the prompt and Cisinski (2006, Definition 3.3.2). For this definition strong saturation, including closure under retracts, is a theorem (Cisinski, 2006, Proposition 4.2.4). We do not silently apply that theorem to the weaker historical definition. The minimum for the literal Loc (1,2,3) is nevertheless also $\mathcal{W}_\infty$: a separate theorem for weak fundamental localizers establishes this, as explained in A.19.

A.2. The exact two excision axioms of XII. [Known]

Let X be a small category, $U, V \subseteq X$ full sieves, $U \cup V = X$, and $S = U \cap V$. Every arrow of X lies in U or V, so the square

$$\begin{array}{ccc} S & \xrightarrow{i_1} & U \\ {\scriptstyle i_2}\downarrow & & \downarrow{\scriptstyle j_2} \\ V & \xrightarrow[j_1]{} & X \end{array} \tag{A.MV}$$

is a pushout and a pullback in **Cat**. XII, manuscript pp. 75–76, calls W(7 bis) the assertion

$$i_1 \in \mathcal{W} \iff j_1 \in \mathcal{W}, \qquad i_2 \in \mathcal{W} \iff j_2 \in \mathcal{W}.$$

The second equivalence follows from the first by exchanging U and V. W(7), manuscript pp. 76–77, concerns a strict natural transformation $f : X_\bullet \to Y_\bullet$ between two such squares. With 0 the intersection, $1, 2$ the pieces, and 3 the union, it says that any two of

$$f_0 \in \mathcal{W}, \qquad (f_1, f_2 \in \mathcal{W}), \qquad f_3 \in \mathcal{W}$$

imply the third. These are quantifications over all small categories and all such maps of covers, including empty intersections and empty pieces (Grothendieck, 1991b, ms. 75–77).

Opposites exchange sieves with cosieves. Since a modern fundamental localizer is invariant under opposites, either formulation gives exactly the corresponding cosieve formulation. This is category-theoretic duality; it does not exchange homotopy pushouts with homotopy pullbacks. XII's comparison with the dual derivator at manuscript pp. 79–80 is a different operation.

A.3. Loc (6), its three cube clauses, and Loc (6′). [Known]

XVI, manuscript pp. 16′–18′, uses the same map $f : X_\bullet \to Y_\bullet$ of MV squares and defines

$$\begin{aligned} \mathrm{Loc}(6a) \quad &: \quad f_0, f_1, f_2 \in \mathcal{W} \implies f_3 \in \mathcal{W}, \\ \mathrm{Loc}(6b) \quad &: \quad f_1, f_2, f_3 \in \mathcal{W} \implies f_0 \in \mathcal{W}, \\ \mathrm{Loc}(6c) \quad &: \quad f_0, f_3 \in \mathcal{W} \implies f_1, f_2 \in \mathcal{W}. \end{aligned}$$

After footnote 13, Loc (6) means $(6a) + (6b)$, expressly excluding $(6c)$. On the single square (A.MV), Loc $(6'a)$ is $i_1 \in \mathcal{W} \Rightarrow j_1 \in \mathcal{W}$, and Loc $(6'b)$ is $j_1 \in \mathcal{W} \Rightarrow i_1 \in \mathcal{W}$. Thus XII W(7 bis) is XVI Loc $(6')$, whereas XII W(7) includes all three clauses $(6a, b, c)$.

The quoted adjective "inhabituelle" at XII, manuscript p. 270, applies to the reverse implication. The change of conventions in XVI is substantive: one cannot replace every occurrence of W(7) by the later label Loc (6) before accounting for retract closure.

A.4. Cofibrations and the four structural formulations. [Known]

For a functor $i : A \to B$, let

$$i_! : A\backslash\mathbf{Cat} \longrightarrow B\backslash\mathbf{Cat}, \qquad (A \to X) \longmapsto (B \to B \amalg_A X).$$

The class $\mathrm{Cof}_{\mathcal{W}}$ consists of the i for which $i_!$ sends every morphism whose underlying functor is in $\mathcal{W}$ to another such morphism. XVI, manuscript pp. 25–27, states:

Loc $(6A)$	Every $f : A \to B$ factors in **Cat** as $A \xrightarrow{i} M \xrightarrow{p} B$, with $i \in \mathrm{Cof}_{\mathcal{W}}$ and $p \in \mathcal{W}$.
Loc $(6A')$	Every Dwyer map belongs to $\mathrm{Cof}_{\mathcal{W}}$.
Loc $(6B)$	Every MV square (A.MV) is $\mathcal{W}$-homotopy cocartesian.
Loc $(6C)$	In every $\mathcal{W}$-homotopy cocartesian square (A.MV), now without any sieve or strict-pushout hypothesis, $j_1 \in \mathcal{W}$ implies $i_1 \in \mathcal{W}$.

Here $\mathcal{W}$-homotopy cocartesian means that the canonical comparison from a specified Grothendieck model of the homotopy pushout to the fourth vertex belongs to $\mathcal{W}$. Under enough cofibrations this agrees with the cofibration-replacement definition of XII, manuscript pp. 261–264.

A Dwyer map, in the convention used here, is an inclusion $A \hookrightarrow B$ of a full sieve such that the inclusion into the cosieve generated by A has a right adjoint restricting to the identity on A. Its opposite is the closed variant. Closure of this class under pushouts and compositions is not closure under retracts: the latter fails (Cisinski, 1999). None of the arguments below assumes retract closure of the class of Dwyer maps.

A.5. Strict fibrations and the auxiliary classes. [Known]

For $p : E \to S$, the class $\mathrm{Fib}_{\mathcal{W}}$ is defined by strict base change:

$$p \in \mathrm{Fib}_{\mathcal{W}} \quad \Longleftrightarrow \quad \big(u : X \to Y \text{ over } S,\ u \in \mathcal{W}\big) \Rightarrow \big(E \times_S X \to E \times_S Y\big) \in \mathcal{W}.$$

We write $\mathcal{W}^u_S$ for morphisms over S whose pullbacks along every $S' \to S$ lie in $\mathcal{W}$. A functor $E \to S$ itself lies in $\mathcal{W}^u$ when all its strict base changes lie in $\mathcal{W}$. The class $\mathcal{F}_{\mathcal{W}}$ consists of Grothendieck fibrations $p : E \to S$ whose reindexing functors $\alpha^* : E_t \to E_s$, for every $\alpha : s \to t$ in S, lie in $\mathcal{W}$.

XVI, manuscript pp. 27–32, calls p weakly $\mathcal{W}$-fibrant if, for every $T' \to T$ over S with T, T' contractible in the category-homotopy sense, $E \times_S T' \to E \times_S T$ is in $\mathcal{W}$. This defines $\mathrm{Fib}^{\mathrm{weak}}_{\mathcal{W}}$. Finally, $\mathrm{Parf}_{\mathcal{W}}$ means both $\mathcal{W}$-smooth and $\mathcal{W}$-proper. The manuscript has

$$\mathrm{Parf}_{\mathcal{W}} \subseteq \mathrm{Fib}^{\mathrm{weak}}_{\mathcal{W}}, \quad \mathcal{F}_{\mathcal{W}} \subseteq \mathrm{Fib}^{\mathrm{weak}}_{\mathcal{W}}, \quad \mathrm{Fib}_{\mathcal{W}} \subseteq \mathrm{Fib}^{\mathrm{weak}}_{\mathcal{W}}.$$

These are different classes; a Grothendieck fibration is not, by definition, a member of $\mathrm{Fib}_{\mathcal{W}}$.

A.6. The exact fibration axioms. [Known]

XVI, manuscript pp. 38–45, defines Loc (7) to mean Loc $(7a)$:

$$\mathrm{Loc}(7a) : \quad \mathcal{F}_{\mathcal{W}} \subseteq \mathrm{Fib}_{\mathcal{W}}.$$

Loc $(7b)$ says that, whenever $p : X \to S$ and $q : Y \to S$ belong to $\mathcal{F}_{\mathcal{W}}$, a functor $f : X \to Y$ over S with $f \in \mathcal{W}$ lies in $\mathcal{W}^u_S$. Equivalently in this situation, every strict fibre map $f_s : X_s \to Y_s$ lies in $\mathcal{W}$. Loc $(7c)$ says

$$\mathcal{F}_{\mathcal{W}} \cap \mathcal{W} \subseteq \mathcal{W}^u.$$

For a reference localizer $V \subseteq \mathcal{W}$, Loc $(7D_V)$ is

$$\mathrm{Parf}_V \cap \mathcal{W} \subseteq \mathcal{W}^u.$$

Loc $(7D)$ without a subscript means $V = \mathcal{W}$. The weakest reference version singled out by the manuscript is D_{W_ω}, with its historical minimum W_ω; the modern version used in this paper is $D_{\mathcal{W}_\infty}$.

Loc $(7F)$ means that every functor $f : X \to Y$ admits a factorization

$$X \xrightarrow{u} E \xrightarrow{p} Y, \qquad u \in \mathcal{W}, \quad p \in \mathrm{Fib}_{\mathcal{W}}\,.$$

For fundamental localizers, all $X \to e$ belong to $\mathrm{Fib}_{\mathcal{W}}$, by product stability. Thus the associated category of fibrant objects has all objects fibrant. The word "cofibrations" in the initial formulation of $(7F)$ on manuscript p. 39 is a variance error; the subsequent formulations explicitly say fibrations.

A.7. The four other numbered conditions used in the audit. [Known]

Loc (4), XVI manuscript p. 8, is

$$\coprod_{i\in I} u_i \in \mathcal{W} \quad \Longleftrightarrow \quad u_i \in \mathcal{W} \text{ for every } i \in I$$

for every small family, including $I = \varnothing$. Loc (5), manuscript p. 12, is $\mathcal{W} \subseteq \mathcal{W}_0$, where $\mathcal{W}_0$ consists of functors inducing bijections on connected components. Loc (8), manuscript pp. 71–72, is closure of $\mathcal{W}$ under all small filtered colimits in the arrow category of **Cat**; Loc (8 bis) asks only for sequential colimits. Loc (9), manuscript pp. 74–75, says that $\mathcal{W}$ is an accessible full subcategory of the arrow category, with its inclusion accessible. It does not mean preservation of every filtered colimit.

The manuscript's cardinal convention in §5 is not the convention in which "$\aleph_0$-accessible" means finitely accessible. Its countable-subcategory calculation should therefore not be paraphrased as finite presentability of all weak-equivalence data. Part C uses the modern existence of a regular accessibility cardinal.

A.8. Brown's lemma in the form needed in XVI 3.2.1. [Proved]

Let $\mathcal{M}$ have an initial object 0 and let $C, \mathcal{W} \subseteq \mathrm{Fl}(\mathcal{M})$. Assume that $\mathcal{W}$ contains identities and satisfies two out of three; C contains isomorphisms and all $0 \to X$, is closed under composition and pushout, and all pushouts along maps in C exist; every map factors as a map in C followed by a map in $\mathcal{W}$; and pushouts of maps in $C \cap \mathcal{W}$ belong to $C \cap \mathcal{W}$. Then $C \subseteq \mathrm{Cof}_{\mathcal{W}}$. Neither strong saturation nor closure of C under retracts is needed.

Proof. We first prove the elementary functor lemma. If $F : \mathcal{M} \to \mathcal{N}$ sends $C \cap \mathcal{W}$ into a class V containing identities and satisfying two out of three, it sends $\mathcal{W}$ into V. For $u : X \to Y$ in $\mathcal{W}$, factor

$$[u, \mathrm{id}_Y] : X \amalg Y \xrightarrow{j} Z \xrightarrow{r} Y, \qquad j \in C, r \in \mathcal{W}.$$

Let $j_X : X \to Z$ and $j_Y : Y \to Z$ be the two composites with the coproduct injections. They lie in C since all objects are cofibrant, and in $\mathcal{W}$ because $rj_X = u$ and $rj_Y = \mathrm{id}_Y$. Thus $F(j_X), F(j_Y) \in V$, and $F(r)F(j_Y) = \mathrm{id}_{F(Y)}$ implies $F(r) \in V$. Hence $F(u) \in V$.

For an object A, the full subcategory of $A\backslash\mathcal{M}$ on $A \to X$ in C satisfies the same cofibrant-object axioms. Its coproduct is pushout over A. An absolute factorization of an under-A map gives a relative factorization since composites of cofibrations are cofibrations. For any $s : A \to A'$, pushout defines a functor from this category to the corresponding category under A'; it preserves $C \cap \mathcal{W}$ by the pushout assumption. The functor lemma therefore proves that

$$h : X \to Y \text{ under } A, \quad A \to X, A \to Y \in C, \quad h \in \mathcal{W} \implies A' \amalg_A h \in \mathcal{W}. \tag{A.8.1}$$

Now let $i : A \to B$ be in C and $u : A \to A'$ in $\mathcal{W}$. We must prove that $B \to B \amalg_A A'$ belongs to $\mathcal{W}$. The preceding factorization of u writes it as a trivial cofibration followed by the retraction of a trivial cofibration. Pushout of the first factor is in $\mathcal{W}$ by assumption. We can therefore reduce, by an actual factorization of the pushout map and two out of three, to the case that u has a section $v : A' \to A$ in $C \cap \mathcal{W}$, with $uv = \mathrm{id}_{A'}$.

Put $B_1 = A \amalg_{A'} B$, where the attaching maps are $v : A' \to A$ and $iv : A' \to B$. Write $l : A \to B_1$ and $k : B \to B_1$ for the coprojections. Both l and i lie in C, and $k \in C \cap \mathcal{W}$. There are two functors, or morphisms in $\mathcal{M}$,

$$f = [i, \mathrm{id}_B] : B_1 \to B, \qquad g = [ivu, \mathrm{id}_B] : B_1 \to B.$$

They exist because iv agrees with $ivuv$, and both retract k, so both belong to $\mathcal{W}$. The map f is a morphism from l to i under A. By (A.8.1), its pushout along u belongs to $\mathcal{W}$. Associativity of ordinary pushouts identifies $A' \amalg_A B_1$ canonically with B. Under this identification the comparison $B_1 \to A' \amalg_A B_1$ is g. We obtain a commutative square

$$\begin{array}{ccc} B_1 & \xrightarrow{f} & B \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle q} \\ B & \xrightarrow[u_! f]{} & A' \amalg_A B, \end{array}$$

with $f, g, u_! f \in \mathcal{W}$. Consequently $q \in \mathcal{W}$. Stability of C under pushout applies this conclusion after any initial cobase change of i, proving precisely the defining condition of $i \in \mathrm{Cof}_{\mathcal{W}}$. □

This is the dual of the argument in Brown (1973, §4, Lemmas 1–2, pp. 428–429); we have supplied the maps because the phrase "Brown's lemma" alone can conceal the relative-category step.

A.9. The consequences of enough cofibrations and their duals. [Proved]

Let $\mathcal{W}$ contain identities and satisfy two out of three. Suppose pushouts exist and every morphism of $\mathcal{M}$ factors as a $\mathcal{W}$-cofibration followed by a $\mathcal{W}$-equivalence. Then

$$\mathrm{Cof}_{\mathcal{W}} \cap \mathcal{W} = \{i : \text{every pushout of } i \text{ belongs to } \mathcal{W}\}.$$

Moreover, a $\mathcal{W}$-equivalence between objects cofibrant under a fixed A stays in $\mathcal{W}$ after arbitrary cobase change $A \to A'$. Dually, enough $\mathcal{W}$-fibrations implies that a $\mathcal{W}$-equivalence between objects fibrant over S lies in $\mathcal{W}^u_S$, and $\mathrm{Fib}_{\mathcal{W}} \cap \mathcal{W} = \mathcal{W}^u$.

Proof. For the equality, the right side implies the left by the identity pushout and two out of three in a morphism of pushout diagrams. Conversely, let $i : A \to B$ lie in $\mathrm{Cof}_{\mathcal{W}} \cap \mathcal{W}$, and factor $s : A \to A'$ as $A \xrightarrow{j} A_0 \xrightarrow{w} A'$, where j is a $\mathcal{W}$-cofibration and $w \in \mathcal{W}$. The pushout $A_0 \to B_0 = B \amalg_A A_0$ of i lies in $\mathcal{W}$ because j preserves $\mathcal{W}$. The pushout $B_0 \to B' = B_0 \amalg_{A_0} A'$ of w lies in $\mathcal{W}$ because $A_0 \to B_0$ is a $\mathcal{W}$-cofibration. The commutative square with horizontal maps $A_0 \to B_0$ and $A' \to B'$ then gives $A' \to B' \in \mathcal{W}$ by two out of three.

For the relative assertion, factor $A \to A'$ in the same way. Pushout along j preserves the given map $X \to Y \in \mathcal{W}$. Pushout along w gives equivalences on both the source and target because the respective structural maps are cofibrations. Two out of three proves the assertion. Reversing all arrows proves the two fibration statements. This is exactly the valid formal argument of XII Proposition 16, manuscript pp. 231–232, and XVI 4.4.1. □

A.10. The elementary implications between the excision clauses. [Proved]

Using only two out of three and the definitions of MV squares, one has

$$(6a) \Rightarrow (6'a), \qquad (6b) \Rightarrow (6'b), \qquad (6c) \Rightarrow (6'a) + (6'b).$$

Consequently XII W(7) implies W(7 bis), as Proposition 6 at manuscript p. 78 correctly states.

Proof. For the first implication map the degenerate square (S, S, V, V) to (S, U, V, X). Its four vertex maps are $(\mathrm{id}_S, i_1, \mathrm{id}_V, j_1)$. For the second use the map from (S, U, V, X) to (U, U, X, X), with vertex maps $(i_1, \mathrm{id}_U, j_1, \mathrm{id}_X)$. Both are strict maps of sieve-cover squares, and all maps have the indicated sources and targets. The first cube gives $(6c) \Rightarrow (6'b)$, and the second gives $(6c) \Rightarrow (6'a)$. Exchanging the two pieces gives the symmetric formulations. In XVI's prose after the second cube, the repeated $f_1 = j_1$ must read $f_2 = j_1$. □

A.11. The missing cofibration premise and its repair. [Proved]

For the modern fundamental-localizer convention, Loc $(6a)$, $(6'a)$, $(6A)$, $(6A')$, and $(6B)$ all hold. The equivalences asserted in XVI Proposition 3.6 and its footnote 19 are therefore valid in this convention. The proof gap identified in XII footnote 136 is real in the argument as originally written: termwise replacement by maps in $\mathcal{W}$ does not by itself show that ordinary pushout preserves them. The Dwyer pushout comparison and the gluing proof in Part B provide that missing premise without reverse excision.

Proof. Part B constructs the specified homotopy-pushout model and proves that objectwise $\mathcal{W}$-equivalences of spans induce $\mathcal{W}$-equivalences of these models, using relative Theorem A. For MV covers the nerve is the union of two simplicial subsets, so the comparison to the ordinary categorical union is in $\mathcal{W}_\infty \subseteq \mathcal{W}$. This gives $(6a)$ and $(6B)$. For a pushout along a Dwyer map the corresponding nerve comparison is again in $\mathcal{W}_\infty$; the same span argument proves $(6A')$. The manuscript's cylinder, formed with the product of the endpoint inclusion $\{0, 2n\} \hookrightarrow \Phi_n$ with the entire source category, then gives the factorization in $(6A)$. It is not a pushout-product construction. Every comparison here is a specified functor or specified nerve comparison, not an inference from an abstract isomorphism of homotopy types.

The historically proposed route from full $(6')$ to Dwyer cofibrations is also sound once its premises are verified: apply A.8 with C the Dwyer maps, using stability under composition and pushout, the same cylinder factorization, and the preservation of trivial Dwyer maps under pushout obtained by the full MV argument. What was missing in XII was exactly the verification of these premises before ordinary pushout was used. Brown's lemma proves the implication; it does not say that an arbitrary sieve is a cofibration. □

A.12. The reversible equivalences and the role of retracts. [Proved]

For a modern fundamental localizer the following are equivalent:

$$\mathrm{Loc}(6b), \quad \mathrm{Loc}(6'b), \quad \mathrm{Loc}(6C), \quad \mathrm{Loc}(6), \quad \mathrm{Loc}(6'), \quad \mathrm{Loc}(6c), \quad \mathrm{W}(7).$$

In a historical or abstract formulation where retract closure is not yet available, the proof of $(6a, b) \Rightarrow (6c)$ requires that retracts of $\mathcal{W}$-aspherical objects be $\mathcal{W}$-aspherical. This is weaker than asking that every arrow retract of every member of $\mathcal{W}$ belong to $\mathcal{W}$, but the latter suffices.

Proof. The single-square reverse axiom is equivalent to detection of $\mathcal{W}$-equivalences by their homotopy cofibres, with the empty-source convention made explicit in Part B. Homotopy cofibres of opposite sides of a homotopy-pushout square agree by the canonical iterated-colimit comparison. Therefore cofibres detect those two maps simultaneously, proving $(6C)$. Its restriction to MV squares gives $(6'b)$, and A.10 gives the converse from $(6b)$.

For a map between homotopy-pushout squares, take the pointed homotopy cofibre at each of its four vertices. The resulting square is a homotopy-pushout square: form its homotopy colimit as an iterated colimit of the cube and interchange the two colimits. The constant-point square is cocartesian, so no extra term appears. Detection by cofibres turns the assertions about f_i into assertions about four aspherical objects Z_i. If Z_1, Z_2, Z_3 are aspherical, reverse excision in their square implies that Z_0 is aspherical. This proves $(6b)$.

If instead Z_0, Z_3 are aspherical, their specified cone points include a section $e \to Z_0$, which is in $\mathcal{W}$ by two out of three with $Z_0 \to e$. The map of spans $(Z_1 \leftarrow e \to Z_2) \to (Z_1 \leftarrow Z_0 \to Z_2)$, using that section and the identity on both outside vertices, gives a specified $\mathcal{W}$-equivalence from the homotopy wedge to Z_3. The forgetful functor from pointed spaces creates this homotopy pushout because the span indexing category is contractible. Its two intermediate vertices are retracts, in the localized category, of the homotopy wedge representing the fourth vertex: collapse the other wedge summand to the chosen point. Strong saturation and closure under retracts imply that both are aspherical. Thus $(6c)$ follows. Conversely $(6c)$ implies both single-square clauses by A.10, and the forward clause is already available. This proves all stated equivalences. Part F explains why modern relative Theorem A supplies the necessary retract property automatically. □

The cofibres in this proof are pointed objects, even though the original categories and functors are unpointed. The point is the specified cone vertex. Forgetting it before defining the wedge retractions would omit data needed by the argument.

A.13. The formal fibration implication scheme. [Proved]

Let $\mathcal{F}$ be a class of arrows in a category with the relevant pullbacks. Define

$$\begin{array}{rcl} \mathrm{A}(\mathcal{F}) & : & \mathcal{F} \subseteq \mathrm{Fib}_{\mathcal{W}}, \\ \mathrm{B}(\mathcal{F}) & : & \text{every } \mathcal{W}\text{-map over } S \text{ between } \mathcal{F}\text{-objects lies in } \mathcal{W}^u_S, \\ \mathrm{C}(\mathcal{F}) & : & \mathcal{F} \cap \mathcal{W} \subseteq \mathcal{W}^u. \end{array}$$

If every map factors as a $\mathcal{W}$-map followed by a map in $\mathcal{F}$, then $\mathrm{A}(\mathcal{F}) \Rightarrow \mathrm{B}(\mathcal{F})$. If identities belong to $\mathcal{F}$, then $\mathrm{B}(\mathcal{F}) \Rightarrow \mathrm{C}(\mathcal{F})$. If in addition $\mathcal{F}$ is closed under pullback and composition and contains all terminal maps, then

$$\mathrm{A}(\mathcal{F}) \iff \mathrm{B}(\mathcal{F}) \iff \mathrm{C}(\mathcal{F}).$$

These are the precise premises of XVI Lemmas 4.4.1 and 4.4.3.

Proof. The factorization premise and $\mathrm{A}(\mathcal{F})$ give enough $\mathcal{W}$-fibrations. Apply the dual of A.9 to prove $\mathrm{B}(\mathcal{F})$. To obtain $\mathrm{C}(\mathcal{F})$, take the target over S to be $\mathrm{id}_S : S \to S$. Finally, under the additional hypotheses, $\mathrm{C}(\mathcal{F})$ says that the trivial members of $\mathcal{F}$ are stable under pullback. The dual of A.8 gives $\mathcal{F} \subseteq \mathrm{Fib}_{\mathcal{W}}$. □

For $V \subseteq \mathcal{W}$, the factorization by a V-equivalence followed by a V-perfect map is the path-category input of the manuscript. Applying A.13 to Parf_V gives the three equivalent versions of D_V. In particular D_V implies $(7F)$. Passing to a smaller reference class weakens each of $\mathrm{A}_V, \mathrm{B}_V, \mathrm{C}_V, D_V$, simply by inclusion of the classes on which the conditions are tested.

A.14. Replacement transfers the weak-fibration formulations. [Proved]

Suppose two classes $\mathcal{F}, \mathcal{G}$ have the following comparison property: each $p : X \to S$ in $\mathcal{F}$ factors as

$$X \xrightarrow{j} \overline{X} \xrightarrow{\overline{p}} S, \qquad j \in \mathcal{W}_S^u, \quad \overline{p} \in \mathcal{G}.$$

Then $\mathrm{A}(\mathcal{G}) \Rightarrow \mathrm{A}(\mathcal{F})$ and $\mathrm{C}(\mathcal{G}) \Rightarrow \mathrm{C}(\mathcal{F})$. If the replacements are functorial over S on morphisms between objects in $\mathcal{F}$, then also $\mathrm{B}(\mathcal{G}) \Rightarrow \mathrm{B}(\mathcal{F})$.

Proof. Given $T' \to T$ in $\mathcal{W}$ over S, compare its pullbacks along $X \to S$ and $\overline{X} \to S$. The two vertical comparison maps are in $\mathcal{W}$ because j is universally in $\mathcal{W}$; the bottom horizontal map is in $\mathcal{W}$ by $\mathrm{A}(\mathcal{G})$. Two out of three gives the top map. For C, if $p \in \mathcal{W}$, then $\overline{p} \in \mathcal{W}$ since $j \in \mathcal{W}$, so both factors remain in $\mathcal{W}$ after any base change. For B, functoriality supplies a commutative square $j_Y f = \overline{f} j_X$ over S. If $f \in \mathcal{W}$, then $\overline{f} \in \mathcal{W}$, hence $\overline{f} \in \mathcal{W}_S^u$; the same square after arbitrary pullback proves $f \in \mathcal{W}_S^u$. □

This proves XVI Lemma 4.4.9 and identifies exactly why its last assertion requires more than a separate factorization for each object. In the manuscript the replacement is the functorial Grothendieck-fibration replacement $X \to \Phi_S(X)$; its universal V-equivalence for weakly V-fibrant $X \to S$ gives Corollary 4.4.10. This supplies the equivalence between using $\mathcal{F}_V$ and using $\mathrm{Fib}_V^{\mathrm{weak}}$ in each of $\mathrm{A}_V, \mathrm{B}_V, \mathrm{C}_V$.

A.15. The complete list of additional arrows asserted in XVI §4. [Known]

Besides the formal implications just proved, XVI, manuscript pp. 45–70, asserts the following. For $V \subseteq \mathcal{W}$ one has

$$\mathrm{A}_V \Rightarrow \mathrm{B}_V \Rightarrow \mathrm{C}_V \Rightarrow D_V \Rightarrow (7F),$$

and the same chain with $V = \mathcal{W}$. It asserts

$$\begin{aligned} \mathrm{A}_{\mathcal{W}} &\iff \big(\mathrm{C}_{\mathcal{W}} + \mathcal{F}_{\mathcal{W}} \text{ closed under composition}\big) \\ &\iff \big(\mathrm{C}_{\mathcal{W}} + \mathrm{Fib}_{\mathcal{W}}^{\mathrm{weak}} \text{ closed under composition}\big). \end{aligned}$$

It asserts $\mathrm{A}_V \Leftrightarrow \mathrm{B}_V \Leftrightarrow \mathrm{C}_V$ if $\mathcal{F}_V$ is closed under composition, in particular if V satisfies Loc (7). Its summary 4.4.13 additionally asserts that sequential-colimit closure of V makes all four reference conditions equivalent; sequential-colimit closure of both V and $\mathcal{W}$ makes all eight conditions equivalent; and strong saturation plus Loc (3 bis) for $\mathcal{W}$ gives $\mathrm{B}_{\mathcal{W}} \Rightarrow \mathrm{A}_{\mathcal{W}}$. The printed summary also says that $(7F)$ for V suffices to identify the first three reference conditions; its earlier argument at 4.4.6 had used the stronger condition Loc (7).

The composition equivalences follow from A.13, A.14, and the identity $\mathcal{F}_{\mathcal{W}} = \mathrm{Catfib} \cap \mathrm{Fib}_{\mathcal{W}}^{\mathrm{weak}}$. Once $\mathrm{Fib}_{\mathcal{W}}^{\mathrm{weak}} \subseteq \mathrm{Fib}_{\mathcal{W}}$, equality holds, so composition closure follows. The reverse implications use Brown's lemma with the stated composition hypothesis. The sequential-colimit arguments use the path replacement, not reverse cofibration excision. The strong-saturation implication is attributed in the manuscript to XII Proposition 1, manuscript p. 54, using locally constant diagrams; it is not a consequence of two out of three alone. Parts C–E settle these implications for the modern localizers considered here and distinguish $(7F)$ from them. In particular, the proposed converse $(7F) \Rightarrow D_{\mathcal{W}_\infty}$ is false even for an accessible, strongly saturated fundamental localizer satisfying Loc $(4, 5, 6a, 8, 9)$: $\mathcal{W}_0$ is a counterexample, as already recognized by XVI 4.16 (the heading following 4.4.15).

A.16. The weaker historical axiom system is not identified with the modern one here. [Open]

This audit does not prove that the literal historical Loc $(1, 2, 3)$ with absolute Theorem A implies relative Theorem A, Loc (3bis). Consequently it does not use the modern strong-saturation theorem to settle the original eight-condition implication diagram for every class satisfying only that weaker definition. The unresolved statement here is precisely that implication between axiom systems; a first nonterminal test base is $[1]$, since the relative axiom over e is tautological. The minimum $W_\omega = \mathcal{W}_\infty$ is already settled by the distinct weak-localizer theorem in A.19; it is not part of this residual obligation.

The attempted transfer would require that a morphism of Grothendieck constructions, objectwise in the proposed weak localizer, belong to it. Absolute Theorem A tests aspherical comma categories, not arbitrary $\mathcal{W}$-equivalences of fibres, and therefore does not supply this step as stated. All results in this paper that use relative Theorem A say so through the conventions. In particular, Lead 4 itself includes Loc (3 bis), so this unresolved historical comparison does not affect the determination requested there.

A.17. All eight fibration variants collapse in the modern convention. [Proved]

Let $V \subseteq \mathcal{W}$ be modern fundamental localizers. Each of the eight conditions $\mathrm{A}_{\mathcal{W}}, \mathrm{B}_{\mathcal{W}}, \mathrm{C}_{\mathcal{W}}, D_{\mathcal{W}}, \mathrm{A}_V, \mathrm{B}_V, \mathrm{C}_V, D_V$ is equivalent to

$$\mathcal{W} = \mathcal{W}_\infty \qquad \text{or} \qquad \mathcal{W} = \mathrm{Fl}(\mathbf{Cat}).$$

No accessibility or filtered-colimit hypothesis is needed for this conclusion. Thus every conditional arrow in A.15 is valid for the modern convention, including the summary's $(7F)$-for-V formulation, although that formulation is not what the earlier manuscript proof establishes.

Proof. The formal chain in A.13, the replacement in A.14, and $\mathcal{W}_\infty \subseteq V \subseteq \mathcal{W}$ imply that each condition entails $D_{\mathcal{W}_\infty}$. E.4 proves that this condition makes the class of $\mathcal{W}$-equivalences stable under homotopy pullback. If $\mathcal{W}$ contains a map which is not a weak homotopy equivalence, its homotopy fibres, and then successive based loop spaces of a noncontractible fibre, show that $\mathcal{W}$ contains a nonempty disconnected aspherical space or the map $\varnothing \to e$. Pulling back one point inclusion along a point in a different component gives $\varnothing \to e \in \mathcal{W}$ in the first case as well. Product stability and two out of three then make every map belong to $\mathcal{W}$. Otherwise minimality gives $\mathcal{W} = \mathcal{W}_\infty$. For $\mathcal{W}_\infty$, the fibration and path-replacement theorems identify weak fibrations with strict homotopical base-change maps; this gives Loc (7), hence all eight conditions. The total localizer satisfies them directly. The proof details and all comparisons with models in **Cat** are in E.5. □

A.18. The coproduct consequences accompanying the implication diagram. [Proved]

Write Loc $(4a_0)$ for preservation of finite coproducts of equivalences, Loc $(4a)$ for preservation of all small coproducts, and Loc $(4b)$ for reflection of equivalences by coproducts of families with a common indexing set. The assertions surrounding XVI §3 and used in §4 are

$$(6a) \Rightarrow (4a_0), \qquad (6a) + (8) \Rightarrow (4a), \qquad (7) \Rightarrow (4b).$$

Consequently $(6a) + (7) + (8)$ implies Loc (4). In the modern convention $(4a)$ already follows from relative Theorem A; no reverse excision is needed for any of these assertions.

Proof. Apply $(6a)$ to the disjoint-union square with empty intersection to obtain the binary case, and iterate; the empty coproduct is $\mathrm{id}_{\varnothing}$. Every small coproduct is the filtered colimit of its finite subcoproducts, which proves the second implication. For the third, view the functor $\coprod_i X_i \to \coprod_i Y_i$ over the discrete category I. Both structural maps to I are Grothendieck fibrations with identity transition functors, hence belong to $\mathcal{F}_{\mathcal{W}}$. By A.13, Loc (7) makes the given $\mathcal{W}$-map universally a $\mathcal{W}$-equivalence over I. Pulling back at $i \in I$ gives $X_i \to Y_i \in \mathcal{W}$. Finally, the modern forward coproduct assertion is the objectwise Grothendieck-construction theorem over the discrete indexing category. This also covers arbitrarily many empty summands. □

A.19. The literal historical minimum is also the weak-homotopy localizer. [Known]

Let W_ω denote the intersection of all classes satisfying the literal XVI Loc $(1, 2, 3)$, where Loc (3) is absolute Theorem A. Then

$$W_\omega = \mathcal{W}_\infty.$$

This equality does not require identifying every weak fundamental localizer with a modern fundamental localizer.

Proof. In Cisinski (2006, §1.4.12, p. 56; Definition 6.1.12, pp. 240–241), a weak fundamental localizer is a class satisfying identities, two out of three, the split-retract cancellation rule

$$ri = \mathrm{id}_A, \quad ir \in \mathcal{W} \quad \Longrightarrow \quad i \in \mathcal{W},$$

the terminal-object axiom, and absolute Theorem A. Literal Loc (1) implies that cancellation rule: apply its clause about gf and fg with $f = i$, $g = r$. Thus every historical localizer in this intersection is a weak fundamental localizer in Cisinski's precise sense. His Theorem 6.1.18, pp. 242–243, states that every such class contains $\mathcal{W}_\infty$. Conversely $\mathcal{W}_\infty$ satisfies the literal historical axioms, including the cancellation rule for both composites, so the intersection is exactly $\mathcal{W}_\infty$.

For clarity about the scope of the external theorem, its proof first establishes the weak-localizer asphericity of horn inclusions by gluing and the asphericity of their transfinite cell attachments. It then obtains all weakly contractible spaces as aspherical objects. Proposition 6.1.16 passes from those objects to all $\mathcal{W}_\infty$-equivalences using Brown's lemma and the proper Thomason model of $\mathcal{W}_\infty$. It does not assume relative Theorem A for the target weak localizer. This is therefore a valid separate resolution of the historical minimum, rather than an application of the modern minimum theorem with an unchecked hypothesis. □

B Cofibres, reversible excision, and integral homology

B.1. The baseline used in this part. [Known]

For the modern fundamental localizers fixed on the first page, $\mathcal{W}_\infty \subseteq \mathcal{W}$, and $\mathcal{W}$ is strongly saturated and closed under retracts in $\mathbf{Cat}^{[1]}$. Neither assertion requires accessibility. Consequently a specified zigzag of weak homotopy equivalences of categories preserves $\mathcal{W}$-asphericity, and a commutative square whose vertical arrows are in $\mathcal{W}_\infty$ transfers membership of its horizontal arrows in $\mathcal{W}$ in both directions.

The first assertion is Cisinski (2004, Theorem 2.2.11, p. 136); the second is Cisinski (2006, Proposition 4.2.4, p. 183). For the last assertion, write the equality of the two composites in the square and apply two out of three. For a zigzag, apply this argument to each of its arrows.

We do not identify this modern definition with the weaker printed XVI Loc (1)–(3); Part A records the discrepancy.

B.2. Forward gluing is automatic under relative Theorem A. [Proved]

Let J be a small category, $F, G : J \to \mathbf{Cat}$ covariant functors, and $\eta : F \Rightarrow G$ a natural transformation with $\eta_j \in \mathcal{W}$ for every j. The functor $\int_J \eta : \int_J F \to \int_J G$ belongs to $\mathcal{W}$. In particular every modern fundamental localizer satisfies Loc (6 a), and the augmentation from the Grothendieck construction of a sieve-cover span to its union is in $\mathcal{W}_\infty$.

Proof. An object of $\int_J F$ is (j, x), $x \in F(j)$; an arrow $(j, x) \to (k, y)$ is (α, v) with $\alpha : j \to k$ and $v : F(\alpha)x \to y$. Composition is

$$(\beta, w)(\alpha, v) = (\beta\alpha, w\, F(\beta)(v)).$$

Over an object $s \in J$, evaluation

$$E_s : (\int_J F)/s \longrightarrow F(s), \qquad (j, x, \alpha : j \to s) \longmapsto F(\alpha)x$$

is left adjoint to $I_s(y) = (s, y, \mathrm{id}_s)$, with identity counit. The unit has component (α, id). These adjunctions give specified nerve homotopy equivalences, naturally in F. The square with horizontal maps induced by η and vertical maps E_s commutes. Thus $(\int_J \eta)/s \in \mathcal{W}$ by two out of three, and relative Theorem A proves the assertion. This is also the explicit argument behind Cisinski (2004, Proposition 1.2.4).

For a span $U \leftarrow K \rightarrow V$, its Grothendieck construction has three layers K, U, V. The subcategories $K \cup U$ and $K \cup V$ are full sieves, each retracting by the displayed adjunction onto its outer layer. Their intersection is K. Their nerves include NK by simplicial monomorphisms and cover the nerve of the entire construction, so that union is a homotopy pushout. The retractions give a map from the span $(K \cup U) \leftarrow K \rightarrow (K \cup V)$ to $U \leftarrow K \rightarrow V$, identity on the middle category and a weak equivalence on each arm. Homotopy-pushout invariance therefore supplies the specified weak-equivalence comparison with the double mapping cylinder. No equality of the nerve with a simplicial replacement is asserted. For arbitrary indexing categories, the general Grothendieck-construction comparison used below is Thomason's homotopy-colimit theorem, in the natural form proved by Chachólski and Scherer (2001, Theorem 26.8 and Corollary 26.9).

Now suppose $K = U \cap V$ and $X = U \cup V$ is a cover by full sieves. A chain in X lies in a piece containing its last object. Hence $NX = NU \cup_{NK} NV$ literally, and both attaching maps are simplicial monomorphisms. Realization gives inclusions of CW subcomplexes; the augmentation from the double mapping cylinder to this union is a homotopy equivalence. Thus the specified functor

$$q : \int (U \leftarrow K \rightarrow V) \longrightarrow X$$

which forgets each layer tag belongs to $\mathcal{W}_\infty$. A map between two cover squares induces a natural transformation of their spans. If its three input maps are in $\mathcal{W}$, the first paragraph and the two augmentation functors show that its union map is in $\mathcal{W}$. This is exactly Loc (6 a). Empty pieces cause no change: the same statements hold for the empty simplicial subcomplex and the disjoint union. □

B.3. A functorial categorical homotopy cofibre. [Proved]

For every functor $g : A \to B$ define M_g on disjoint layers A and B, with their original internal arrows and

$$\mathrm{Hom}_{M_g}(a, b) = \mathrm{Hom}_B(g(a), b), \qquad \mathrm{Hom}_{M_g}(b, a) = \varnothing.$$

Composition of crossing arrows is induced by g and composition in B. Adjoin an object t, with $\mathrm{Hom}(a,t) = \{*\}$ for every $a \in A$ and no arrows between t and the B-layer. Call the resulting category $K(g)$. There is a functorial homotopy-pushout square of spaces

$$\begin{array}{ccc} BA & \xrightarrow{Bg} & BB \\ \downarrow & & \downarrow \\ * & \longrightarrow & BK(g), \end{array}$$

where t supplies the basepoint. Thus $K(g)$ represents the unpointed homotopy cofibre, based at its cone point. A commutative square of functors induces a specified functor of these categories. If both vertical functors are in $\mathcal{W}$, the induced functor on $K(-)$ is in $\mathcal{W}$.

Proof. The formulas define a category: the only mixed associativity equations are the functoriality of g or associativity in B. The retraction $r_g : M_g \to B$, equal to g on the first layer and the identity on the second, is left adjoint to the inclusion $s_g : B \hookrightarrow M_g$. Its unit is the crossing arrow $a \to g(a)$ corresponding to $\mathrm{id}_{g(a)}$, and its counit is the identity. In particular $r_g \in \mathcal{W}_\infty$ and $r_g i_g = g$, where $i_g : A \hookrightarrow M_g$ is the other inclusion.

The full subcategories M_g and $A^{\triangleright} = A \cup \{t\}$ cover $K(g)$ by sieves and meet in A. The argument of B.2 identifies their union with the homotopy pushout of the specified span. The map of spans $(r_g, \mathrm{id}_A, A^{\triangleright} \to e)$ is objectwise a weak homotopy equivalence. It identifies that pushout with the displayed square, including the prescribed map Bg, rather than merely its source and target homotopy types.

For a commutative square $vg = g'u$ with $u : A \to A'$ and $v : B \to B'$, the induced functor sends the layers by u, v, sends t to t', and sends a crossing arrow $g(a) \to b$ by v. This is well-defined because $vg = g'u$. It is the map of the Grothendieck constructions of the spans $(B \leftarrow A \to e)$ and $(B' \leftarrow A' \to e)$. Apply B.2. In particular no condition on components, endomorphisms, or idempotents was used.

If $A = \varnothing$, then $K(g) = B \amalg e$. If also $B = \varnothing$, then $K(g) = e$. There is no functor from a nonempty A to an empty B. These are precisely the homotopy-pushout conventions needed in degrees zero and minus one; the empty source is not silently replaced by a chosen basepoint. □

B.4. The manuscript cylinder and the exact cofibration hypothesis. [Proved]

Let $\mathrm{Cof}_{\mathcal{W}}$ mean the functors $i : A \to B$ for which $B \amalg_A (-) : A\backslash\mathbf{Cat} \to B\backslash\mathbf{Cat}$ preserves every $\mathcal{W}$-map. Every classical Dwyer map is in $\mathrm{Cof}_{\mathcal{W}}$. Every functor $g : A \to B$ has a functorial factorization $g = p_g j_g$, with j_g a Dwyer map and $p_g \in \mathcal{W}_\infty$. The ordinary pushout $e \amalg_A \mathrm{Cyl}(g)$ is a categorical homotopy cofibre of g.

Proof. A Dwyer map is a full sieve inclusion $A \subseteq B$ such that its inclusion into the cosieve V generated by A has a right adjoint $r : V \to A$ restricting to the identity. The pushout comparison

$$\int (B \leftarrow A \to C) \longrightarrow B \amalg_A C$$

is in $\mathcal{W}_\infty$: this is the Dwyer pushout theorem, Thomason (1980, Proposition 4.3). Its hypotheses concern a Dwyer map, not an arbitrary sieve. Given $C \to C'$ in $\mathcal{W}$ under A, apply B.2 to the map of spans and use these two comparison functors. This proves the full undercategory definition of $\mathrm{Cof}_{\mathcal{W}}$, including preliminary cobase changes.

Take the exact interval of XII ms. 228,

$$I = \Phi_2 = (0 \to 1 \leftarrow 2 \to 3 \leftarrow 4).$$

The endpoint inclusion $\{0, 4\} \hookrightarrow I$ is a Dwyer map: its generated cosieve has the disjoint components $0 \to 1$ and $4 \to 3$, each retracting onto its endpoint by a right adjoint. Taking its product with A preserves the explicit adjunction. Put

$$\mathrm{Cyl}(g) = B \amalg_{A\times\{0\}} (A \times I).$$

The functor $p_g : \mathrm{Cyl}(g) \to B$ is induced by id_B and $g\,\mathrm{pr}_A$, and j_g is the inclusion of the free endpoint $A \times \{4\}$. Dwyer maps are closed under cobase change and composition, as follows directly from their right-adjoint description (or Thomason, 1980, Proposition 4.3 and Lemma 5.3(1)). Applying this to the two-endpoint inclusion, then the second summand $A \hookrightarrow B \amalg A$, proves that j_g is Dwyer. The map $B \to \mathrm{Cyl}(g)$ is the cobase change of the Dwyer weak equivalence $A \times \{0\} \to A \times I$. The pushout theorem makes it a weak homotopy equivalence. Since its composite with p_g is id_B, $p_g \in \mathcal{W}_\infty$.

The same theorem identifies the indicated ordinary pushout along j_g with the homotopy pushout of $B \leftarrow A \to e$. A specified comparison zigzag with $K(g)$ is obtained from the map of span Grothendieck constructions induced by p_g and from their augmentation functors. Every arrow in that zigzag is in $\mathcal{W}_\infty$. This justifies using either categorical model in tests of $\mathcal{W}$-asphericity.

Nothing here uses a pushout-product. The product $A \times \{0, 4\} \to A \times I$, its cobase change, and its composite with a summand inclusion are the three operations actually used by the manuscript. Loc (6 a) supplies the forward gluing in the manuscript's weaker framework; for our modern localizer it is already B.2. No converse excision is needed to construct a cofibre or to obtain enough $\mathcal{W}$-cofibrations. □

B.5. The exact reversible-cofibre criterion: Lead 1. [Proved]

For a modern fundamental localizer $\mathcal{W}$, the following conditions are equivalent: Loc (6 b); its single-square reverse implication Loc (6′ b); the biconditional Loc (6 C) for every $\mathcal{W}$-homotopy-cocartesian square; and

$$g \in \mathcal{W} \quad \Longleftrightarrow \quad K(g) \longrightarrow e \in \mathcal{W} \qquad \text{for every small-category functor } g. \tag{1}$$

Under these conditions Loc (6 a,b,c), W(7), and W(7 bis) all hold. Accessibility is unnecessary. The implication from left to right in (1) holds for every fundamental localizer.

Proof. If $g \in \mathcal{W}$, the map of spans from $(B \leftarrow A \to e)$ to $(B \leftarrow B \to e)$ induces $K(g) \to K(\mathrm{id}_B)$ in $\mathcal{W}$ by B.2. The latter target is weakly contractible; this proves the unconditional direction.

Suppose Loc (6′ b) holds and $K(g)$ is $\mathcal{W}$-aspherical. In the actual sieve-cover square

$$\begin{array}{ccc} A & \xrightarrow{i_g} & M_g \\ \downarrow & & \downarrow \\ A^{\triangleright} & \xrightarrow{j} & K(g), \end{array}$$

the functor j is in $\mathcal{W}$, since its source has a terminal object and its target is $\mathcal{W}$-aspherical. Reverse excision gives $i_g \in \mathcal{W}$. Then $g = r_g i_g \in \mathcal{W}$. This proof uses no assertion that i_g is itself a $\mathcal{W}$-cofibration; it need not be one.

Conversely suppose (1). For a homotopy-pushout square of spaces, the map between the cofibre of its top edge and the cofibre of its bottom edge is a weak equivalence. Indeed both are the colimit of the same expanded diagram after adjoining a cone point, with the two orders of taking pushouts interchanged. In the categorical models just constructed these are the augmentation maps from the iterated span Grothendieck construction: objects are triples consisting of the two span indices and an object in the corresponding category, and the canonical interchange of the indices preserves the arrow-composition formula in B.2. Thus the comparison is attached to the square, with no choice of an abstract isomorphism of homotopy types.

For a $\mathcal{W}$-homotopy-cocartesian square $(A \to B, A \to C, B \to D, C \to D)$, factor its comparison as $P = \int(B \leftarrow A \to C) \to D \in \mathcal{W}$. The preceding weak comparison and B.3 give a zigzag of $\mathcal{W}$-maps between $K(A \to B)$ and $K(C \to D)$. The latter step uses the commutative square $C \to P \to D$ with identity on C. Their asphericities agree, and (1) gives Loc (6 C). Sieve-cover squares are such squares by B.2, giving Loc (6′ a,b).

We spell out the cube implication, since a single-edge statement must not be substituted for it. For a morphism $f_i : X_i \to Y_i$ of two homotopy-pushout squares, put $C_i = \mathrm{cofib}((BX_i)_+ \to (BY_i)_+)$ in pointed spaces. Its underlying space is the homotopy cofibre in B.3. Adding a disjoint basepoint and taking a pointed cofibre are colimit functors. Consequently the four C_i form a homotopy-pushout square of pointed spaces. The forgetful functor to spaces creates these pushouts, whose indexing category is weakly contractible, so their underlying square is also homotopy cocartesian. The comparison is the same iterated-colimit comparison just specified. If $f_1, f_2, f_3 \in \mathcal{W}$, then C_1, C_2, C_3 are $\mathcal{W}$-aspherical. Applying forward gluing to $C_1 \leftarrow C_0 \to C_2$ and its map to $* \leftarrow C_0 \to *$ makes

$$C_3 \longrightarrow \Sigma C_0$$

a $\mathcal{W}$-equivalence. Thus ΣC_0 is $\mathcal{W}$-aspherical. Applying (1) to $C_0 \to *$ gives asphericity of C_0, and applying it to f_0 gives $f_0 \in \mathcal{W}$. Here statements about spaces are interpreted by their simplex categories; the natural last-vertex comparisons of Cisinski (2004, Proposition 2.2.3 and Corollary 2.2.4) transfer the specified comparisons to **Cat**. This proves Loc (6 b). The implication Loc (6 b)⇒Loc (6′ b) is the degenerate cube in Part A.

Finally, if $f_0, f_3 \in \mathcal{W}$, then C_0, C_3 are $\mathcal{W}$-aspherical. The basepoint section $* \to C_0$ is in $\mathcal{W}$, since its composite with $C_0 \to *$ is the identity. The induced map of spans $(C_1 \leftarrow * \to C_2) \to (C_1 \leftarrow C_0 \to C_2)$ and forward gluing give the specified $\mathcal{W}$-map $C_1 \vee C_2 \to C_3$, so that wedge is $\mathcal{W}$-aspherical. Each summand is a retract, by collapsing the other summand to the specified basepoint. Retract closure from B.1 makes each C_i, $i = 1, 2$, aspherical. The criterion gives Loc (6 c). All these cofibre spaces have a specified basepoint, including when an original category is empty. The argument never chooses a point in an empty source. □

B.6. The two-cofibre argument: Lead 2. [Proved]

Every modern fundamental localizer satisfying Loc (6 b), or only Loc (6′ b), contains $\mathcal{W}_{\mathbb{Z}}$. The conclusion does not require adding Loc (6 a) as an independent axiom.

Proof. Let $g : A \to B$ induce an isomorphism on integral homology in every degree, including degree zero. If $A = \varnothing$, then $H_0(BB; \mathbb{Z}) = 0$, so $B = \varnothing$ and g is the identity. Otherwise BA and BB are nonempty and $\pi_0(Bg)$ is a bijection: $H_0(Bg)$ is the free abelian-group map on the actual component function, whose matrix has exactly one 1 in each column; invertibility forces that function to be a bijection, also for infinite sets.

The homotopy cofibre $C = BK(g)$ is connected. More generally its components consist of one component containing the cone point and all components of BB met by BA, together with

each unmet component of BB. The reduced cofiber exact sequence, or the augmented cellular mapping-cone complex, now gives $\widetilde{H}_*(C;\mathbb{Z}) = 0$. Thus its unreduced suspension ΣC is simply connected by van Kampen and integrally acyclic by the suspension homology sequence. Both are CW spaces. If ΣC had a first nonzero homotopy group, the Hurewicz theorem would identify it with a nonzero homology group. Therefore it is weakly contractible; Whitehead's theorem also makes it contractible. These are the standard consequences in Hatcher (2002, Theorem 4.32 and Corollary 4.33).

The categorical cofibre $K(K(g) \to e)$ has precisely that weak homotopy type by B.3, so its map to e belongs to $\mathcal{W}_\infty \subseteq \mathcal{W}$. The criterion B.5, applied first to $K(g) \to e$ and then to g, gives $g \in \mathcal{W}$. The argument permits arbitrarily many source components; only their bijective correspondence with target components was used. The independent input is reverse excision. Forward gluing and the cofibre construction already follow from relative Theorem A and minimality, as proved above. □

B.7. The finite acyclic category, with a complete certificate. [Proved]

There is a finite category A with 35 objects and 159 arrows, all identities included, for which BA is connected and integrally acyclic and $\pi_1(BA) \neq 1$.

Proof. Use the words $w_0 = aaaaaBBB$ and $w_1 = bbbBABA$, where capitals mean inverses; their lengths are 8 and 7. Define a semisimplicial set with vertices v, c_0, c_1, loop edges a, b at v, radial edges $r_{k,i} : c_k \to v$ for $i \in \mathbb{Z}/|w_k|$, and triangles $t_{k,i}$. For a positive letter ℓ in position i set

$$(d_0 t_{k,i}, d_1 t_{k,i}, d_2 t_{k,i}) = (\ell, r_{k,i+1}, r_{k,i});$$

for ℓ^{-1} set this triple to $(\ell, r_{k,i}, r_{k,i+1})$. The endpoints of a radial edge are (v, c_k), and those of a loop edge are (v, v). The semisimplicial identities follow by taking endpoints; there are no higher simplices.

An object of A is one of these simplices. An arrow $\sigma \to \tau$ is an order-preserving injection $\theta : [\dim \sigma] \hookrightarrow [\dim \tau]$ with $\theta^* \tau = \sigma$. Composition is composition of injections. This specifies all arrows and composites, including distinct occurrences of the same face. The nerve is the barycentric subdivision of this semisimplicial realization: a nondegenerate nerve simplex is a flag of proper face injections, whose barycentres give the corresponding subdivided simplex. These maps agree with every face identification.

The realization is the cone-triangulation of the presentation complex

$$\langle a, b \mid a^5 b^{-3},\ b^3 (ab)^{-2} \rangle.$$

Its cellular boundary in degrees two and one is

$$d_2 = \begin{pmatrix} 5 & -2 \\ -3 & 1 \end{pmatrix}, \qquad \det d_2 = -1,$$

and $d_1 = 0$. This proves integral acyclicity. To certify nontrivial fundamental group, map a to $(1\ 4\ 3\ 2\ 5)$ and b to $(1\ 2\ 3)$ in A_5, composing right to left. The relators map to the identity: $a^5 = b^3 = 1$ and $ab = (1\ 5)(3\ 4)$. The image of a is nonidentity. This is the finite model of the acyclic complex in Hatcher (2002, Example 2.38, p. 142); the categorical specification above rechecks the supplied TOTASPH example.

The executable `code/finite_examples.py` constructs every arrow and composite and checks every identity and associativity equation. It finds normalized nerve-chain ranks $(35, 124, 90)$ and boundary ranks $(34, 90)$. Every nonzero Smith factor is 1. Thus the images of the boundaries are primitive, and the integral homology is $\mathbb{Z}$ in degree zero and zero otherwise.

The script checks the two permutations separately. The full arrow and composition tables, integer invariants, and output are included; the computer check is not being used to infer contractibility from acyclicity. □

B.8. The reverse axiom for $\mathcal{W}_\infty$, including Dwyer cobase changes. [Refuted]

The assertions that $\mathcal{W}_\infty$ satisfies W(7), W(7 bis), Loc (6 b), or Loc (6 c) are false. XII Theorem 2 at ms. 78 and XVI Theorem 5.4 insofar as it includes Loc (6) are false. The reverse edge implications can fail even for a pushout along a classical Dwyer map.

Proof. Take A of B.7. Adjoin two terminal cone points u, t, with one arrow from each object of A to each cone point and no arrows between the cone points. Call this category X, and put $U = A \cup \{u\}$ and $V = A \cup \{t\}$. The two pieces are full sieves, cover X, and meet in A. Their nerves are cones, while BX is the unreduced suspension of BA. It is simply connected and acyclic, hence contractible by the argument in B.6. Thus U, V, X are weakly contractible and A is not. The map of this cover square to the constant point square violates the cube reverse implication. The edge $U \to X$ is in $\mathcal{W}_\infty$ while $A \to V$ is not, violating W(7 bis). The one-point complement t has $U/t \cong A$, refuting the unqualified one-point converse. The program verifies the 37 objects and 231 arrows, both sieve conditions, and integral homology of X.

For the Dwyer statement use $g : A \to e$ and the factorization of B.4. The sieve $j_g : A \to \mathrm{Cyl}(g)$ is Dwyer, its target is weakly contractible, and $j_g \notin \mathcal{W}_\infty$. The pushout square

$$\begin{array}{ccc} A & \xrightarrow{j_g} & \mathrm{Cyl}(g) \\ \downarrow & & \downarrow \\ e & \longrightarrow & e \amalg_A \mathrm{Cyl}(g) \end{array}$$

has weakly contractible lower-right vertex, because it represents ΣBA. Both right and bottom arrows are weak equivalences, whereas both left and top arrows are not. This refutes both biconditionals printed in XII ms. 228, footnote 124, when their additional conservative axioms are omitted.

For completeness the finite model used in the code has objects y and (a, i) for $i = 1, 2, 3, 4$. Within and between the layers the arrows are those of $A \times \{1 \leftarrow 2 \to 3 \leftarrow 4\}$; from y to each $(a, 1)$ there is a unique arrow. There are no other arrows. The pushout replaces the entire fourth layer by a new object t, with a unique arrow $t \to (a, 3)$. Composition inside product layers uses composition in A; composition after either new arrow is the unique new arrow. There are 141 objects and 1149 arrows before this quotient, and 107 objects and 867 arrows after it. The code checks the quotient functor and all category axioms. The generated cosieve of the fourth layer consists of the third and fourth layers; forgetting their index is the displayed right adjoint, whose Hom-set bijections are checked. This is an explicit finite counterexample, not an appeal to an unspecified cofibrant replacement. □

B.9. The two variances of the one-point collage. [Proved]

For a covariant functor $H : A \to \mathbf{Set}$, let A_H be obtained by adjoining x with $\mathrm{Hom}(x, a) = H(a)$, no arrows $a \to x$, and $\mathrm{End}(x) = \{\mathrm{id}\}$. If $E = \mathrm{El}_A(H)$ and $\pi : E \to A$, there is a literal isomorphism of simplicial sets

$$NA_H \cong NA \amalg_{NE} (\Delta^0 * NE).$$

Thus BA_H is the homotopy cofibre of $B\pi$. Dually, for $H : A^{\mathrm{op}} \to \mathbf{Set}$, adjoining a last object t with $\mathrm{Hom}(a, t) = H(a)$ gives the right-cone formula with $NE * \Delta^0$. Under the equivalent

axioms in B.5, a full sieve $U \subset X$ with one complementary object t and $\mathrm{End}_X(t) = \{\mathrm{id}\}$ satisfies

$$U \to X \in \mathcal{W} \quad \Longleftrightarrow \quad U/t \to e \in \mathcal{W}.$$

Proof. A nondegenerate simplex not in NA begins with x, followed by an element $h \in H(a_0)$ and a string $a_0 \to \cdots \to a_n$. These are exactly the data of a simplex of NE; successive elements are determined by H. The first face forgets h and is $N\pi$, while all other faces and degeneracies agree with the cone. This proves the literal simplicial isomorphism, including repeated identities. Realization of $NE \hookrightarrow \Delta^0 * NE$ is a CW cofibration, so this ordinary pushout is a homotopy pushout. Passing to opposites gives the right-cone formula, with its variance changed as stated.

For the sieve use $H(u) = \mathrm{Hom}_X(u, t)$, whose category of elements is exactly U/t. The square $(U/t \to U, U/t \to e, U \to X, e \to X)$ is a homotopy-pushout square by the formula. Loc (6 C) identifies membership of its top and bottom parallel pair, or of its other parallel pair. The latter says $U \to X \in \mathcal{W}$ iff $U/t \to e \in \mathcal{W}$. This also includes $U/t = \varnothing$, when the attached point is isolated. The one-point statements from the supplied TOTASPH answer are therefore verified with their full stated axiom, and their failure without it is B.8. □

B.10. An explicit nontrivial acyclic group. [Proved]

Every small group H embeds in a small acyclic group $M(H)$ of cardinality at most $\max(|H|, \aleph_0)$; for infinite H equality holds. In particular $M(\mathbb{Z})$ is countable, nontrivial, and acyclic. Its one-object category has no nonidentity idempotents.

Proof. Form $m(H)$ from $H \times H$ by two HNN extensions with stable letters d, s and relations

$$d^{-1}(h, 1)d = (h, h), \qquad s^{-1}(h, 1)s = (1, h) \quad (h \in H).$$

The associated subgroups in both extensions are isomorphic to H. The normal-form theorem for HNN extensions embeds the base group; equivalently a reduced word with stable letters cannot equal a nontrivial base-group word unless an adjacent stable-letter pair pinches through its specified associated subgroup. Thus these are injections. The resulting group is generated by the first copy of H, d, and s; it has the relations $h^d = hh^s$ and $[h', h^s] = 1$. Iterate these injections and take the union $M(H) = \bigcup_{r \geq 0} m^r(H)$. Cardinality and the embedding assertion follow from words of finite length in the indicated generators.

We include the homology argument rather than relying on the earlier AI answer. Let $f : L \to H$ kill positive homology over a field k below degree n. The homomorphism $q : H \times H \to m(H)$ given by $q(h, h') = h(h')^s$ is defined by the commutation relations. Its restrictions to the two factors and to the diagonal are conjugate to the inclusion $j : H \to m(H)$. In the Künneth decomposition of $(f \times f)_* \Delta_* z$, $z \in H_n(L; k)$, all mixed positive-degree terms vanish. The remaining terms are $f_* z \otimes 1 + 1 \otimes f_* z$. Applying q_* and conjugacy invariance gives $(jf)_* z = 2(jf)_* z$, hence $(jf)_* z = 0$. Induction shows that n consecutive mitoses kill field homology through degree n. Every bar cycle in the union occurs at a finite stage, so all positive field homology of $M(H)$ vanishes. Rational vanishing makes its integral positive homology torsion. The universal-coefficient sequence with $\mathbb{F}_p$ in the next degree says that this torsion group has no elements of order p, for any prime p. It is zero.

This is the mitosis proof checked against Baumslag et al. (1980, Proposition 4.1, Theorem 4.2, and Lemma 5.4). For any group an equation $h^2 = h$ implies $h = 1$ by multiplication with h^{-1}, proving the final statement. No simplicity assertion is needed for the counterexamples; the stronger arbitrary-cardinal simple-overgroup assertion in CATHOT is not used here. □

B.11. The one-object test required after CATHOT. [Proved]

Let G be any nontrivial integrally acyclic small group, and let $\mathsf{B}G$ denote its one-object category. For every fundamental localizer satisfying Loc (6 b), $\mathsf{B}G \to e$ belongs to $\mathcal{W}$, although it does not induce an isomorphism on fundamental groups. Thus such a localizer cannot require π_1-isomorphisms even on one-object sources.

Proof. The classifying nerve realizes to the usual $K(G,1)$: the nerve of the translation category with object set G and a unique arrow between each pair is contractible, and its free G-quotient is $N\mathsf{B}G$. Its fundamental group is therefore G. The two-cone category on $\mathsf{B}G$ has objects $*, u, t$, endomorphisms G at $*$, unique arrows $* \to u$ and $* \to t$, identities at u, t, and no other arrows. The composites of any group element with either crossing arrow are that same crossing arrow. Its realization is ΣBG, which is simply connected and acyclic, hence contractible. Apply B.5 to $\mathsf{B}G \to e$, or apply B.6 directly. The same cover by the two cones gives the infinite-arrow, three-object version of B.8. All its comma categories at the new terminal points are explicitly the indicated group category or a terminal cone; no idempotent issue is involved. □

B.12. The amalgam is a homological suspension model. [Proved]

Let $H \hookrightarrow G$ be an injective homomorphism of small groups with G integrally acyclic. The double mapping cylinder of $BG \leftarrow BH \to BG$ is a classifying space for $D = G *_H G$. Collapsing its two vertex spaces gives a specified map $BD \to \Sigma BH$ inducing an integral homology isomorphism. In particular this holds when H is acyclic, but it does not require that assumption, and it does not assert a weak homotopy equivalence.

Proof. The amalgam normal form embeds both vertex groups and identifies the fundamental group of this graph of spaces with D. The universal cover is a tree of contractible vertex and edge spaces over the Bass–Serre tree. Attaching a terminal vertex of a finite subtree preserves contractibility, so each finite subtree union is contractible. A cellular sphere and a cellular nullhomotopy problem meet finitely many cells and therefore a finite subtree; the entire cover is weakly contractible. This proves the asserted classifying-space model. One may use mapping cylinders of the classifying-space maps so that every edge-space attachment is a cofibration.

The quotient map to ΣBH collapses both copies of BG to the two suspension points and is the identity on the intervening cylinder of BH. The natural Mayer–Vietoris sequences for these two homotopy pushouts compare by the identities on BH and by $BG \to *$ on both vertex spaces. These vertex maps are integral homology isomorphisms, including degree zero. The exact-sequence comparison proves that the quotient induces isomorphisms in every degree. This verifies the assertion from CATHOT B.8 and identifies the precise map. For acyclic nontrivial H, the suspension is contractible while D contains G; this illustrates why only homology equivalence is claimed. □

B.13. The upper interval is not the reversible class. [Refuted]

The assertion that $\mathcal{W}_{\mathbb{Z}} \subseteq \mathcal{W}$ suffices for Loc (6 b) is false. In particular $\mathcal{W}_0$ contains $\mathcal{W}_{\mathbb{Z}}$ and fails the cofiber criterion.

Proof. Let g be the fold from the discrete two-object category $\{a_0, a_1\}$ to e. It is not in $\mathcal{W}_0$. The category $K(g)$ has four objects a_0, a_1, b, t, its four nonidentity arrows are $a_i \to b$ and $a_i \to t$, and composition only involves identities. Its realization is the four-edge circle, hence is nonempty and connected and therefore $\mathcal{W}_0$-aspherical. The code checks these data and the

Betti numbers $(1, 1)$. The inclusion $\mathcal{W}_{\mathbb{Z}} \subseteq \mathcal{W}_0$ follows from the degree-zero argument in B.6. Thus a minimality theorem for the reverse axiom is not an upward-closure theorem for that axiom. □

C Homology localizers and truncated homotopy types

C.1. The families and their model structures. [Known]

For a $\mathcal{U}$-small spectrum E, we use the unreduced convention

$$\mathcal{W}_E = \{u : A \to B \mid E \wedge \Sigma^\infty(BA)_+ \longrightarrow E \wedge \Sigma^\infty(BB)_+ \text{ is an equivalence}\}.$$

For a unital ring R, not assumed commutative and including the zero ring, $\mathcal{W}_R$ means $\mathcal{W}_{HR}$. For an integer $n \geq 0$, $\mathcal{W}_n$ consists of the functors inducing a bijection on π_0 and isomorphisms on all based π_i, $1 \leq i \leq n$. These are accessible fundamental localizers. Their associated simplicial localizations have monomorphisms as cofibrations. On **Cat** they admit the left proper Thomason model structure, with the usual Thomason cofibrations. For $\mathcal{W}_n$ this structure is also right proper.

The existence assertions are Cisinski (2006, Theorem 9.4.2, Corollary 9.4.3, Theorem 5.2.15). For the truncations, the defining sphere generator and properness are stated in Cisinski (2006, 9.2.1, Corollary 9.2.15, Proposition 9.2.17). In particular the sphere generator is $\Delta/\partial\Delta^{n+2}$, of homotopy type S^{n+1}; the shift in its index is essential. We use these existence theorems with their proofs in the cited source, rather than re-proving the small-object argument. The remaining axiom assertions below have explicit proofs.

C.2. Sums, connectedness, filtered colimits, and accessibility. [Proved]

Each $\mathcal{W}_n$, $n \geq 0$, and each $\mathcal{W}_E$ satisfies Loc (4), (8), and (9). Each $\mathcal{W}_n$ satisfies Loc (5). For $\mathcal{W}_E$, Loc (5) holds exactly when $E \not\simeq 0$. Thus it holds for $\mathcal{W}_R$ exactly when $R \neq 0$.

Proof. The nerve of a disjoint union is the disjoint union of the nerves. Generalized homology of a disjoint union is the direct sum of the unreduced homology groups. A block-diagonal map of direct sums is an isomorphism if and only if each block is. This proves both implications of Loc (4), including arbitrary small index sets and the empty family. The same assertion for $\mathcal{W}_n$ follows by taking components and based homotopy groups component by component.

Suppose $E \neq 0$ and choose $k \in \mathbb{Z}$ with $\pi_k E \neq 0$. If a map of spaces $f : X \to Y$ omits a component Y_j, the summand $E_k(Y_j)$ contains a copy of $\pi_k E$, split by any point and the augmentation, and cannot be hit by $E_k(f)$. If distinct components $X_i, X_{i'}$ map into the same component of Y, choose one point in each. Their images are joined by a path. For nonzero $v \in \pi_k E$, the two point inclusions define the nonzero element $(v, -v)$ in the corresponding direct-sum summands of $E_k(X)$, and its image is zero. Thus an E-equivalence is bijective on components. This also proves that $\varnothing \to X$ is an E-equivalence only when $X = \varnothing$. For $E = 0$ every functor belongs to $\mathcal{W}_E$, and the fold $e \amalg e \to e$ violates Loc (5).

The q-simplices of a nerve are finite strings of composable arrows. Consequently N commutes with filtered colimits. Filtered colimits of simplicial sets model their filtered homotopy colimits: every finite simplicial sphere, disk, and homotopy factors through one stage, and equalities between finitely many simplices hold at a later stage. This proves commutation of components and based homotopy groups with the filtered colimit and establishes Loc (8) for $\mathcal{W}_n$. Suspension spectra and smashing with E preserve homotopy colimits, and stable homotopy groups commute with filtered homotopy colimits. This proves Loc (8) for $\mathcal{W}_E$.

Loc (9) is the accessibility assertion of C.1; here accessibility is for the full subcategory of $\mathbf{Cat}^{[1]}$, as in XVI, manuscript pp. 74–78. □

C.3. The cofibration columns. [Proved]

Every $\mathcal{W}_E$, including $E = 0$, satisfies Loc (6 a), (6 b), and (6 c). Every $\mathcal{W}_n$, $n \geq 0$, satisfies Loc (6 a) and fails both Loc (6 b) and Loc (6 c).

Proof. For a covering by two sieves $X_1, X_2 \subset X_3$, with intersection X_0, every simplex of NX_3 lies in one of NX_1, NX_2. Indeed the last object of its chain lies in one sieve, and closure under sources then places every earlier object in that sieve. The nerve square is therefore a pushout along simplicial monomorphisms, hence a homotopy pushout. After applying $E \wedge \Sigma^\infty(-)_+$ it yields the cofiber sequence of spectra

$$E \wedge \Sigma^\infty(BX_0)_+ \longrightarrow (E \wedge \Sigma^\infty(BX_1)_+) \oplus (E \wedge \Sigma^\infty(BX_2)_+) \longrightarrow E \wedge \Sigma^\infty(BX_3)_+.$$

A map of these sequences satisfies two out of three for equivalences. Equivalence of the middle direct-sum map is equivalent to equivalence of its two diagonal blocks. These statements prove precisely the three implications (6 a), (6 b), and (6 c). No connectivity or nonemptiness assumption was made.

For $\mathcal{W}_n$, homotopy pushouts preserve n-equivalences because $\tau_{\leq n} : \mathcal{S} \to \mathcal{S}_{\leq n}$ is a left adjoint. This proves (6 a) on the same nerve squares. To disprove the other two clauses, let P_n be the poset of nonempty proper subsets of $\{0, \ldots, n+1\}$, ordered by inclusion. Its objects, arrows, and composition are thus completely specified: there is one arrow for each inclusion and composites are inclusions. Its nerve is the barycentric subdivision of $\partial\Delta^{n+1}$ and realizes to S^n. Adjoin two distinct maximal objects $+$ and $-$, each above every object of P_n, with no arrow between them. Denote the resulting poset by $S(P_n)$ and its two cones by $C_+(P_n), C_-(P_n)$. These cones are sieves, cover $S(P_n)$, have intersection P_n, and have terminal objects. Their union realizes to the unreduced suspension S^{n+1}.

Map this cover to the cover $e = e \cup e$. The maps from both cones and from $S(P_n)$ to e are n-equivalences, whereas $P_n \to e$ is not. This is the full cube counterexample to (6 b). For (6 c), map the cover $C_-(P_n) = P_n \cup C_-(P_n)$ into $S(P_n) = C_+(P_n) \cup C_-(P_n)$. The intersection map is id_{P_n}, the union map $C_-(P_n) \hookrightarrow S(P_n)$ is an n-equivalence, and the first piece map $P_n \hookrightarrow C_+(P_n)$ is not. The construction includes $n = 0$, where P_0 has two objects and $S(P_0)$ realizes to a circle. The program `code/ce_sphere_cover.py` checks the finite category laws, sieve conditions, and nerves of the first four instances. □

C.4. A nullity obstruction valid for every nonzero spectrum. [Proved]

Let $E \neq 0$, and let A be an E-aspherical space. Then A is nonempty and connected, and $\mathrm{Hom}(\pi_1(A, a), \mathbb{Z}) = 0$ for every $a \in A$. For a free group F, the constant-map inclusion $BF \to \mathrm{Map}(A, BF)$ is an equivalence. Thus BF is null with respect to every E-aspherical space.

Proof. Connectedness and nonemptiness follow from C.2. If a homomorphism $\pi_1(A, a) \to \mathbb{Z}$ is nonzero, its image is $m\mathbb{Z}$ for some $m \geq 1$. Dividing by m gives a surjection $\chi : \pi_1(A, a) \twoheadrightarrow \mathbb{Z}$. Choose a loop mapping to 1. The corresponding pointed maps $s : S^1 \to A$ and $h : A \to K(\mathbb{Z}, 1) = S^1$ satisfy $hs \simeq \mathrm{id}_{S^1}$, since pointed maps to $K(\mathbb{Z}, 1)$ are classified by their homomorphisms on fundamental groups. Therefore ΣE is a retract of $E \wedge \Sigma^\infty A$, where the latter uses the chosen basepoint. This last spectrum is zero by E-asphericity, a contradiction.

Any nontrivial subgroup H of a free group is free: it is the fundamental group of the connected covering graph associated with H. A nontrivial free group admits a surjection to $\mathbb{Z}$. Consequently the image of any homomorphism $\pi_1(A, a) \to F$ is trivial. Mapping to the 1-type BF depends only on the fundamental groupoid of A. Since A is connected, this groupoid is pointedly equivalent to the one-object groupoid $B\pi_1(A, a)$. The pointed mapping groupoid then has the homomorphisms $\pi_1(A, a) \to F$ as objects and conjugacies with conjugating element forced to be 1 as morphisms. Thus $\mathrm{Map}_*(A, BF)$ is the discrete space of these homomorphisms, and is therefore contractible. Evaluation at a has this mapping space as fiber, so the constant-map section is an equivalence. □

C.5. No nonzero homology localizer has enough intrinsic fibrations. [Proved]

For every spectrum $E \neq 0$, $\mathcal{W}_E$ fails Loc (7 F). In particular this holds for $\mathcal{W}_R$ for every nonzero unital ring R, with no restriction on its characteristic or torsion.

Proof. Let $F = \langle x, y \rangle$ and let

$$\alpha : F \longrightarrow F, \qquad \alpha(x) = x, \quad \alpha(y) = yxyx^{-1}y^{-1}.$$

The completely specified group categories, the proof that $B\alpha$ is an integral homology equivalence, and the finite S_3 witness proving that α is not surjective are given in F.8. A graph model $BF \simeq S^1 \vee S^1$ shows directly that its reduced suspension-spectrum map is the identity matrix: the exponent-sum matrix of α is I_2. Hence $B\alpha$ is an E-equivalence for every spectrum, including nonconnective spectra. By C.4, both copies of BF are local for nullification at all E-aspherical objects. This nullification exists and is accessible by Cisinski (2006, Proposition 6.1.8), since $\mathcal{W}_E$ is accessible. A localization is fully faithful on its local objects, so it cannot invert this non-equivalence between them. Thus $\mathcal{W}_E$ is not generated by its aspherical categories. The equivalence of aspherical generation and Loc (7 F), proved in E.2, completes the argument. □

C.6. Failure of the stronger fibration axioms. [Proved]

For $E \neq 0$, and for every finite $n \geq 0$, both Loc (7) and the weakest Loc (7 $D_{\mathcal{W}_\infty}$) fail for $\mathcal{W}_E$ and $\mathcal{W}_n$. For $E = 0$ all the fibration clauses hold.

Proof. Here is a direct counterexample to the weakest clause for $\mathcal{W}_E$. With F and α as in C.5, define a groupoid $\mathcal{P}$ with object set F, and with $\mathrm{Hom}_{\mathcal{P}}(g, g') = F$. Write $h : g \to g'$ for the arrow labelled by h. Its composite with $h' : g' \to g''$ is labelled $h'h$. Define

$$p : \mathcal{P} \longrightarrow BF, \qquad p(h : g \to g') = g'\alpha(h)g^{-1}.$$

This is a functor because the formula respects multiplication. The functor $j : BF \to \mathcal{P}$ at object 1, sending h to its label, is an equivalence of categories, and $pj = B\alpha$. Thus $p \in \mathcal{W}_E$. For an arrow k in the base, the label-1 arrow $g \to kg$ lifts it. These lifts and their inverses make p both a Grothendieck fibration and opfibration, with isomorphic transport functors; in particular $p \in \mathrm{Parf}_{\mathcal{W}_\infty}$. Its strict fiber over the unique object has components $F/\alpha(F)$, since its arrows satisfy $g'\alpha(h)g^{-1} = 1$. There are at least two components, because α is not surjective. This fiber is not E-aspherical by C.2. Hence p is in $\mathrm{Parf}_{\mathcal{W}_\infty} \cap \mathcal{W}_E$ and is not universally in $\mathcal{W}_E$, exactly the failure of Loc (7 $D_{\mathcal{W}_\infty}$).

For $\mathcal{W}_n$, the map $* \to K(\mathbb{Z}, n+1)$ is an n-equivalence, but its homotopy fiber $K(\mathbb{Z}, n)$ is not n-aspherical (for $n = 0$ this fiber is the discrete set $\mathbb{Z}$). Represent the map by a functor between simplex categories and apply the $\mathcal{W}_\infty$-equivalence–$\mathrm{Parf}_{\mathcal{W}_\infty}$ factorization described

in E.4. The resulting perfect functor remains in $\mathcal{W}_n$, its strict fibers model those homotopy fibers, and so it violates the same clause. Loc (7) implies this clause by Part A. When $E = 0$, all functors belong to $\mathcal{W}_E$, every functor is in $\mathrm{Fib}_{\mathcal{W}_E}$, and the identity factorization proves (7 F). □

C.7. The complete axiom table for the named families. [Proved]

With the conventions of C.1, the following table is exact. In its $7D$ column we use the weakest version $7D_{\mathcal{W}_\infty}$; failure there also means failure of all stronger $7D$ versions.

	4	5	6a	6b	6c	7	7D	7F	8	9
$\mathcal{W}_R,\ R \neq 0$	+	+	+	+	+	−	−	−	+	+
$\mathcal{W}_E,\ E \neq 0$	+	+	+	+	+	−	−	−	+	+
$\mathcal{W}_n,\ 0 \leq n < \infty$	+	+	+	−	−	−	−	+	+	+
$\mathcal{W}_0$	+	+	+	−	−	−	−	+	+	+
$\mathcal{W}_E,\ E = 0$	+	−	+	+	+	+	+	+	+	+

Proof. All entries except $7F$ for $\mathcal{W}_n$ are proved in C.1–C.6. The sphere generator of C.1 makes $\mathcal{W}_n$ aspherically generated, so E.2 proves that remaining entry. Repeating the $\mathcal{W}_0$ row records explicitly the case singled out in XVI, manuscript pp. 68–70 and 91. In particular the answer to the manuscript's question about $\mathcal{W}_n$ and (7 F) is affirmative, while the answer for every nonzero homology theory is negative. □

C.8. An intrinsic description of $\mathrm{Fib}_{\mathcal{W}_0}$. [Proved]

Let $p : T \to B$ be a functor. For a component $B_j \subset B$, put $T_j = T \times_B B_j$, and for $b \in B_j$ let T_b be the strict fiber. Then $p \in \mathrm{Fib}_{\mathcal{W}_0}$ if and only if both of the following conditions hold. First, $\pi_0(T_b) \to \pi_0(T_j)$ is a bijection for every b. Second, for every arrow $\beta : b \to b'$ and every $c \in \pi_0(T_j)$, there exists an arrow of T over β whose source, and hence target, lies in c. Equivalently, for every functor $X \to B$ the canonical map is a bijection

$$\pi_0(T \times_B X) \longrightarrow \pi_0(T) \times_{\pi_0(B)} \pi_0(X).$$

Proof. Suppose $p \in \mathrm{Fib}_{\mathcal{W}_0}$. Pulling back the $\mathcal{W}_0$-map $e_b \to B_j$ proves the first condition. Pulling back either endpoint inclusion $e \to [1]$, where $[1] \to B$ represents β, shows that both fibers induce bijections on components into $T \times_B [1]$. Its only arrows between its two layers are the arrows of T over β. The first condition labels the fiber components by $c \in \pi_0(T_j)$; therefore each such label must have an arrow between the layers. This proves the second condition.

Conversely, fix a component X_a mapping into B_j. For each $x \in X_a$, the objects over x labelled by c form a nonempty connected category. An arrow $x \to x'$ in X_a has a lift joining these two categories by the second condition. A finite zigzag in X_a therefore joins all objects with a given label c in $T \times_B X_a$. Distinct labels cannot be joined, since their images in T_j have distinct components. This proves the displayed bijection, naturally in $X \to B$. A $\mathcal{W}_0$-equivalence over B induces a bijection on its right-hand side, and hence on its left-hand side. Empty fibers and empty components obey the same argument: if T_j is empty, all the relevant fibers are empty and both conditions are vacuous there. □

C.9. The n-equivalence fibration class. [Proved]

Let $n \geq 0$ and $p : T \to B$ be a functor. Then $p \in \mathrm{Fib}_{\mathcal{W}_n}$ if and only if, for every $X \to B$, the canonical comparison is an equivalence in $\mathcal{S}_{\leq n}$:

$$\tau_{\leq n} B(T \times_B X) \longrightarrow \tau_{\leq n} BT \times_{\tau_{\leq n} BB} \tau_{\leq n} BX. \tag{C.9.1}$$

The right-hand pullback is a homotopy pullback of n-types. In particular every $\mathcal{W}_n$-Thomason fibration belongs to $\mathrm{Fib}_{\mathcal{W}_n}$. This supplies a functorial factorization for (7 F) at every finite n.

Proof. Right properness in C.1 implies that a strict pullback along a model fibration preserves all weak equivalences. It also implies that such strict pullbacks represent derived pullbacks, by taking a fibrant replacement of the base and comparing the two pullback diagrams by right properness. Thus Thomason fibrations have the asserted property.

For $p \in \mathrm{Fib}_{\mathcal{W}_n}$, factor p over B as a $\mathcal{W}_n$-equivalence $T \to \widehat{T}$ followed by a Thomason fibration $\widehat{T} \to B$. Both T and $\widehat{T}$ are objects of the Brown fibration category over B with intrinsic fibrations $\mathrm{Fib}_{\mathcal{W}_n}$. The dual Brown lemma of Part A says that a weak equivalence between two such fibrations is preserved by every strict base change. Consequently $T \times_B X \to \widehat{T} \times_B X$ is a $\mathcal{W}_n$-equivalence. The preceding model-category calculation for $\widehat{T}$ proves (C.9.1), including its canonical map. Conversely, if (C.9.1) is an equivalence for every X, apply it to the source and target of any $\mathcal{W}_n$-map over B. Their right-hand sides are equivalent, so the strict pullback map is a $\mathcal{W}_n$-map. This is precisely the definition of $\mathrm{Fib}_{\mathcal{W}_n}$. This is the specialization to $\mathcal{W}_n$ of the intrinsic weak-fibration criterion in Cisinski (2006, Proposition 6.2.3); its properness hypothesis is supplied by C.1. □

C.10. Bounds for the homological fibration classes. [Proved]

For any spectrum E, $\mathrm{Fib}_{\mathcal{W}_E}$ contains all product projections $B \times K \to B$, all inclusions of unions of components, and the closure of these maps under composition, strict base change, and retracts. More generally it contains every functor p for which Np universally preserves E-equivalences of simplicial sets under strict pullback. An upper bound is the class Q_E of functors $p : T \to B$ such that, for every $m \geq 0$, every $\sigma : [m] \to B$, and every $i \in [m]$, the strict fiber inclusion

$$T_{\sigma(i)} \longrightarrow T \times_B [m]$$

belongs to $\mathcal{W}_E$: one has $\mathrm{Fib}_{\mathcal{W}_E} \subseteq Q_E$. For every $E \neq 0$ this upper inclusion is strict. Indeed $\mathrm{Fib}_{\mathcal{W}_E}$ omits the explicit three-sheeted action-groupoid covering described below, although that covering lies in $\mathrm{Parf}_{\mathcal{W}_\infty} \cap Q_E$. For a fixed p, membership in $\mathrm{Fib}_{\mathcal{W}_E}$ can be tested on a set of bounded-cardinality $\mathcal{W}_E$-maps over its base, rather than on a proper class.

Proof. For a product projection the pullback map is $u \times \mathrm{id}_K$. The suspension spectrum of its nerve, after adjoining a basepoint, is $\Sigma^\infty(Bu)_+ \wedge \Sigma^\infty(BK)_+$. Smashing an E-equivalence with the second factor preserves equivalence. Component inclusions follow from Loc (4), and the three closure properties follow directly from the definition of Fib and closure of $\mathcal{W}_E$ under retracts. The nerve preserves strict pullbacks, proving the simplicial sufficient condition. It is often called a sharpness condition; we do not identify it with the unrelated assertion that a map is merely a fibration in a left Bousfield model structure that is not right proper. For the upper bound, the functor $[0] \xrightarrow{i} [m]$ over B is in $\mathcal{W}_\infty \subseteq \mathcal{W}_E$, so its pullback along any $p \in \mathrm{Fib}_{\mathcal{W}_E}$ has to be in $\mathcal{W}_E$. This is exactly the defining condition of Q_E, with all its quantifiers.

Let $F = \langle x, y \rangle$, $S = \{1, 2, 3\}$, and let $\rho : F \to S_3$ send x to (12) and y to (13). Define the action groupoid $S//F$ to have objects S and arrows labelled $g \in F$ from s to $\rho(g)s$, with composition given by multiplication in F. Its projection $p : S//F \to BF$ is a discrete fibration and opfibration, with bijective transports. The action is transitive, so its total groupoid is connected. Pulling back along $B\alpha$ from C.5 replaces the action by $\rho\alpha$, which sends both generators to (12). Its orbits are $\{1, 2\}$ and $\{3\}$. Thus the pullback map is not a $\mathcal{W}_0$-map, hence not a $\mathcal{W}_E$-map for $E \neq 0$, although $B\alpha \in \mathcal{W}_E$. The group laws and these orbit calculations are checked in `code/f_free_group_witness_output.json`. To check that this same covering

lies in Q_E, pull it back along $\sigma : [m] \to BF$. Transport from vertex 0 identifies the resulting category with the disjoint union of three copies of $[m]$. Every strict fiber inclusion is therefore a weak homotopy equivalence. Thus the upper bound is strictly larger than the intrinsic fibration class.

Finally fix B. Accessibility of $\mathcal{W}_E \subset \mathbf{Cat}^{[1]}$ gives a regular cardinal κ and a set $\mathcal{T}$ of $\mathcal{W}_E$-maps whose κ-filtered colimits exhaust $\mathcal{W}_E$; choose κ so that the inclusion preserves these colimits. The cardinalities of the objects and arrows in all sources and targets of members of $\mathcal{T}$ have a common bound, since $\mathcal{T}$ is a set. Equip each member of $\mathcal{T}$ with every possible functor from its target to B; these choices again form a set. For a given u over B, any chosen presentation by $\mathcal{T}$ becomes a presentation over B by composing the stage-to-target maps with the specified map to B. Strict pullback along p commutes with these filtered colimits, because objects and arrows are computed by finite limits of sets. Loc (8) therefore reduces the test to that set of bounded stages over B. This is an exact set-sized test, not a claimed finite decision procedure. □

C.11. The cofibration threshold at $\mathcal{W}_1$. [Proved]

Every functor is a $\mathcal{W}_0$-cofibration. If a fundamental localizer $\mathcal{W}$ contains $\mathcal{W}_1$, every functor injective on objects is a $\mathcal{W}$-cofibration. Hence Loc (6 A) holds in this range. These facts do not license replacing the products with endpoint inclusions in XII, manuscript pp. 228–234, by pushout-products.

Proof. The functor $\pi_0 : \mathbf{Cat} \to \mathbf{Set}$ is left adjoint to the discrete-category inclusion and preserves all pushouts. Pushout in sets preserves an isomorphism along every map. This proves the first assertion.

For the second, groupoid completion $\Pi_1 : \mathbf{Cat} \to \mathbf{Gpd}$ preserves pushouts and does not change objects. The folk model structure on groupoids has equivalences as weak equivalences, object injections as cofibrations, and all objects cofibrant and fibrant. Its pushout along an object injection computes the homotopy pushout. One can see the relevant gluing assertion directly by representing morphisms in the groupoid pushout as alternating words and cancelling adjacent arrows in each factor. Consequently, for a span in **Cat** with one leg injective on objects, the canonical comparison from its Grothendieck homotopy colimit to its ordinary pushout is a $\mathcal{W}_1$-equivalence. The comparison is therefore in $\mathcal{W}$. Homotopy colimits preserve objectwise $\mathcal{W}$-equivalences, and two out of three now shows that the given object injection preserves every $\mathcal{W}$-map under cobase change. A categorical mapping cylinder provides a factorization by an object injection followed by a categorical homotopy equivalence, proving (6 A). The last sentence is a statement about the specific construction: none of these assertions states that the pushout-product of two designated maps is a cofibration for a localizer below $\mathcal{W}_1$. □

D The minimal reversible localizer and the proposed simple types

D.1. Minimality of integral homology: the valid part of Lead 3. [Proved]

The class $\mathcal{W}_{\mathbb{Z}}$ of all functors inducing integral homology isomorphisms on nerves is a fundamental localizer satisfying Loc (6 a,b,c). It is the smallest modern fundamental localizer satisfying Loc (6 b), and is strictly larger than $\mathcal{W}_\infty$.

Proof. Identities, two out of three, and the weak retraction axiom follow by applying the functor $H_*(-;\mathbb{Z})$. A terminal object contracts the nerve. To verify relative Theorem A without

presupposing that this class is fundamental, take a triangle $A \xrightarrow{u} B \to C$ with each $A/c \to B/c$ a homology equivalence. The augmentations

$$\int_{c\in C}(A/c) \longrightarrow A, \qquad \int_{c\in C}(B/c) \longrightarrow B$$

are weak homotopy equivalences. For example the first sends $(c, a, \gamma : vu(a) \to c)$ to a; it is right adjoint to the section $a \mapsto (vu(a), a, \mathrm{id})$, since a morphism out of this section is uniquely determined by the arrow in A. The augmentations commute with u. The simplicial replacement of either diagram has the first-quadrant homology spectral sequence

$$E^1_{p,q} = \bigoplus_{c_0\to\cdots\to c_p} H_q(B(A/c_0);\mathbb{Z}) \implies H_{p+q}\left(B\int_C (A/c);\mathbb{Z}\right),$$

and likewise for B. This is the normalized double chain complex of the replacement; its filtration is finite in each total degree. The induced map is an isomorphism on the E^1 page, hence on homology. The two augmentations prove that $u \in \mathcal{W}_{\mathbb{Z}}$.

For a sieve cover $X = X_1 \cup X_2$ with intersection X_0, every nondegenerate chain lies in a piece containing its last vertex. Thus there is a natural short exact sequence of free chain complexes

$$0 \to C_*(NX_0;\mathbb{Z}) \xrightarrow{(i_*,-j_*)} C_*(NX_1;\mathbb{Z}) \oplus C_*(NX_2;\mathbb{Z}) \to C_*(NX;\mathbb{Z}) \to 0.$$

For a map of such squares, take the algebraic cones of the four induced chain maps. They form the same short exact sequence, with the middle term the direct sum of the two piece cones. The homology long exact sequence says that acyclicity of any two of the intersection cone, the middle cone, and the union cone implies acyclicity of the third. A direct sum of two complexes is acyclic exactly when each summand is acyclic. This proves all three cube implications, including every empty-piece case, using unreduced chains. The reverse-edge version follows from the degenerate cube.

Every other fundamental localizer satisfying the reverse axiom contains $\mathcal{W}_{\mathbb{Z}}$ by B.6. Strictness over $\mathcal{W}_\infty$ is witnessed by the functor $A \to e$ in B.7. The assertion that $\mathcal{W}_\infty$ satisfies the complete manuscript list including Loc (6 b) is therefore false, not merely unsupported by its proposed proof. □

D.2. The incompatibility of the two poles, with the connectivity correction. [Proved]

There is no modern fundamental localizer satisfying Loc (5), Loc (7D$_{\mathcal{W}_\infty}$), and Loc (6 b). If Loc (5) is omitted, the unique fundamental localizer satisfying both Loc (7D$_{\mathcal{W}_\infty}$) and Loc (6 b) is Fl(**Cat**).

Proof. The fibration classification proved in Part E says that Loc (7D$_{\mathcal{W}_\infty}$) leaves exactly $\mathcal{W}_\infty$ and the total class. The first fails Loc (6 b) by B.8. The total class satisfies every stability implication, including the two in this statement, but it fails Loc (5), since $\varnothing \to e$ has a nonbijective component map. This last point corrects the wording of Lead 3: the total class is the answer only after dropping the connectedness requirement. It is not an exceptional solution satisfying that requirement. □

D.3. A nontotal reversible localizer kills no sphere. [Proved]

If a fundamental localizer satisfies Loc (6 b) and makes any sphere S^n, $n \geq 0$, aspherical, it is the total class. In particular no nontotal reversible localizer contains $\mathcal{W}_1$.

Proof. The cofiber criterion applied to $X \to e$ identifies asphericity of X with asphericity of its unreduced suspension, for every category X. The categorical models of spheres supplied by simplex categories permit repeated application. Starting with S^n, we obtain asphericity of S^0. But $K(\varnothing \to e) = e \amalg e$ represents S^0; one more application gives $\varnothing \to e \in \mathcal{W}$.

For clarity, this forces totality under the stated axioms. Fundamental localizers are stable under product with any category C: for $u : A \to B$ in $\mathcal{W}$, the slices over $c \in C$ of $u \times C$ identify with $u \times (C/c)$; the projections to u are homotopy equivalences because C/c has a terminal object. Relative Theorem A gives $u \times C \in \mathcal{W}$. Applying this to $\varnothing \to e$ gives $\varnothing \to C \in \mathcal{W}$ for every C. Two out of three on $\varnothing \to A \to B$ gives every functor $A \to B$. Finally $\mathcal{W}_1$ contains $S^2 \to e$ by its defining homotopy-group test. □

D.4. Both readings of the stability intersection: Lead 4. [Proved]

Let $\widetilde{\mathcal{W}}$ be the intersection of all modern fundamental localizers satisfying the stability list XVI Loc (1 ter, 2, 3 bis, 4, 6, 7, 8), with no prior inclusion in $\mathcal{W}_\infty$ imposed. If Loc (6) includes its reverse half, then $\widetilde{\mathcal{W}} = \mathrm{Fl}(\mathbf{Cat})$. If it means only Loc (6 a), then $\widetilde{\mathcal{W}} = \mathcal{W}_\infty$. If the definition additionally insists, as the prose of XVI ms. 87 does, that $\widetilde{\mathcal{W}} \subseteq \mathcal{W}_\infty$, there is no such localizer in the first reading.

Proof. Loc (7) implies Loc ($7\mathrm{D}_{\mathcal{W}_\infty}$), as made precise in Parts A and E. With the reverse half of Loc (6), D.2 forces each member of the intersection to be total. The total class itself satisfies the entire stability list, so the intersection is total. This class is not a subclass of $\mathcal{W}_\infty$, proving the last sentence.

For the forward reading, minimality B.1 gives the lower bound $\mathcal{W}_\infty$. This class satisfies strong saturation, the terminal-object and relative Theorem A axioms, disjoint-union preservation and reflection, forward gluing, the ordinary homotopy-fibration axiom, and filtered-colimit stability. The first six assertions are established in Parts A–C; filtered-colimit stability follows because each finite simplicial homotopy problem factors through a stage of a filtered diagram of nerves. Thus $\mathcal{W}_\infty$ belongs to the family being intersected, giving the opposite inclusion. No conjecture about simple homotopy can yield a strict intermediate answer under either reading. □

D.5. What can and cannot retain simple homotopy information. [Proved]

No modern fundamental localizer on all small categories is a strict subclass of $\mathcal{W}_\infty$. Even weakening relative Theorem A to the absolute version while retaining weak saturation and terminal asphericity does not produce such a subclass. A finer equivalence relation can instead be retained on finite CW or finite simplicial complexes by keeping elementary expansions, collapses, and the simple structures as geometric data. Such a structure is not a fundamental localizer. Merely localizing the category of all finite CW complexes and all cellular maps at simple homotopy equivalences does not retain the simple homotopy classification.

Proof. The first assertion is B.1, and the stronger weak-localizer minimum is A.19. For the last assertion, the endpoint inclusions $X \to X \times [0,1]$ of a finite CW complex are simple homotopy equivalences: collapse the prisms one cell at a time. Hence localization at simple homotopy equivalences identifies homotopic cellular maps. If $f : X \to Y$ is any cellular homotopy equivalence with cellular homotopy inverse g, then gf and fg become identities there, making f invertible whether or not it is simple. Thus a homotopy-category localization of this unrestricted mapping category loses the distinction in question.

There is also an immediate obstruction at the saturation axiom. Whenever f is a nonsimple homotopy equivalence and g a homotopy inverse, both gf and fg are homotopic to identities and are simple, while f is not. Thus the printed Loc (1 c), and a fortiori the fundamental-localizer package, cannot hold for the class of simple equivalences in this mapping category. This is a conditional obstruction for every such f, not a claim that Whitehead torsion is an invariant of an undecorated homotopy type.

A suitable weaker structure is the groupoid generated by elementary expansions, elementary collapses, and cellular isomorphisms, retaining the allowed simple structures and the relations one chooses between these geometric operations. Its components classify simple homotopy types by definition. Alternatively one retains torsion in a category of structured finite complexes, instead of declaring all homotopy inverses invertible through the fundamental-localizer axioms. The classical definitions and the based chain construction of the torsion lie in Lurie (2014, Definition 2, Example 3, and Definition 17); the Whitehead group is

$$\mathrm{Wh}(\pi) = K_1(\mathbb{Z}[\pi])/\langle [\pm g] : g \in \pi \rangle.$$

It is not merely an unspecified cohomological invariant in a group denoted $K(\mathbb{Z}[\pi_1])$. This distinguishes the valid geometric suggestion in XVI ms. 88–96 from the impossible proposed fundamental localizer. □

D.6. The finite-poset observation survives without the conjecture. [Proved]

Let P be the face poset of the boundary of a triangle, consisting of its three vertices and three edges, and let sd P be the poset of nonempty chains in P. The last-element map sd $P \to P$ is in $\mathcal{W}_\infty$ but has no inverse up to zigzags of natural transformations of functors. Thus strict categorical homotopy is finer than weak homotopy of nerves; this fact does not produce a fundamental localizer smaller than $\mathcal{W}_\infty$.

Proof. The two nerves realize to circles. The canonical subdivision homeomorphism and the usual linear homotopy within each simplex identify realization of the last-element map with a degree-one map. It is consequently a weak homotopy equivalence. The nerve of P is a cycle with six edges, while that of sd P is a cycle with twelve edges. Any order-preserving map $P \to$ sd P induces a simplicial map taking each of the six edges to one edge or a vertex. A map of nonzero degree between these oriented circles must traverse every one of the twelve target edges, so no such simplicial map has nonzero degree. A natural transformation gives an actual homotopy on nerves, as does any finite zigzag. Thus no order-preserving map in the reverse direction can be a categorical homotopy inverse of the last-element map. The program `code/poset_subdivision.py` enumerates every order-preserving map and verifies the vanishing degree; its data and output are attached. This checks the relevant observation at XVI ms. 95 while leaving minimality intact. □

E Fibration axioms, modalities, and exactness

E.1. The accessible localization attached to a localizer. [Known]

For an accessible fundamental localizer $\mathcal{W}$, let $L : \mathcal{S} \rightleftarrows \mathcal{S}_{\mathcal{W}} : \iota$ be the accessible reflective localization of spaces corresponding to $\mathcal{W}$. A functor u is in $\mathcal{W}$ if and only if $L(Bu)$ is an equivalence. We identify $\mathcal{S}_{\mathcal{W}}$ with its full subcategory of local spaces. Limits of local spaces are computed in $\mathcal{S}$; colimits in $\mathcal{S}_{\mathcal{W}}$ are obtained by applying L to colimits in $\mathcal{S}$. In particular $*$ is local. These conventions concern the infinity-category, not just its homotopy category.

The passage from fundamental localizers to localizers of simplicial sets and the preservation of accessibility are Cisinski (2006, Theorem 4.2.15, Remark 4.2.16). The associated model structures are left Bousfield localizations of the Kan model structure. Their derived localizations give the stated reflective localization of spaces. The unit is denoted $\eta_X : X \to LX$ for every space X. No such reflector is being assumed for an arbitrary nonaccessible class.

E.2. Enough intrinsic fibrations mean aspherical generation. [Proved]

For an accessible fundamental localizer $\mathcal{W}$, let $\mathcal{W}^{\mathrm{asph}}$ be the fundamental localizer generated by $\{A \to e \mid A \text{ is } \mathcal{W}\text{-aspherical}\}$. The following conditions are equivalent: Loc (7 F); $\mathcal{W} = \mathcal{W}^{\mathrm{asph}}$; $\mathcal{W}$ is generated by a small set of maps $A \to e$; the simplicial localization of E.1 is right proper; and the Thomason model structure on **Cat** for $\mathcal{W}$ is right proper. Thus Loc (7 F) is a properness condition, and does not follow merely from existence of a model structure.

Proof. Suppose first that (7 F) holds. The class $\mathrm{Fib}_{\mathcal{W}}$ is stable under composition, pullback, and retracts. All maps to e belong to it because fundamental localizers are stable under products. We verify explicitly the only potentially hidden part of the Brown structure. Let $p : T \to B$ belong to $\mathrm{Fib}_{\mathcal{W}} \cap \mathcal{W}$, and let $g : X \to B$ be arbitrary. Factor g as $X \xrightarrow{i} Z \xrightarrow{q} B$, with $i \in \mathcal{W}$ and $q \in \mathrm{Fib}_{\mathcal{W}}$. Then $p_Z : T \times_B Z \to Z$ is in $\mathcal{W}$, since q preserves $\mathcal{W}$ under pullback. The comparison $T \times_B X \to T \times_B Z$ is also in $\mathcal{W}$, because $p_Z \in \mathrm{Fib}_{\mathcal{W}}$ and it is the pullback of i. The identity between their composites over B, followed by two out of three, implies $p_X \in \mathcal{W}$. Thus trivial intrinsic fibrations are universally in $\mathcal{W}$, as required for the Brown lemma.

For $b \in B$, pull p back along $B/b \to B$. Its total category is T/b. The resulting map belongs to $\mathcal{W}$, and B/b has a terminal object. Thus T/b is $\mathcal{W}$-aspherical. By definition its map to e belongs to $\mathcal{W}^{\mathrm{asph}}$. The Theorem A axiom for $\mathcal{W}^{\mathrm{asph}}$ now gives $p \in \mathcal{W}^{\mathrm{asph}}$. The dual Brown lemma, applied to the identity functor with target weak equivalences $\mathcal{W}^{\mathrm{asph}}$, implies that every map in $\mathcal{W}$ belongs to $\mathcal{W}^{\mathrm{asph}}$. The opposite inclusion holds from the definition of the generators. This direction does not require accessibility.

For accessible $\mathcal{W}$, Cisinski (2006, Theorem 6.1.11) identifies $\mathcal{W} = \mathcal{W}^{\mathrm{asph}}$ with generation by a set of categories and with properness of the associated simplicial model structure. Cisinski (2006, Theorem 5.2.15) identifies this with right properness of its Thomason model structure. In a right proper model structure every model fibration preserves weak equivalences under strict pullback. Consequently its factorization of an arbitrary functor as a trivial cofibration followed by a fibration is a factorization as $\mathcal{W}$ followed by $\mathrm{Fib}_{\mathcal{W}}$. This proves (7 F). The comparison with intrinsic fibrations is also stated in Cisinski (2006, Example 6.1.14). □

E.3. The precise modality and local-cartesian dictionary. [Proved]

For the accessible localization L of E.1, the conditions of E.2 are also equivalent to each of the following: L is a nullification; its local spaces form a modality; L is locally cartesian in the sense that

$$L(X \times_Z A) \longrightarrow LX \times_Z A \tag{E.3.1}$$

is an equivalence whenever A, Z are local and $X \to Z \leftarrow A$ is a diagram of spaces; and the presentable infinity-category $\mathcal{S}_{\mathcal{W}}$ is locally cartesian closed. Here a modality means the stable orthogonal factorization system whose right maps have local homotopy fibers and whose left maps have L-aspherical homotopy fibers. Its left class need not equal the class of maps inverted by L. Nor is its right class identified with the maximal intrinsic class $\mathrm{Fib}_{\mathcal{W}}$ in **Cat**: every terminal projection is an intrinsic fibration, whereas a terminal map is in the modality's right class exactly when its source is local.

Proof. Generation by maps $A_i \to *$ means precisely localization for which a space Y is local if and only if the constant-map maps $Y \to \mathrm{Map}(A_i, Y)$ are equivalences. This is nullification. For completeness its local spaces are closed under dependent sums over a local base. If X and the spaces Y_x are local, then for each generator A_i there are natural equivalences

$$\mathrm{Map}\left(A_i, \int_{x\in X} Y_x\right) \simeq \int_{f\in \mathrm{Map}(A_i,X)} \mathrm{Map}_{/X}(A_i, Y) \simeq \int_{x\in X} \mathrm{Map}(A_i, Y_x) \simeq \int_{x\in X} Y_x.$$

In the middle expression the map $A_i \to X$ is f; after the constant-map equivalence for X, it is constant at x. The equivalences are inverse to constant maps. Thus the reflective subcategory is closed under dependent sums, which is the modality condition. This is the argument of Rijke et al. (2020, Theorem 2.19). The equivalence with the stable factorization-system formulation is established there in Theorems 1.34 and 1.54, with uniqueness in Theorem 1.53, and is the definition used by Anel et al. (2020, Section 3.4).

Conversely, an accessible modality presented by a set of maps $u_i : A_i \to B_i$ is presented by a set of objects: take all homotopy fibers of the unit maps $A_i \to LA_i$ and $B_i \to LB_i$. Their indexing spaces are small in the fixed universe. These fibers are L-aspherical. A space Y null for them is local for both unit maps. Indeed, for either unit $q : U \to V$ and its fiber diagram $v \mapsto U_v$, the canonical identification $U \simeq \mathrm{colim}_{v\in V} U_v$ gives

$$\mathrm{Map}(U, Y) \simeq \lim_{v\in V} \mathrm{Map}(U_v, Y), \qquad \mathrm{Map}(V, Y) \simeq \lim_{v\in V} Y.$$

The constant-map equivalences $Y \to \mathrm{Map}(U_v, Y)$ are natural in v, including its paths and all higher coherences. Taking their homotopy limit therefore proves that $\mathrm{Map}(V, Y) \to \mathrm{Map}(U, Y)$ is an equivalence. This is a limit over the full base space, with its monodromy, rather than a product over its set of points. In the commuting square formed by u_i and its units, Lu_i is an equivalence. Such a space is therefore u_i-local. This proves equality of the local objects and hence of the localizations. It is exactly Rijke et al. (2020, Lemma 2.20).

For a modality the units have L-aspherical fibers. In (E.3.1) the map $X \times_Z A \to LX \times_Z A$ is a base change of η_X and therefore has L-aspherical fibers. It is inverted by L. Its target is local, being a limit of local objects, so it exhibits the target as the localization of its source. This proves local cartesianness. Conversely apply (E.3.1) to the unit $X \to LX$ and a point $* \to LX$. Every unit fiber is L-aspherical. This implies closure of local objects under dependent sums: for $X \to Z$ with local Z and all fibers local, (E.3.1) identifies those fibers with the fibers of $LX \to Z$, so the unit is a fiberwise equivalence. This is the modality condition. The terminology and related equivalences agree with Macpherson (2021, Propositions 3.1 and 3.3).

Local cartesianness implies local cartesian closedness as follows. In a slice over a local Z, colimits are obtained by L. For local $A \to Z$, (E.3.1), together with universality of colimits in spaces, shows that pullback to A preserves colimits in the local slices. The adjoint functor theorem therefore provides its right adjoint, which is the dependent product.

The converse uses the special ambient category $\mathcal{S}$ and should not be asserted for an arbitrary reflective subcategory of an arbitrary infinity-topos without qualification. Every $X \to Z$ in $\mathcal{S}/Z$ is the colimit of its diagram of points $* \xrightarrow{x} Z$, indexed by the infinity-groupoid X. These points are local objects of the slice when Z is local. The slice inclusion $\mathcal{S}_{\mathcal{W}}/Z \to \mathcal{S}/Z$ has left adjoint L_Z, taking $X \to Z$ to $LX \to Z$: its adjunction follows by taking the fibers of the mapping-space adjunction over the prescribed map to the local space Z. Applying L_Z to the point diagram therefore proves that its colimit in the local slice is $LX \to Z$. If $\mathcal{S}_{\mathcal{W}}$ is locally cartesian closed, pullback along local $A \to Z$ preserves colimits in its slices. It consequently sends the colimit $LX \to Z$ of that diagram to the colimit of its fibers A_x. The latter are local because the inclusion of the local category preserves limits. Computed in spaces, their colimit

is $X \times_Z A$. Localizing this colimit proves (E.3.1). This argument checks the claimed converse in exactly the setting of the question. □

E.4. Perfect categorical models and arbitrary homotopy pullbacks. [Proved]

For any fundamental localizer $\mathcal{W}$, Loc (7 $D_{\mathcal{W}_\infty}$) is equivalent to stability of the induced $\mathcal{W}$-equivalences of spaces under arbitrary homotopy pullback. No accessibility hypothesis is needed for this equivalence. The test class $\mathrm{Parf}_{\mathcal{W}_\infty}$ is essential: replacing homotopy pullbacks by arbitrary strict pullbacks of categories would be stronger and false even for $\mathcal{W}_\infty$.

Proof. We specify the comparison used in passing between models. Every functor $v : C \to B$ has a factorization

$$C \xrightarrow{i_v} P(v) \xrightarrow{p_v} B, \qquad i_v \in \mathcal{W}_\infty, \quad p_v \in \mathrm{Parf}_{\mathcal{W}_\infty}. \tag{E.4.1}$$

This is the categorical path factorization recalled in XVI, manuscript pp. 28–29 and 89–90. An independently constructed factorization with the same properties follows from Cisinski (2006, Theorem 5.3.14), applied to $\mathcal{W}_\infty$: factor as a trivial cofibration followed by a strongly proper and smooth Thomason bifibration. We use this latter verified construction, so do not assume that every detail of the manuscript's infinite-path presentation is correct.

For $p \in \mathrm{Parf}_{\mathcal{W}_\infty}$, every strict base-change square along p is a homotopy pullback after applying B. To check the scope of this assertion, properness and smoothness are defined by the fiber-to-comma comparisons in Cisinski (2006, Definition 5.3.1); both classes are closed under base change and composition by Corollary 5.3.6 and Proposition 5.3.8 and their duals. Their intersection is locally constant by Corollary 6.4.8. Pull back p along an arbitrary $[m] \to B$. For any $k \in [m]$, properness identifies the strict fiber with the comma category over k, and local constancy identifies this comma category with the one over m. The latter is the total category, because m is terminal in $[m]$. Thus every point-to-$[m]$ pullback induces a weak equivalence on the total categories. This is exactly the finite-ordinal condition in Proposition 6.4.14(iii), which identifies p as a $\mathcal{W}_\infty$-weak fibration. Finally Proposition 6.2.3 identifies such weak fibrations with the maps whose every strict base-change square is homotopy cartesian. This establishes the comparison for every base category, not only for points.

The dual Brown argument in Part A turns $\mathrm{Parf}_{\mathcal{W}_\infty} \cap \mathcal{W} \subset \mathcal{W}^u$ into $\mathrm{Parf}_{\mathcal{W}_\infty} \subset \mathrm{Fib}_{\mathcal{W}}$, using (E.4.1) and closure of $\mathrm{Parf}_{\mathcal{W}_\infty}$ under composition and base change. If $u : A \to B$ belongs to $\mathcal{W}$ and $v : C \to B$ is arbitrary, replace v by (E.4.1). Then

$$A \times_B P(v) \longrightarrow P(v)$$

belongs to $\mathcal{W}$, and its nerve models the homotopy pullback of Bu along Bv. The map Bi_v is the specified comparison to the original base and is a weak homotopy equivalence. Minimality and two out of three transfer membership in $\mathcal{W}$ across that comparison.

This suffices for arbitrary diagrams of spaces: present a space by a simplicial set, and replace it by its category of simplices. The canonical realization map $B(\Delta/K) \to |K|$ is a weak equivalence, naturally in K. A homotopy-coherent cospan of spaces may be represented by a strict cospan of simplicial sets using Kan replacements and a path object for its homotopies. Applying $\Delta/-$ gives a strict cospan of small categories and a levelwise weak-equivalent diagram. Its derived pullback is the specified pullback of spaces, by invariance of the limit under equivalence of diagrams. Thus the preceding argument proves homotopy-pullback stability in full scope.

Conversely, if $\mathcal{W}$-equivalences of spaces have that stability, take $p \in \mathrm{Parf}_{\mathcal{W}_\infty} \cap \mathcal{W}$ and any strict base change. The established nerve comparison identifies it with a homotopy pullback of Bp, hence with a $\mathcal{W}$-equivalence. This is exactly the universally-$\mathcal{W}$ conclusion in (7 $D_{\mathcal{W}_\infty}$). □

E.5. The fibration pole has two possibilities. [Proved]

For a fundamental localizer in the convention of this paper, Loc (7 $D_{\mathcal{W}_\infty}$) holds if and only if $\mathcal{W} = \mathcal{W}_\infty$ or $\mathcal{W} = \mathrm{Fl}(\mathbf{Cat})$. Therefore all eight variants $A, B, C, D, A_V, B_V, C_V, D_V$ of XVI, Section 4.4, for $\mathcal{W}_\infty \subset V \subset \mathcal{W}$, are equivalent in this modern axiom system and have precisely these two solutions. Loc (5) excludes the second solution. This strengthens the conclusion $\mathcal{W} \subset \mathcal{W}_\infty$ of XVI, Theorem 6.1, by combining it with minimality.

Proof. Assume the weakest clause and suppose some $f \in \mathcal{W}$ is not a weak homotopy equivalence. By E.4, a homotopy fiber F of Bf is $\mathcal{W}$-aspherical and not weakly contractible. If it is empty, $\varnothing \to *$ belongs to $\mathcal{W}$. Otherwise, every chosen point $* \to F$ belongs to $\mathcal{W}$ by two out of three. Pulling it back along itself shows that ΩF is $\mathcal{W}$-aspherical. Iterating this argument, either F already has more than one component, or a nonzero homotopy group produces an iterated loop space Z with at least two components. If there were no such homotopy group and F were connected, F would be weakly contractible, contrary to its selection. Choose points z_0, z_1 in distinct components of Z. Both $* \to Z$ maps are in $\mathcal{W}$, while their homotopy pullback is empty. Again E.4 gives $\varnothing \to * \in \mathcal{W}$.

Now $\varnothing \to X$ is in $\mathcal{W}$ for every category X: it is the product of $\varnothing \to e$ with id_X. For every $u : X \to Y$, the equality $u \circ (\varnothing \to X) = (\varnothing \to Y)$ and two out of three give $u \in \mathcal{W}$. Thus $\mathcal{W}$ is total. If no $f \in \mathcal{W} \setminus \mathcal{W}_\infty$ exists, minimality gives $\mathcal{W} = \mathcal{W}_\infty$.

The total class satisfies all the clauses. For $\mathcal{W}_\infty$, Quillen B for a Grothendieck fibration with weakly equivalent transition functors identifies every base-change square with a homotopy pullback, so Loc (7 a) holds. The implications in Part A now show that both extreme classes satisfy all eight variants and that any one of those variants implies the weakest one. Finally $e \amalg e \to e$ proves that the total class fails Loc (5). □

E.6. Lead 5 with its exact hypotheses. [Proved]

For an accessible fundamental localizer, the following are equivalent: Loc (7); Loc (7 $D_{\mathcal{W}_\infty}$); L preserves finite limits; and L is either the identity localization of spaces or the localization to the terminal infinity-category. These equivalences do not identify Loc (7 F) with left exactness. Instead (7 F) has the strictly weaker meanings in E.2 and E.3.

Proof. Write $\mathcal{E}_L$ for the maps of spaces inverted by L. If L preserves finite limits, $\mathcal{E}_L$ is stable under homotopy pullback. Conversely suppose it is stable under homotopy pullback. In a cospan $X \to Z \leftarrow Y$, factor the canonical map to the pullback of its localized cospan as

$$X \times_Z Y \longrightarrow X \times_{LZ} Y \longrightarrow LX \times_{LZ} Y \longrightarrow LX \times_{LZ} LY.$$

The last two maps are pullbacks of units, hence are in $\mathcal{E}_L$. The first map is the pullback of the diagonal $Z \to Z \times_{LZ} Z$. This diagonal is in $\mathcal{E}_L$, because its projection back to Z is a pullback of η_Z and its composite with the diagonal is the identity. Thus all three maps are in $\mathcal{E}_L$. Their target is local, being a limit of local objects; it is therefore the localization of the source. Since $*$ is local, L preserves finite limits. Combining this calculation with E.4 proves the exactness dictionary, and E.5 supplies the classification.

There is also a proof of the classification entirely in spaces. If a left exact reflector is not the identity, some unit $X \to LX$ has a noncontractible homotopy fiber. Left exactness makes that fiber and all its iterated loop spaces L-aspherical. The argument in E.5 then shows $L\varnothing \simeq *$. For any X, left exactness applied to $X \times \varnothing = \varnothing$ gives $LX \times L\varnothing \simeq L\varnothing$, whence $LX \simeq *$. This proves that there are exactly two left exact reflective localizations of spaces; accessibility is needed here for constructing L from $\mathcal{W}$, not for this last classification argument.

To see the strict distinction from (7 F), use $L = \tau_{\leq n}$, $n \geq 0$. It is nullification at S^{n+1} and satisfies (7 F). The homotopy pullback $* \times_{K(\mathbb{Z},n+1)} * \simeq K(\mathbb{Z}, n)$ is not preserved by L, since the localized base is a point. Thus it is a modality with locally cartesian closed local category, but it is not left exact. This example also explains why the reflective equivalences cannot be substituted for the left class of a general modality. The same distinction is explicit in Rijke et al. (2020, Theorem 3.1 and the discussion after Corollary 3.9) and in the congruence formalism of Anel et al. (2022). □

E.7. The three classes on XII, manuscript pp. 178–187. [Proved]

For $n \geq 2$ and a map of spaces $f : X \to Y$, use the manuscript's definitions with their missing range made explicit: a_n means that every homotopy fiber is nonempty, connected, and has $\pi_i = 0$ for $1 \leq i \leq n$; b_n means that $\tau_{\leq n} f$ is an equivalence; and c_n means that

$$H^i(Y; M) \longrightarrow H^i(X; f^*M)$$

is an isomorphism for $0 \leq i \leq n$, for every local system of sets at $i = 0$, groups at $i = 1$, and abelian groups at $i \geq 2$. Then a_n is the left class of the n-truncation modality, whereas b_n is its class of reflective equivalences. Precisely,

$$a_n \subset b_n \subset c_n \subset a_{n-1}, \qquad a_n = (b_n)^{\text{univ}},$$

where universal means stable membership under all homotopy base changes. Moreover $b_2 = c_2$, while $b_n \subsetneq c_n$ for every $n \geq 3$. Thus the manuscript's expectation that all displayed implications are strict is false at $n = 2$. The final suggestion on manuscript p. 187 that a_n itself defines the localizer of n-types is also false: a_n fails two out of three. That localizer is b_n.

Proof. The long exact sequence of a homotopy fiber gives

$$f \in a_n \quad \Longleftrightarrow \quad f \in b_n \text{ and } \pi_{n+1}(X, x) \to \pi_{n+1}(Y, f(x)) \text{ is surjective at every } x.$$

This includes the requirement that every component of Y has a fiber. The n-connected–n-truncated factorization system factors a map by truncating each homotopy fiber; the first factor has n-connected fibers and the second has n-truncated fibers. Both classes are stable under base change. The reflective equivalences are the b_n, and their universal subclass is exactly a_n: necessity is the test by points of Y, and sufficiency follows from the unchanged fibers after a base change.

If $f \in b_n$, it induces an equivalence of fundamental groupoids. For a local system M, the twisted Eilenberg–MacLane space over this fundamental groupoid representing $H^i(-; M)$ is an n-type when $i \leq n$. Mapping into it therefore depends only on $\tau_{\leq n}$. This proves $b_n \subset c_n$, including the torsor interpretation at degree one.

Suppose $f \in c_n$. Degree zero with all constant sets implies a bijection on components, so we work separately on each connected component. Degree one with all constant groups says that

$$\mathrm{Hom}(\pi_1 Y, G)/G \longrightarrow \mathrm{Hom}(\pi_1 X, G)/G$$

is bijective for every group G, with conjugation as the quotient action. Put $\phi = \pi_1 f$. For $G = \pi_1 X$, surjectivity supplies $r : \pi_1 Y \to \pi_1 X$ with $r\phi$ conjugate to the identity; conjugate r so that $r\phi = \mathrm{id}$. For $G = \pi_1 Y$, injectivity says that ϕr is conjugate to id, since both restrict to ϕ on $\pi_1 X$. Hence ϕr is an automorphism, and ϕ is surjective as well as injective.

Identify this common group with Γ. Lift f to an equivariant map $\widetilde{f} : \widetilde{X} \to \widetilde{Y}$ of universal covers. Choose injective $\mathbb{Z}[\Gamma]$-modules I. For the free equivariant singular-chain complexes, injectivity gives

$$H^i(X; I) = \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(H_i(\widetilde{X}), I),$$

and the corresponding formula for Y. These identifications are natural in f. The degree-i assertions of c_n for all injectives imply that $H_i(\widetilde{f})$ is an isomorphism for $i \leq n$: an injective cogenerator detects the kernel and cokernel of a module map. Relative Hurewicz, applied inductively to the simply connected mapping-cylinder pair of $\widetilde{f}$, now gives vanishing relative homotopy through degree n. Equivalently its homotopy fiber is $(n-1)$-connected. This proves $c_n \subset a_{n-1}$. At $n = 2$, ordinary Hurewicz identifies the second homotopy groups of the simply connected covers with their second homology groups. Thus c_2 also implies b_2.

For strictness, $* \to K(\mathbb{Z}, n+1)$ is in b_n and not in a_n. Its retraction $K(\mathbb{Z}, n+1) \to *$ and their composite id_* are both in a_n, proving the asserted failure of two out of three. For $n \geq 3$, the inclusion

$$S^2 \vee S^{n-1} \longrightarrow S^2 \times S^{n-1}$$

is in c_n: both spaces are simply connected and their cellular cochain complexes agree through degree n for every abelian coefficient group. It is not in b_n. The product is obtained by attaching its top $(n+1)$-cell to the wedge by the Whitehead product $[\iota_2, \iota_{n-1}] \in \pi_n(S^2 \vee S^{n-1})$. This attaching map is nonzero, for otherwise the product would be homotopy equivalent to $S^2 \vee S^{n-1} \vee S^{n+1}$, whose positive-dimensional cup products vanish, whereas the product has a nonzero cross cup product in degree $n+1$. Its nonzero class is killed by the inclusion, as required. Finally $S^n \to *$ belongs to a_{n-1} but fails c_n by degree n cohomology. These examples also show explicitly why c_n, for $n \geq 3$, is neither of the two designated classes of the n-truncation modality; at $n = 2$ it is its reflective equivalence class. The finite cellular complexes and their integral cup-product certificates for $3 \leq n \leq 8$ are checked in `code/ce_truncation_cochains.py`, with the complete output in its accompanying JSON file. The code retains the two distinct degree-two generators when $n = 3$; it does not replace the unstable attaching-map argument by a homology calculation. □

The local coefficients in this last proof may be described covariantly as left $\mathbb{Z}[\Gamma]$-modules or, with the manuscript's presheaf convention, as right modules using the inverse action. The equivariant-chain and cochain identifications use one of these choices consistently; no change in variance is being hidden in the comparison.

F Stable detection, its Galois connection, and its losses

F.1 The pointed spectrum attached to an unpointed map

F.1. The spectrum in the proposed assignment. [Proved]

For a map $f : X \to Y$ of $\mathcal{U}$-small spaces, put

$$C_f = Y \amalg^h_X *, \qquad D(f) = \mathrm{cofib}(\Sigma^\infty_+ X \xrightarrow{\Sigma^\infty_+ f} \Sigma^\infty_+ Y) \in \mathrm{Sp}.$$

The cone point gives C_f a specified basepoint. There is a natural equivalence $D(f) \simeq \Sigma^\infty C_f$, with *reduced* suspension spectrum on the right. Moreover, $D(C_f \to *) \simeq \Sigma D(f)$. Both assertions include empty and disconnected X, Y.

Proof. Adjoining a disjoint basepoint is a left adjoint from spaces to pointed spaces. It therefore sends the defining homotopy pushout for C_f to the corresponding pointed homotopy pushout, with a copy of S^0 at the cone vertex. Collapsing that copy of S^0 yields the pointed cofiber of $f_+ : X_+ \to Y_+$. Applying the colimit-preserving functor Σ^∞ gives the asserted equivalence. Equivalently, this is the pasting law for the two pushout squares which first attach the cone

and then collapse its vertex together with the disjoint basepoint. For the second assertion the specified cone point splits the augmentation $\Sigma^\infty_+ C_f \to \mathbb{S}$. Its fiber is $\Sigma^\infty C_f$; its cofiber is its suspension. For $X = \varnothing$ the first formula reads $D(f) = \Sigma^\infty_+ Y$ and $C_f = Y_+$, so that no unmentioned nonemptiness assumption has entered the calculation. □

F.2. Localizing subcategories give reversible excision. [Proved]

Let $\mathcal{L}$ be a full replete localizing subcategory of $\mathrm{Ho}(\mathrm{Sp}_{\mathcal{U}})$: it is closed under all shifts, distinguished triangles, retracts, and $\mathcal{U}$-small coproducts. Define

$$R(\mathcal{L}) = \{u : A \to B \text{ in } \mathbf{Cat} \mid D(Bu) \in \mathcal{L}\}.$$

Then $R(\mathcal{L})$ is a strongly saturated fundamental localizer in the relative-Theorem-A convention of this paper. It satisfies Loc (6 a), Loc (6 b), and Loc (6 c), and for every functor u it satisfies

$$u \in R(\mathcal{L}) \quad \Longleftrightarrow \quad C_{Bu} \text{ is } R(\mathcal{L})\text{-aspherical.}$$

The asphericity of a space means that of any category-of-simplices model of that space; the comparison weak equivalences make the definition independent of the model.

Proof. The class $\mathcal{L}$ is closed under homotopy colimits of spectra. Indeed, sequential homotopy colimits are cofibers of the shift-minus-identity map on a coproduct. Geometric realizations are sequential homotopy colimits of their skeletal filtrations, whose successive cofibers are suspensions of coproducts. The simplicial replacement expresses an arbitrary small homotopy colimit in these terms. Cofiber commutes with homotopy colimits. It follows that maps with cofiber in $\mathcal{L}$ are closed under homotopy colimits in the arrow category. The octahedral triangle for composable maps gives two out of three; functoriality of cofiber gives closure under arrow retracts. The Verdier quotient by the thick subcategory $\mathcal{L}$ detects precisely these maps: a map becomes invertible exactly when its cone belongs to $\mathcal{L}$. The composite $\mathbf{Cat} \xrightarrow{B} \mathcal{S} \xrightarrow{\Sigma^\infty_+} \mathrm{Sp} \to \mathrm{Ho}(\mathrm{Sp})/\mathcal{L}$ therefore detects $R(\mathcal{L})$ and proves its strong saturation. This quotient can be taken in the larger universe; no smallness of its hom sets in $\mathcal{U}$ is assumed.

If A has a terminal object, then BA is contractible, so $A \to e$ is in $R(\mathcal{L})$. To check the relative Theorem A, take a commutative triangle $A \xrightarrow{u} B \to C$ and suppose $A/c \to B/c$ belongs to $R(\mathcal{L})$ for every $c \in C$. Postcomposition makes $c \mapsto A/c$ and $c \mapsto B/c$ covariant diagrams on C. There are natural weak equivalences

$$\mathrm{hocolim}_{c \in C}\, B(A/c) \longrightarrow BA, \qquad \mathrm{hocolim}_{c \in C}\, B(B/c) \longrightarrow BB.$$

For clarity, the category-theoretic projection $p : \int_C (A/c) \to A$ has left adjoint $a \mapsto (v(a), a, \mathrm{id}_{v(a)})$, where $v : A \to C$ is the structure functor. The adjunction supplies the weak equivalence after the usual nerve comparison for the Grothendieck construction. The construction is natural in u. Applying Σ^∞_+, taking the cofiber of the comparison, and using homotopy-colimit closure proves that $D(Bu) \in \mathcal{L}$.

For a homotopy pushout square of spaces with vertices X_0, X_1, X_2, X_3, the suspension spectra form the distinguished triangle

$$\Sigma^\infty_+ X_0 \longrightarrow \Sigma^\infty_+ X_1 \oplus \Sigma^\infty_+ X_2 \longrightarrow \Sigma^\infty_+ X_3 \longrightarrow \Sigma\Sigma^\infty_+ X_0.$$

For a map of such squares, with vertical maps f_i, taking cofibers gives the triangle $D(f_0) \to D(f_1) \oplus D(f_2) \to D(f_3)$. The three choices of two known zero terms in the quotient prove respectively Loc (6 a), (6 b), and (6 c); in the last case each summand is a retract of the middle term. Mayer–Vietoris squares of sieve covers become homotopy pushouts on nerves. The

same proof applies to $R(\mathcal{L})$-cocartesian squares, since their defining comparison is inverted in the quotient. Finally, F.1 and closure under shifts identify the two sides of the asserted cofiber criterion. □

F.3. Accessibility and the size of the assignments. [Proved]

If $\mathcal{L}$ is generated by a $\mathcal{U}$-small set of spectra, then $R(\mathcal{L})$ is accessible. If a fundamental localizer $\mathcal{W}$ is generated by a $\mathcal{U}$-small set S of functors, then

$$T(\mathcal{W}) := \mathrm{loc}\{D(Bu) : u \in \mathcal{W}\} = \mathrm{loc}\{D(Bs) : s \in S\}.$$

Without these set-generation hypotheses the order-theoretic assignments still exist in the larger universe; we make no assertion that the corresponding localizations are accessible in $\mathcal{U}$.

Proof. A set-generated stable localizing subcategory has a presentable stable quotient, obtained by localization at the set of all integer shifts of its generators mapped to zero. The quotient functor $q : \mathrm{Sp} \to \mathrm{Sp}/\mathcal{L}$ is accessible and colimit preserving; existence can equivalently be obtained by the transfinite construction in Casacuberta et al. (2014, Theorem 3.7), applied to the coproduct of the shifted generators. The inverse image of equivalences under the accessible arrow functor induced by $q\Sigma^\infty_+ : \mathcal{S} \to \mathrm{Sp}/\mathcal{L}$ is accessible: it is the iso-comma category of this arrow functor and the accessible inclusion of equivalences. Choose a regular cardinal for which this category and its inclusion into the arrow category are accessible. A small collection of representatives of its presentable arrows generates the whole class under filtered colimits. The class is strongly saturated and closed under all colimits, so these representatives generate it as a localization class. Transport to **Cat** uses the nerve/category-of-simplices correspondence; the accessibility statement is precisely the corresponding part of Cisinski (2006, Theorem 4.2.15). For $\mathcal{W}$ generated by S, let $\mathcal{L}_S = \mathrm{loc}\{D(Bs) : s \in S\}$. By F.2, $R(\mathcal{L}_S)$ is fundamental and contains S, hence contains $\mathcal{W}$. Thus $T(\mathcal{W}) \subseteq \mathcal{L}_S$. The reverse inclusion follows from $S \subseteq \mathcal{W}$. □

F.4. The Galois connection. [Proved]

Order fundamental localizers and full localizing subcategories of $\mathrm{Ho}(\mathrm{Sp})$ by inclusion. With T and R as above,

$$T(\mathcal{W}) \subseteq \mathcal{L} \quad \Longleftrightarrow \quad \mathcal{W} \subseteq R(\mathcal{L}).$$

Consequently RT is an increasing idempotent closure operator on fundamental localizers, while TR is a decreasing idempotent operator on localizing subcategories. The assignments restrict to the accessible and set-generated objects described in F.3.

Proof. The left-hand inclusion holds exactly when $D(Bu) \in \mathcal{L}$ for every $u \in \mathcal{W}$, which is the right-hand inclusion. Both operations preserve inclusions. Applying the equivalence to the identities gives $\mathcal{W} \subseteq RT(\mathcal{W})$ and $TR(\mathcal{L}) \subseteq \mathcal{L}$. Applying R and T to these two inequalities in turn gives $RTR = R$ and $TRT = T$, and hence the two idempotence assertions. □

F.5. The exact image and the fibers of stable detection. [Proved]

Let $\mathscr{D}$ denote the class of spectra equivalent to $\Sigma^\infty K$ for a pointed $\mathcal{U}$-small space K. Then

$$TR(\mathcal{L}) = \mathrm{loc}(\mathcal{L} \cap \mathscr{D}).$$

The stable fixed objects are exactly the localizing subcategories generated by the suspension spectra which they contain. The unstable fixed objects are exactly the localizers $\mathcal{W} = RT(\mathcal{W})$,

equivalently those of the form $R(\mathcal{L})$. On these two fixed-object posets, T and R are inverse order isomorphisms. Furthermore,

$$R(\mathcal{L}) = R(\mathcal{M}) \quad \Longleftrightarrow \quad \mathcal{L} \cap \mathscr{D} = \mathcal{M} \cap \mathscr{D}.$$

Thus each nonempty fiber of R has the least member $TR(\mathcal{L})$; its other members are exactly the localizing subcategories with the same intersection with $\mathscr{D}$. Each such fiber is order convex.

Proof. By F.1, every $D(Bu)$ belongs to $\mathscr{D}$. Conversely every pointed space K is, as a pointed space up to natural equivalence, the homotopy cofiber of its basepoint map $* \to K$. Replace this map by the functor on categories of simplices of its singular simplicial sets. The natural realization comparisons are weak equivalences at its source and target and hence on its cofiber. Therefore every spectrum in $\mathcal{L} \cap \mathscr{D}$ occurs, up to equivalence, among the generators of $TR(\mathcal{L})$. This proves the formula. The fixed-object assertions follow from F.4. The same realization argument shows that equality of the two map classes tests every object of $\mathscr{D}$ and therefore is equivalent to equality of the intersections. If $\mathcal{L}_0 \subseteq \mathcal{L}_1 \subseteq \mathcal{L}_2$ and the two outside members have the same intersection with $\mathscr{D}$, the middle one has that intersection as well. This proves order convexity without asserting that the fiber has a greatest member. □

F.6. The zero stable subcategory detects integral homology. [Proved]

One has $R(0) = \mathcal{W}_{\mathbb{Z}}$. In fact

$$T(\mathcal{W}) = 0 \quad \Longleftrightarrow \quad \mathcal{W} \subseteq \mathcal{W}_{\mathbb{Z}}.$$

In particular, every fundamental localizer between $\mathcal{W}_\infty$ and $\mathcal{W}_{\mathbb{Z}}$ has the same image 0 under T, and $RT(\mathcal{W}) = \mathcal{W}_{\mathbb{Z}}$ throughout this interval.

Proof. The spectrum $D(Bu)$ is connective, since it is the cofiber of a map of connective spectra. A connective spectrum V with $H\mathbb{Z} \wedge V = 0$ must be zero. Otherwise, choose its least nonzero homotopy group $\pi_n V$. The stable Hurewicz theorem identifies this first nonzero homotopy group with $H_n(V;\mathbb{Z})$, a contradiction. Thus

$$D(Bu) = 0 \quad \Longleftrightarrow \quad H\mathbb{Z} \wedge D(Bu) = 0 \quad \Longleftrightarrow \quad u \in \mathcal{W}_{\mathbb{Z}}.$$

The last equivalence is the exact homology sequence of the cofiber of $\Sigma^\infty_+ Bu$, including degree zero. A localizing subcategory generated by a family is zero precisely when each member of that family is zero, which proves the second assertion. □

F.7. The proposed bijection with all localizing subcategories. [Refuted]

The assignments in Lead 6 are not inverse bijections, even with accessibility and set generation. Let

$$\mathcal{L}_{\mathbb{Z}} = \{V \in \mathrm{Ho(Sp)} : H\mathbb{Z} \wedge V = 0\}.$$

Then $\mathcal{L}_{\mathbb{Z}} \neq 0$, but $R(\mathcal{L}_{\mathbb{Z}}) = R(0) = \mathcal{W}_{\mathbb{Z}}$ and $TR(\mathcal{L}_{\mathbb{Z}}) = 0$.

Proof. The equality of the two map classes is the connective argument of F.6. A nonzero Morava spectrum $K(n)$ at any prime, with $n > 0$, belongs to $\mathcal{L}_{\mathbb{Z}}$: one has $H\mathbb{Z} \wedge K(n) = 0$, while $\pi_* K(n) = \mathbb{F}_p[v_n, v_n^{-1}] \neq 0$. This standard example is recorded explicitly in Lawson (2019, Example 9.20). Hence the two stable subcategories are distinct. For an immediately singly generated counterexample, use $\mathrm{loc}(K(n)) \subseteq \mathcal{L}_{\mathbb{Z}}$ in place of $\mathcal{L}_{\mathbb{Z}}$; it is nonzero and has exactly the same intersection with connective spectra as 0. The kernel of a homological localization of

spectra is also generated by a set; equivalently Bousfield's smallness argument gives a single coproduct of generators. The set-generation assertion is also recalled in Hovey and Palmieri (1999, Section 2, discussion of Lemma 1.13 of Bousfield). Thus imposing accessibility does not remove this counterexample. The loss occurs because the input cofibers are suspension spectra, and in particular connective spectra; no space map probes a nonzero $H\mathbb{Z}$-acyclic connective spectrum. □

F.2 An explicit unstable distinction invisible to the stable assignment

F.8. A homology equivalence between free-group categories. [Proved]

Let $F = \langle x, y \rangle$ be the free group on two generators. The one-object category BF has arrows the reduced words in x, x^{-1}, y, y^{-1}, identity the empty word, and composition concatenation followed by free reduction. The endomorphism

$$\alpha : F \longrightarrow F, \qquad \alpha(x) = x, \quad \alpha(y) = yxyx^{-1}y^{-1}$$

defines a functor $B\alpha : BF \to BF$ which is an integral homology equivalence, and indeed an E-homology equivalence for every spectrum E, but is not a weak homotopy equivalence. The group F has no nontrivial perfect subgroup.

Proof. The graph with one vertex and two oriented loops is a $K(F, 1)$: its universal covering graph is the tree of reduced words. Therefore $H_0(BF; \mathbb{Z}) = \mathbb{Z}$, $H_1(BF; \mathbb{Z}) = \mathbb{Z}^2$, and all higher groups vanish. Counting exponents in the two words shows that $H_1(B\alpha; \mathbb{Z})$ is the identity matrix. The functor is therefore in $\mathcal{W}_{\mathbb{Z}}$, and F.6 says that its spectral cofiber is zero. Smashing this cofiber with any E proves the generalized homology assertion.

To prove failure of surjectivity without relying on a word-search heuristic, let $h_0, h_1 : F \to S_3$ be the unique homomorphisms with

$$h_0(x) = h_1(x) = (12), \qquad h_0(y) = (12), \qquad h_1(y) = (13).$$

They are distinct. The braid relation for the two transpositions gives $h_0\alpha(y) = h_1\alpha(y) = (12)$, and equality on x is immediate. Thus $h_0\alpha = h_1\alpha$. Surjectivity of α would force $h_0 = h_1$, a contradiction. In particular the induced fundamental-group map is not an isomorphism. The program `code/f_free_group_witness.py` checks all compositions and all 216 associativity triples of the six-arrow target category BS_3, the identity and inverse laws, both evaluations, and the exponent matrix; its complete output is attached. The program supplements the universal property proof; it does not pretend to enumerate the infinite category BF.

Finally, any subgroup of F is the fundamental group of a connected covering of the two-loop graph. Choosing a maximal tree in that covering expresses the subgroup as a free group on its remaining edges. A nontrivial free group has nonzero free abelianization, so no such subgroup is perfect. □

F.3 Plus construction and acyclic nullification

F.9. The plus localizer has the same aspherical objects as integral homology. [Proved]

Let P_+ be nullification at the universal acyclic space of Berrick and Casacuberta (1999, Theorem 4.3), applied componentwise to all spaces, with $P_+(\varnothing) = \varnothing$. It is the plus construction with respect to the largest perfect subgroup of each component's fundamental group. Put

$\mathcal{W}_+ = \{u : A \to B \mid P_+(Bu) \text{ is a weak equivalence}\}$. Then $\mathcal{W}_+$ is an accessible fundamental localizer, and

$$\mathcal{W}_\infty \subsetneq \mathcal{W}_+ \subsetneq \mathcal{W}_{\mathbb{Z}}.$$

Its aspherical categories are precisely the nonempty connected integrally acyclic categories, hence are exactly the $\mathcal{W}_{\mathbb{Z}}$-aspherical categories. Nevertheless $\mathcal{W}_+$ fails Loc (6 b), and $T(\mathcal{W}_+) = 0$.

Proof. The cited theorem supplies a single small connected acyclic space whose nullification realizes the indicated plus construction. Localization at one space-to-point map is accessible. Its equivalences form a strongly saturated class closed under homotopy colimits; the argument with the diagrams A/c in F.2 proves the relative Theorem A after transfer to categories. Thus it defines a fundamental localizer.

The plus-construction unit $X \to P_+X$ is an integral homology isomorphism on each component. Naturality and two out of three imply $\mathcal{W}_+ \subseteq \mathcal{W}_{\mathbb{Z}}$. If X is nonempty, connected, and integrally acyclic, then $\pi_1 X$ is perfect, so P_+X is simply connected and integrally acyclic. Hurewicz, in the least possible nonzero degree, and Whitehead imply that P_+X is weakly contractible. Conversely, $P_+X \simeq *$ and the homology isomorphism of the unit imply that X is connected and acyclic. The empty space is not made into a point by this localization.

The acyclic noncontractible category of B.7, or the group category of B.10, proves that the first inclusion is strict. For the second inclusion use F.8. Since F has no nontrivial perfect subgroup, BF is plus-local, and the two unit maps for $B\alpha$ are weak equivalences. Thus $P_+(B\alpha)$ is a weak equivalence if and only if $B\alpha$ is, which it is not. But its homotopy cofiber is integrally acyclic and connected and is therefore $\mathcal{W}_+$-aspherical. The cofiber criterion of B.5 proves the failure of Loc (6 b). Finally $T(\mathcal{W}_+) = 0$ follows from F.6. □

F.10. Nullifying an individual acyclic space does not supply reversible excision. [Proved]

For every connected integrally acyclic small space A, let P_A be localization at $A \to *$, and let $\mathcal{W}_{P_A}$ be its class of equivalences on categories. Then

$$\mathcal{W}_\infty \subseteq \mathcal{W}_{P_A} \subseteq \mathcal{W}_+ \subsetneq \mathcal{W}_{\mathbb{Z}}, \qquad T(\mathcal{W}_{P_A}) = 0.$$

Every such $\mathcal{W}_{P_A}$ fails Loc (6 b). These assertions include contractible A, for which $\mathcal{W}_{P_A} = \mathcal{W}_\infty$.

Proof. By Berrick and Casacuberta (1999, Theorem 3.3), the unit $X \to P_A X$ is a plus construction with respect to a particular perfect normal subgroup of $\pi_1 X$ and is an integral homology isomorphism. The same theorem characterizes the local objects by the vanishing of the subgroup swept out by maps from A. Any plus-local space has fundamental group with no nontrivial perfect subgroup and is therefore A-null. Equivalently, apply the construction to the universal perfect radical; this gives the inclusion of equivalence classes $\mathcal{W}_{P_A} \subseteq \mathcal{W}_+$. The other inclusions and $T = 0$ follow from F.9 and F.6. If Loc (6 b) held, the minimality theorem B.6 would force $\mathcal{W}_{\mathbb{Z}} \subseteq \mathcal{W}_{P_A}$, contradicting the strict upper inclusion. For contractible A, every space is A-null, so the localization is the identity. □

F.4 Retracts and the question on manuscript page 76

F.11. The precise direct-factor condition. [Proved]

The retract condition relevant to Loc (1 bis) is closure in the arrow category, not merely closure under retracts of domains or codomains. Explicitly, if $u : A \to B$ and $v : C \to D$ admit

commuting squares $(i_A, i_B) : u \to v$ and $(r_A, r_B) : v \to u$ with $r_A i_A = \mathrm{id}_A$ and $r_B i_B = \mathrm{id}_B$, then $v \in \mathcal{W}$ must imply $u \in \mathcal{W}$. Under the cofiber criterion of B.5, this condition is equivalent to closure of the class of $\mathcal{W}$-aspherical categories under retracts. For stable detection, closure of $\mathcal{L}$ under retracts implies this condition for $R(\mathcal{L})$. Conversely, retract closure of $R(\mathcal{L})$ alone does not recover all spectral retract data of $\mathcal{L}$.

Proof. Functoriality of the categorical mapping-cone model sends an arrow retraction to a retraction of its cones. Thus closure of aspherical objects under retracts, together with the cofiber criterion, gives closure of arrows under retracts. Conversely, a retraction $A \to C \to A$ gives an arrow retraction of $A \to e$ from $C \to e$, so arrow retract closure gives object retract closure without any cofiber hypothesis. After applying B, $(-)_+$, and cofiber, an arrow retraction gives a retraction $D(Bu) \to D(Bv) \to D(Bu)$ of spectra. This proves the assertion for $R(\mathcal{L})$. The last assertion follows already from F.7: the entire stable kernel is not determined by the map class. More precisely only its intersection with the suspension spectra and the retractions induced there can be tested by this assignment.

For the modern fundamental localizers used in this paper, the arrow retract condition is automatic: Cisinski (2006, Proposition 4.2.4, p. 183) proves strong saturation from Definition 3.3.2, and a functor whose image is invertible is closed under arrow retracts. This theorem uses the relative Theorem A; it is not a license to omit Loc (1 bis) in a formal argument assuming only the weaker absolute Loc (3) of the manuscript. □

F.12. Intersection and union aspherical imply both pieces aspherical. [Proved]

Let $\mathcal{W}$ be a fundamental localizer in this paper's convention. Let $U, V \subseteq X$ be full sieves with $U \cup V = X$, and put $K = U \cap V$. If $K \to e$ and $X \to e$ belong to $\mathcal{W}$, then $U \to e$ and $V \to e$ belong to $\mathcal{W}$. Thus row XIIba-A-053 has an affirmative answer in the specified convention. Full W(7 bis), and in particular its reverse half, is unnecessary for this implication.

Proof. If $K = \varnothing$, the hypothesis is $\varnothing \to e \in \mathcal{W}$. Products preserve every fundamental localizer (Cisinski, 2006, Sorites 3.3.8(2)). Taking products with an arbitrary category Y gives $\varnothing \to Y \in \mathcal{W}$; two out of three then gives every functor in $\mathcal{W}$. The conclusion follows.

Suppose $K \neq \varnothing$, and choose an object $k \in K$. Since $K \to e$ is in $\mathcal{W}$, its section $k : e \to K$ is in $\mathcal{W}$ by two out of three. Let I be the span category with three objects and arrows $0 \to 1$, $0 \to 2$. Let

$$P = \int_I (U \leftarrow e \to V), \qquad Q = \int_I (U \leftarrow K \to V).$$

The maps from e select k in each piece. The natural transformation which is $k : e \to K$ at 0 and the identity on the two arms induces a functor $j : P \to Q$ over I. This functor is in $\mathcal{W}$. To verify the last assertion directly from the available axioms, for a covariant diagram $F : I \to \mathbf{Cat}$ the inclusion of $F(i)$ into $(\int_I F)/i$ has the left adjoint which pushes $(j, x, j \to i)$ forward to $(i, F(j \to i)x)$. It is therefore a weak homotopy equivalence. Apply this observation to the two diagrams. The induced functor on each comma category over i is in $\mathcal{W}$ by two out of three, because the map on $F(i)$ is. The relative Theorem A gives $j \in \mathcal{W}$.

The functor $q : Q \to X$ is induced by the three inclusions into X. It is a weak homotopy equivalence. One can check this without a cofibration axiom: Q/x is the Grothendieck construction of $U/x \leftarrow K/x \to V/x$. If $x \in U \setminus V$, the sieve condition on U gives $V/x = K/x$, while U/x has terminal object (x, id_x); its homotopy pushout is therefore contractible. The case $x \in V \setminus U$ is symmetric. If $x \in K$, all three comma categories have terminal object x and the span index is contractible. The nerve comparison for Grothendieck constructions and

Quillen's Theorem A give the asserted weak equivalence. Thus $q \in \mathcal{W}_\infty \subseteq \mathcal{W}$, and $P \to e$ is in $\mathcal{W}$.

There are actual functors $s_U : U \to P$ and $r_U : P \to U$ with $r_U s_U = \mathrm{id}_U$: s_U is the arm inclusion; r_U is the identity on the U arm, is constant at k on the V arm, and sends the central object to k. On arrows from the central object to the U arm it uses the indicated arrow with source k; all arrows wholly on the other arm, and its structural arrows, go to id_k. These formulas respect compositions. There are analogous s_V, r_V. Therefore $U \to e$ and $V \to e$ are arrow retracts of $P \to e$, and F.11 proves the conclusion. □

F.5 What the stable description does, and does not, classify

F.13. The homology examples and the distinction from arbitrary localizing kernels. [Proved]

For any spectrum E, define $\mathcal{A}_E = \{V : E \wedge V = 0\}$. Then $R(\mathcal{A}_E) = \mathcal{W}_E$, where generalized homology of an unpointed space X means $\pi_*(E \wedge \Sigma^\infty_+ X)$ in all integer degrees. No ring structure on E is required. Every full localizing subcategory of Ho(Sp) is a tensor ideal. Nevertheless the definition of $R(\mathcal{L})$ does not require a proof that $\mathcal{L} = \mathcal{A}_E$ for some E.

Proof. Smashing with E preserves cofiber sequences. Thus $E \wedge D(Bu) = 0$ exactly when u induces isomorphisms in all E-homology groups. For the ideal assertion fix $V \in \mathcal{L}$. The spectra Z for which $V \wedge Z \in \mathcal{L}$ form a localizing subcategory containing the sphere. The sphere generates spectra under shifts, cofibers, and coproducts: attach sphere cells successively to hit and then kill homotopy groups, and take the homotopy colimit of the resulting cellular tower. Hence this subcategory is all of Ho(Sp). The construction and proof in F.2 use only the localizing closure properties, not representation as the acyclics of a smash homology theory. □

F.14. Two additional classification assertions not proved by the Galois connection. [Open]

We have not established either of the following assertions in this audit. First, every accessible fundamental localizer satisfying Loc (6 b) is fixed by RT. Equivalently, for such a $\mathcal{W}$ and every functor g,

$$D(Bg) \in \mathrm{loc}\{D(Bu) : u \in \mathcal{W}\} \quad \Longrightarrow \quad g \in \mathcal{W}.$$

Second, a specified arbitrary localizing subcategory of spectra need not be declared a homological Bousfield class merely because it is generated by a set. A concrete test not decided here is

$$\mathrm{loc}(H\mathbb{F}_p) \stackrel{?}{=} \{V : E \wedge V = 0\} \quad \text{for some spectrum } E.$$

We record the latter as an unresolved verification in this audit, not as a new assertion about the present research status of that particular example.

The first missing implication is stronger than the algebraic description of the image in F.5. The cofiber criterion proves invariance under finite suspension and detects ordinary homotopy cofibers; it does not, by itself, supply the comparison needed to transfer an arbitrary stable construction, with desuspensions and infinite coproducts, back to an unstable $\mathcal{W}$-equivalence. The unproved step is closure of the unstable null class under all such stable generation relations. Even for a set-generated reversible localizer, this implication must be proved before replacing "RT-fixed" by "reversible" in the image description. It holds for each of the concrete classes $\mathcal{W}_E$ already in the image by F.13 and for their intersections, by F.5.

There is a useful exact comparison with the model-category literature. Berger and Ratković (2019, Definition 2.1 and Proposition 2.3) call a left proper pointed model category *excisive* when its derived suspension is conservative, and identify this condition with two-out-of-three for maps of cofibre sequences. Their Proposition 2.6 then detects homotopy pushouts by parallel homotopy cofibres. However, the stable embedding in their Remark 2.9 assumes *fully faithful* derived suspension, called *stably excisive*, as well as cofibrant generation. Conservativity alone does not meet that hypothesis, so this theorem does not supply the missing stable-detection implication above.

For the second question, Hovey and Palmieri (1999, Conjecture 9.1 and Proposition 9.2) separate precisely the extra assertion that every localizing subcategory of spectra is a homological Bousfield class. Their Proposition 9.2 reduces the general assertion to principal localizing subcategories; it does not prove it. Set-generated localizing subcategories do admit localization functors, as in Casacuberta et al. (2014, Theorem 3.7 and Proposition 3.8), but that is a different statement. There are actual localizing subcategories which are not homological Bousfield classes in the $H\mathbb{F}_p$*-local* stable category: Wolcott (2015, Proposition 6.4) uses

$$\{V \ H\mathbb{F}_p\text{-local} : [V, \Sigma^n M(p)] = 0 \text{ for all } n \in \mathbb{Z}\}.$$

This is a proper nonzero localizing subcategory there, whereas that category's Bousfield lattice has only two elements. Changing the ambient category back to all spectra changes both coproducts and the relevant tensor product; this example cannot be silently used as a counterexample in Ho(Sp). Our searches and their dates are listed in the accompanying report. The definite failure of the bijection in Lead 6 is F.7, independently of these two unresolved classification assertions.

The second obligation can be made more precise without assuming the general conjecture. Let $a(H\mathbb{F}_p)$ be a wedge of one representative of each Bousfield class annihilating $H\mathbb{F}_p$; Ohkawa's set theorem makes this a set-indexed wedge. Then

$$\operatorname{loc}(H\mathbb{F}_p) \subseteq \ker\big(a(H\mathbb{F}_p) \wedge -\big),$$

and equality is necessary and sufficient for homological representability of $\operatorname{loc}(H\mathbb{F}_p)$. Indeed, the displayed kernel contains the generator and is contained in every smash kernel containing that generator: any such kernel is defined by a spectrum whose Bousfield class occurs in the wedge. Thus this is the smallest possible homological kernel, rather than an unspecified search for a representing spectrum. This is the specialized argument of Hovey and Palmieri (1999, Proposition 9.2). Our additional check of Pützstück (2026, Corollary 1.2) concerns the distinct Hovey–Palmieri *retract conjecture* (their Conjecture 3.12), not their Conjecture 9.1 or the arrow-retract axiom of F.11; it supplies no counterexample to the equality just displayed.

G Consequences for the manuscript

G.1 The scope of the audit and the standing examples

The manuscript locations below refer to the supplied primary editions of Chapters XII and XVI and the XII appendix (Grothendieck, 1991b,a,c). Their manuscript pagination is distinct from the edition's printed-page numbering. Claims about incorrect formulas on XII manuscript pp. 273–274 were additionally checked against the supplied scanned pages.

G.1. The distinction needed to classify the manuscript's uses. [Proved]

In the modern convention, forward homotopy-pushout invariance is available for every fundamental localizer. A statement asserting reverse cancellation for *all* homotopy-pushout

squares is equivalent to the conditions of B.5. The weaker assertion that the intersection and union of two sieves are aspherical only requires forward gluing and retract closure to imply that the pieces are aspherical. It does not imply reverse cancellation for maps of such squares.

Proof. The first two assertions are B.2 and B.5. For the third, let $K = U \cap V$, $X = U \cup V$, with $K \to e$ and $X \to e$ in $\mathcal{W}$. If $\varnothing \to e \in \mathcal{W}$, product stability implies that every $Y \to e$ is in $\mathcal{W}$, so the conclusion holds. Otherwise K is nonempty; choose $k : e \to K$. Two out of three gives $k \in \mathcal{W}$. The map of spans

$$(U \leftarrow e \to V) \longrightarrow (U \leftarrow K \to V)$$

with the identity on U, V induces a specified $\mathcal{W}$-equivalence of Grothendieck constructions. Their second homotopy pushout maps by a $\mathcal{W}_\infty$-equivalence to X. In the first homotopy pushout, both U and V are retracts in $\mathbf{Hot}_{\mathcal{W}}$: collapse the other summand to its indicated basepoint. Since X is aspherical, these retracts are aspherical by strong saturation. The argument uses no reverse implication. The failure of the corresponding assertion about maps is established in G.17 below. This settles the companion's row XIIba-A-053, including the empty-intersection case. □

We use A for the completely specified, checked category of B.7: it has 35 objects and 159 arrows, its classifying space is connected and integrally acyclic, and its fundamental group is nontrivial. For a category T and a fresh object v, $CT = T^{\triangleright}$ denotes the category obtained by adjoining a terminal object v. Its arrows are those of T, the identity of v, and one arrow $t \to v$ for each $t \in T$, with forced compositions. The category $\Sigma T = C_+T \amalg_T C_-T$ has two such terminal cone vertices and no arrows between them. Its two cone subcategories are full sieves. Their nerve union realizes the unreduced suspension of BT. Thus $C_\pm A$ and ΣA are weakly contractible, while A is not. Whenever an infinite example is preferable, A can be replaced by the one-object category BH of the nontrivial integrally acyclic group constructed in B.10. This replacement has no nonidentity idempotents. Its cone comma categories satisfy $(BH)/v \cong BH$ literally. The statements below that use this alternative require no finiteness hypothesis.

G.2. The global reverse claims at the start of XII §5. [Refuted]

XII Théorème 2, manuscript p. 78, and its interpretation of reverse excision as a property of ordinary HOT at pp. 79–80 are false. The forward clauses of W(7) and W(7 bis) hold for $\mathcal{W}_\infty$; the reverse clauses, and the cube clause later called Loc $(6c)$, fail. The conditional implications between those axioms at pp. 76–78 remain valid, with retract closure for the object-to-cube reduction.

Proof. In the sieve-cover square

$$\begin{array}{ccc} A & \longrightarrow & C_+A \\ \downarrow & & \downarrow \\ C_-A & \longrightarrow & \Sigma A, \end{array} \tag{G.2.1}$$

the lower horizontal map is a weak equivalence, whereas the upper horizontal map is not. This refutes W(7 bis) and hence W(7), since A.10 proves the implication W(7)⇒W(7 bis). The finite categorical axioms, sieve conditions, and homology calculations are checked in `finite_examples.py`; the nontrivial fundamental group and weak contractibility of the suspension are proved in B.7. The group version uses BH in the same square. The source's corollary "pieces and union aspherical imply intersection aspherical" is refuted by the same

example. Its forward object corollary is B.2, and its other object corollary is G.1. The assertion about the dual derivator does not repair the claim about HOT: changing a derivator to its opposite changes pushouts to pullbacks, whereas passing from sieves to cosieves by taking opposite categories does not. The relevant companion rows are XIIba-A-049–058 and XIIba-A-062. □

G.2 XII §5: sieve pushouts and total asphericity

G.3. Reverse cancellation remains false along genuine Dwyer maps. [Refuted]

The reverse directions asserted conditionally in XII Proposition 6, manuscript pp. 81–83, Théorème 3, pp. 108–109, and footnote 124, pp. 227–228, do not hold for $\mathcal{W}_\infty$, even when the attaching sieve is a genuine Dwyer inclusion and all the source's boundary and closure conditions hold. Their forward directions are valid. The full conditional statements are valid when the appropriate reverse axiom is assumed.

Proof. Use the actual Φ_2 mapping cylinder of $A \to e$ from B.4, giving a Dwyer inclusion $i : A \hookrightarrow M$, a weak equivalence $M \to e$, and its pushout

$$\begin{array}{ccc} A & \xrightarrow{\ i\ } & M \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ e & \xrightarrow[\ i'\ ]{} & Q = e \amalg_A M. \end{array} \tag{G.3.1}$$

The Dwyer pushout theorem identifies the specified nerve comparison with the homotopy pushout, so $BQ \simeq \Sigma BA$. Therefore $i', q \in \mathcal{W}_\infty$ and $i, p \notin \mathcal{W}_\infty$. The generated cosieve of A in M retracts onto A by a right adjoint; the same is true after pushout. Consequently the closure inclusions required in Proposition 6 are categorical homotopy equivalences. Every genuine Dwyer map also satisfies the "$\mathcal{W}$-dwyerienne" boundary hypothesis used in Théorème 3. The sizes are $|\operatorname{Ob} M| = 141$, $|\operatorname{Fl} M| = 1149$, $|\operatorname{Ob} Q| = 107$, and $|\operatorname{Fl} Q| = 867$; the functor, quotient, sieve, and right-adjoint bijections are checked by the program accompanying B.8.

For the conditional proof, replace the two corners by the specified homotopy-pushout model. The forward implications follow from B.2; the reverse implications follow from B.5. In the more general boundary-weight formulation of Théorème 3, its categorical argument reduces to these comparison squares. No property of arbitrary ordinary pushouts has been substituted for a property of Dwyer pushouts. This records XIIb-a-B002–006, XIIba-C022–024, XIIba-C034, and XIIc-a-C043–045. The implication "a Dwyer map is a $\mathcal{W}$-cofibration" itself survives for every modern fundamental localizer, by A.11. □

G.4. The one-point-complement converse. [Refuted]

The one-point-complement equivalence at XII, manuscript p. 85, and again Corollaries 4–5, pp. 149–151, is valid under W(7 bis), but false for $\mathcal{W}_\infty$. So are reverse formulations in Proposition 7, pp. 84–85, when specialized to a one-point complement. Their direct implication from aspherical comma categories is valid for every fundamental localizer.

Proof. Put $X = \Sigma A$, $U = C_+ A$, and let t be the other cone vertex. Then U is a full sieve, $X \setminus U = \{t\}$ has only its identity, and $U/t \cong A$. Both U and X are weakly contractible, so $U \to X \in \mathcal{W}_\infty$, while U/t is not weakly contractible. For Corollary 5 take $t_0 = t$; its hypothesis on all other complementary objects is vacuous. This also shows why membership

of a sieve inclusion in $\mathcal{W}_\infty$ cannot be replaced by its universal asphericity under changes of the complementary category.

For completeness, the general conditional one-point argument is a homotopy-pushout calculation. For $H : U^{\mathrm{op}} \to \mathbf{Set}$ with $H(u) = \mathrm{Hom}_X(u, t)$, its category of elements is U/t. The nerve of the category formed by adjoining t is the homotopy cone of the projection $U/t \to U$, as in B.9. Reverse cancellation therefore identifies $U \to X \in \mathcal{W}$ with $U/t \to e \in \mathcal{W}$. Without that axiom, the forward direction follows either from the same pushout calculation or directly from Theorem A, since U/x is terminal for $x \in U$. The data and counterexample also apply with $A = BH$. The affected rows are XIIb-a-B012–015, XIIb-b-B050, B054–055, and XIIapp-C060. □

G.5. The total-asphericity gluing argument has a separate error. [Refuted]

XII Proposition 9(b), manuscript pp. 97–98, its retraction-based repair at pp. 98–100, and its conclusion that the outgoing fork is totally aspherical are false even for $\mathcal{W} = \mathcal{W}_R$ with R any nonzero commutative ring. These errors persist when full W(7) is valid. The additional retractions do not make the relevant left Kan extension preserve monomorphisms or intersections.

Proof. Let C have objects x, z, a, b, identities, and the six nonidentity arrows

$$\alpha, \beta : x \rightrightarrows z, \qquad r : z \to a, \quad s : z \to b, \qquad r\alpha = r\beta : x \to a, \quad s\alpha = s\beta : x \to b.$$

There are no further nonidentity arrows or nonforced composites. Let $\Psi \subset C$ be the full outgoing fork on z, a, b, and write $f : \Psi \hookrightarrow C$. For $c = z, a, b$, the category $c\backslash\Psi$ is nonempty and connected. For $c = x$, it has four objects: $\alpha, \beta, r\alpha, s\alpha$, and four nonidentity arrows, one from each of α, β to each of $r\alpha, s\alpha$. It too is connected. Thus f is final in the ordinary covariant sense, and presheaf left Kan extension gives $f_! e_\Psi = e_C$.

The full subcategory $P = \{x, z\}$ is the parallel-arrow category, with $BP \simeq S^1$. The two full subcategories $P \cup \{a\}$ and $P \cup \{b\}$ are its terminal cones, and cover C by sieves. Hence $BC \simeq \Sigma BP \simeq S^2$. Thus $C \to e \notin \mathcal{W}_R$ for $R \neq 0$. This is a final image witnessing that Ψ is not totally $\mathcal{W}_R$-aspherical. Its two arrow subcategories and their intersection $\{z\}$ each have a terminal object, so each is totally aspherical for every fundamental localizer. Each arrow subcategory retracts onto $\{z\}$. Indeed, if a category T has terminal object t and $T \to Y$ is final, the composite $e \xrightarrow{t} T \to Y$ is final, forcing the image of t to be terminal in Y.

The precise preservation error can be read off at x. The subterminal presheaf represented by z on Ψ maps monomorphically into the terminal presheaf represented by a on the arrow category $z \to a$. After the indicated left Kan extension, the relevant map at x identifies the two-element set $\{\alpha, \beta\}$ with a singleton. It is not injective. Retractions of categories do not change this calculation. The complete composition table, all category axioms, the finality commas, and the normalized chain calculation $H_2(C; \mathbb{Z}) = \mathbb{Z}$ are checked in `ag_audit_examples.py`. These calculations settle XIIb-a-B045–049, B051–055 and correct any explanation that attributes their failure solely to the false assertion $\mathcal{W}_\infty \models \mathrm{W}(7)$. □

G.6. The suspension construction used to disprove cofibrancy survives. [Proved]

The counterexample mechanism of XII, manuscript pp. 104–107, survives for $\mathcal{W}_\infty$ when its auxiliary category Y is nonempty and simply connected. Its conclusion that the resulting X' is weakly contractible exactly when Y is weakly contractible has an independent proof. The analogous appendix calculation at manuscript pp. 15–16 has the advertised noncofibrancy conclusion, but its displayed fundamental-group calculation is false.

Proof. In the construction at pp. 104–107, the nerve of X' is the double cone on BY. If BY is simply connected, then ΣBY is weakly contractible if and only if BY is weakly contractible. One implication follows from suspension. For the other, reduced integral homology of BY vanishes by the suspension isomorphism. A simply connected acyclic CW complex is weakly contractible: otherwise a first nonzero homotopy group would be its first nonzero reduced homology group by Hurewicz. Nerve realizations are CW complexes, so this argument applies.

In the appendix, its V_0' is exactly the four-object category C of G.5, after renaming x, z, a, b as $0', 0, a, b$. Thus $BV_0' \simeq S^2$, not S^1, and the assertion $\pi_1(V_0') \simeq \mathbb{Z}$ at p. 15 is false. Adjoining the two cone vertices gives Q with $BQ \simeq S^3$, so the pushed-out terminal-cone inclusion is nevertheless not a weak equivalence. For the later square, $\overline{V} \hookrightarrow X = \Phi_n \times \Phi_m$ is a closed Dwyer inclusion and a weak equivalence: $\overline{V}$ has a terminal object and both factors Φ_n, Φ_m have contractible interval realizations. Thus its pushout $Q \to X'$ is a weak equivalence by the *forward* Dwyer theorem. Two out of three, not reverse excision, now proves $\overline{p} \in \mathcal{W}_\infty \iff q \in \mathcal{W}_\infty$. The closed Dwyer adjoint on its generated sieve $\{0,1,2\}^2$ is coordinatewise $0 \mapsto 0$, $1 \mapsto 1$, $2 \mapsto 1$.

The exact six-object, twenty-arrow double cone is checked in `ag_audit_examples.py`: its normalized chain ranks are $(6, 14, 16, 8)$ and its only positive homology is $H_3 = \mathbb{Z}$. The complete finite $n = m = 2$ attachment and its adjoint are checked in `ag_appendix_square.py`. Thus the proof can use homology in place of the invalid one-point reverse inference or the incorrect fundamental-group calculation. This distinguishes XIIba-C008, C012, C016 and XIIapp-C060–065 from false conclusions. The stronger sentence "nonaspherical whenever $\mathcal{W} \subseteq \mathcal{W}_1$" is false: C is $\mathcal{W}_1$-aspherical, and Q is even $\mathcal{W}_2$-aspherical. □

G.3 XII §6: integral squares, smoothness, and presheaves

G.7. What the integral-square arguments actually require. [Proved]

XII Proposition 13, manuscript pp. 125–126, its cube corollary at pp. 127–128, the homotopy-cocartesian formulations at pp. 132–135, and their proper-integral duals are valid with their stated full reverse assumptions. Their forward clauses, the construction of the integral models, and the comparison with the closed-cover model require no reverse axiom. Their reverse clauses are false for $\mathcal{W}_\infty$.

Proof. For a span $X_1 \leftarrow X_0 \to X_2$, the smooth and proper Grothendieck constructions have specified weak equivalences from the corresponding double mapping-cylinder nerve model. The branch inclusions and the natural transformations of the manuscript exhibit the same coherent cocone. Objectwise $\mathcal{W}$-equivalences induce $\mathcal{W}$-equivalences of these constructions by B.2. This proves the forward cube assertion without reverse excision. The closed-cover replacement is made by the comma categories specified in the text; their branch inclusions have adjoints. It therefore preserves each of the four vertex conditions by actual weak equivalences, not merely by an abstract square isomorphism in $\mathbf{Hot}_{\mathcal{W}}$.

Under the reverse axiom, opposite-edge cancellation is B.5, and the three cube clauses are A.12. For $\mathcal{W}_\infty$, apply the integral construction to the two cones of A. The comparison to (G.2.1) is a weak equivalence, so the failed reverse edge and reverse cube conditions transfer by two out of three. This settles XIIbb-A028–035, A043, XIIb-b-B001, B005–006. Footnote 85, p. 126, correctly removes the reverse assumption from the forward edge implications; it does not establish either converse. □

G.8. The reverse smoothness theorem has an explicit finite counterexample. [Refuted]

The reverse direction of XII Théorème 4(a), manuscript pp. 146–147, and its local version

in Corollary 6(b), p. 157, are false for $\mathcal{W}_\infty$ when the strong W(7 bis) hypothesis is removed. The forward theorem and Corollaries 1–3, pp. 147–149, are valid without reverse excision. The reverse assertions remain valid under their stated full W(7 bis) hypothesis.

Proof. For a category P, let $T(P)$ be P with a new terminal object t, and define $p_P : T(P) \to [1]$ by $P \mapsto 0$, $t \mapsto 1$. The lifting category for the arrow $0 \to 1$ at t is precisely P. All other lifting categories have terminal objects. Hence p_P is $\mathcal{W}$-smooth if and only if P is $\mathcal{W}$-aspherical.

Take $P_0 = A$, $P_1 = C_+A$, $P_2 = C_-A$, and the inclusions $T(P_0) \to T(P_i)$ that identify the new terminal objects. Form the smooth integral I of

$$T(P_1) \longleftarrow T(P_0) \longrightarrow T(P_2),$$

over the base $1 \to 0 \leftarrow 2$, and let $\varphi : I \to [1]$ be induced by the three p_{P_i}. Explicitly its objects are (i, x), and an arrow $(i, x) \to (0, y)$ for $i = 1, 2$ is an arrow $x \to \alpha_i(y)$ in $T(P_i)$; arrows within a fibre are its original arrows, with forced composition. There are no arrows between fibres 1 and 2. At the terminal object of either outside fibre, the exceptional lifting category for $0 \to 1$ is P_1 or P_2, respectively. At the central terminal object, it is the smooth integral of $C_+A \leftarrow A \to C_-A$, whose classifying space is ΣBA and hence is weakly contractible. At any object mapping to 0, and for identity arrows of $[1]$, the lifting category has a terminal object. Thus φ is $\mathcal{W}_\infty$-smooth, whereas p_{P_0} is not.

The program `ag_audit_examples.py` verifies the entire 110-object, 1051-arrow category, all compositions, and the three exceptional lifting categories. The central lifting category has 107 objects and 867 arrows and is identified exactly with the indicated integral. Its simple connectivity follows from the double-cone model; the program independently checks integral acyclicity. The same construction works with BH in place of A.

For the positive direction, the lifting categories for a smooth integral are integrals of the three lifting categories. When the three are aspherical, their integral is aspherical by forward gluing. For the reverse direction under full W(7 bis), the outside lifting categories are first recovered at the outside objects. Then the equivalence "the integral and both outside terms are aspherical if and only if the central term is aspherical" is exactly reverse excision. This gives the required comparison at each arrow of the base. The diamond printed at p. 156 is not strictly commuting in the displayed orientation; its natural transformation supplies the coherent cocone, and the integral comparison is the legitimate homotopy-pushout square. This treats XIIb-b-B035–038, B042–046, XIIbb-C011–020, including the incomplete sentence of footnote 108. □

G.9. The presheaf theorem's converses fail, while its pushout assertion survives. [Refuted]

For a small category X, a pushout square of presheaves on X with one leg a monomorphism gives the homotopy-pushout comparison asserted in XII Théorème 5, manuscript p. 168, for every modern fundamental localizer. The two forward equivalence implications hold. The two converses stated there under strong W(7 bis) are false for $\mathcal{W}_\infty$ without that hypothesis.

Proof. For a presheaf $F : X^{\mathrm{op}} \to \mathbf{Set}$, there is a natural formula

$$N(X/F)_n = \coprod_{x_0 \to \cdots \to x_n} F(x_n).$$

All earlier elements in an element-category simplex are obtained by restriction from its last one. Thus the functor $F \mapsto N(X/F)$ preserves colimits degree by degree and sends monomorphisms to simplicial monomorphisms. Applying it to the given presheaf pushout gives a

simplicial pushout with a cofibration leg. Its comparison from the homotopy pushout is a weak equivalence, and therefore a $\mathcal{W}$-equivalence. This proves the claimed homotopy statement for an arbitrary second leg, including legs that identify elements. Forward homotopy-pushout invariance gives the two forward equivalence implications. No reflection along pushout is used.

For both converses take $X = \Delta$, $K = NA$, the simplicial cone $CK = K \star \Delta^0$, and the pushout $CK \amalg_K CK$. Both attaching maps are monomorphisms. The category of elements Δ/K has objects $([n], \sigma)$ with $\sigma \in K_n$ and arrows $([n], \sigma) \to ([m], \tau)$ the order-preserving maps $\theta : [n] \to [m]$ with $\sigma = \theta^*\tau$. Its realization is naturally weakly equivalent to $|K|$. Consequently the two functors $\Delta/K \to \Delta/CK$ are not weak equivalences, whereas both $\Delta/CK \to \Delta/(CK \amalg_K CK)$ are. This is a completely specified presheaf counterexample obtained from the checked finite category A; the category of all simplices has infinitely many degenerate simplices, which is allowed in the theorem. Under full reverse excision the converses follow from the now-established homotopy-pushout assertion and B.5. This settles XIIbb-C048–052 and confirms the one-way correction in footnote 111. □

G.10. Cofibrancy does not recognize the manuscript's boundary condition. [Refuted]

The assertion at XII, manuscript p. 111, that a $\mathcal{W}$-cofibration sieve with one-point complement is necessarily $\mathcal{W}$-dwyerienne is false. So is its asserted consequence that cofibrancy after every change of the complementary category forces that boundary condition. These failures occur for $\mathcal{W}_\infty$, for all $\mathcal{W}_n$, and for $\mathcal{W}_R$ with $R \neq 0$, including localizers satisfying full W(7).

Proof. Let $U = \{a\}$ and let X have objects a, t, their identities, and exactly two arrows $\alpha, \beta : a \to t$. The sole boundary weight is the two-element presheaf on e. Its category of elements is the discrete two-point category, which is not aspherical for any localizer listed in the statement. Thus the weight fails even the identity test for total asphericity, and $U \hookrightarrow X$ is not $\mathcal{W}$-dwyerienne.

We prove the stronger cofibrancy assertion with all its quantifiers. Change the complement to an arbitrary small category $T \to e$. The resulting category X_T has objects a and the objects of T, the arrows of T, and two arrows $a \to t$ for each $t \in T$; postcomposition preserves the labels α, β. For a functor $e \to C$ choosing $c \in C$, its pushout along $U \hookrightarrow X_T$ is the collage $P(C, c, T)$ with parts C, T and crossing arrows

$$\mathrm{Hom}_{P(C,c,T)}(y, t) = \mathrm{Hom}_C(y, c) \times \{\alpha, \beta\}.$$

The crossing-arrow category is exactly $(C/c) \times \{\alpha, \beta\} \times T$. A functor $f : C \to D$ under e, with $f(c) = d$, induces $C/c \to D/d$; both categories have terminal objects, so this functor is in $\mathcal{W}_\infty$. Products with the fixed category $\{\alpha, \beta\} \times T$ preserve weak equivalences. The gluing comparison proved in G.19 therefore expresses $P(f) : P(C, c, T) \to P(D, d, T)$ as the induced map of homotopy pushouts of a diagram whose three vertex maps are f, this weak equivalence, and id_T. If $f \in \mathcal{W}$, forward gluing gives $P(f) \in \mathcal{W}$. This is precisely the defining condition that $U \hookrightarrow X_T$ be a $\mathcal{W}$-cofibration. When $T = \varnothing$, it is an identity inclusion and the same proof applies.

The four-arrow category and its compositions are included in `ag_small_checks.py`; the universal pushout proof, not a finite sample of C, D, T, proves cofibrancy. This verifies the relevant TOTASPH counterexample independently and corrects XIIba-C037–038. The source's narrower question about *trivial* cofibrations is different: the displayed inclusion is not a weak equivalence for $\mathcal{W}_\infty$ or for nonzero homology localizers, since its nerve is a circle. □

G.4 XII §7 and the previously unaudited end of §8

G.11. The actual cylinder and cofibration formalism survive. [Proved]

The factorizations and enough-cofibration results at XII, manuscript pp. 228–234, and Proposition 16 at pp. 231–232 are valid for every modern fundamental localizer. They use the product of the endpoint inclusion with the source category. They do not require the pushout-product stability sought at pp. 114–115. The reverse clauses in footnote 124 are the false extra assertions isolated in G.3.

Proof. For $n \geq 2$, the construction is

$$Z(f) = Y \amalg_{X\times\{0\}} (X \times \Phi_n), \qquad X \xrightarrow{x\mapsto(x,2n)} Z(f) \xrightarrow{p_f} Y.$$

The map p_f is induced by $f \operatorname{pr}_X$ and id_Y. The endpoint product $X \times \{0, 2n\} \hookrightarrow X \times \Phi_n$ has the Dwyer right-adjoint description of B.4; its pushout gives the required Dwyer inclusion of the other endpoint. The inclusion $Y \to Z(f)$ is a pushout of a trivial Dwyer map and is in $\mathcal{W}_\infty$; its retraction p_f is therefore in $\mathcal{W}_\infty$. This proves the factorization without any cancellation in the reverse direction. Brown's lemma, with the premises verified in A.8–A.11, gives the relative pushout-invariance and couniversal-equivalence assertions of Proposition 16.

By contrast, the map $(U \times \Phi_n) \cup (X \times \{0, 2n\}) \to X \times \Phi_n$ sought at p. 114 is a pushout-product. The finite counterexample and general obstruction are H.6. Thus no inference in this audit uses that desideratum to justify the actual cylinder. The distinction settles XIIc-a-C046, C050, C053 and C062, as well as the issue raised in the prompt after TOTASPH. □

G.12. Pasting homotopy-pushout squares does not use reverse excision. [Proved]

XII Proposition 17, manuscript pp. 234–235, and Proposition 18, pp. 236–239, hold for every modern fundamental localizer. In particular, if the left square in a rectangle is $\mathcal{W}$-homotopy cocartesian, the right square is $\mathcal{W}$-homotopy cocartesian if and only if the outer rectangle is. This is cancellation of *pushout squares*, not reflection of equivalences along pushout.

Proof. For Proposition 17, let $\mathcal{X}_Q$ be the fibred integral of a commutative square Q and $\mathcal{X}$ its subintegral on the first three vertices. The inclusion $i_Y : Y \to \mathcal{X}_Q$ of the fourth fibre is a weak equivalence: the base has its indicated initial object and the fibred replacement has the corresponding adjunction contraction. The natural transformation $i_Y \varphi_Q \Rightarrow i$ written in the source identifies the two maps in **Hot**, where $i : \mathcal{X} \to \mathcal{X}_Q$ is the inclusion. Two out of three gives $i \in \mathcal{W} \iff \varphi_Q \in \mathcal{W}$.

For Proposition 18, write the rectangle as $A \to B \to C$ above $A' \to B' \to C'$. The hypothesis gives a specified equivalence

$$B \amalg^h_A A' \longrightarrow B'.$$

Taking its homotopy pushout with C over B preserves $\mathcal{W}$ by B.2. Homotopy-pushout associativity identifies its source with $C \amalg^h_A A'$. Thus the comparison from this object to C' is in $\mathcal{W}$ exactly when the comparison $C \amalg^h_B B' \to C'$ is, by two out of three. All comparisons use Grothendieck or cylinder models, whose associativity comparison maps are weak equivalences. This proves both directions.

The proof at p. 239 writes a biconditional between two auxiliary maps φ_1 and ψ_1 using W(7 bis). Under the standing hypothesis $\varphi_1 \in \mathcal{W}$, only its forward implication is needed. The remaining comparisons have the contractible comma categories displayed there. Thus its stronger invocation is dispensable; there is no counterexample to Proposition 18 itself. □

G.13. The intervening fibre theory has no reverse-cofibration dependency. [Proved]

The theory of $\mathcal{W}$-fibrations and homotopy-cartesian squares in XII, manuscript pp. 240–261, uses the factorization axiom "enough $\mathcal{W}$-fibrations" and its stated terminal-object hypothesis, not W(7 bis) in reverse. In **Cat**, the all-objects-fibrant hypothesis is automatic from product stability. These results therefore apply whenever Loc $(7F)$ holds, even when Loc $(6b)$ fails.

Proof. For S fixed, the source's replacement category has objects $p : X \to S$ in $\mathrm{Fib}_{\mathcal{W}}$. A factorization $X \to \overline{X} \to S$ makes every object of $\mathbf{Cat}/S$ equivalent to one in this subcategory. The dual of A.9 proves that a $\mathcal{W}$-map between fibrant objects over S remains in $\mathcal{W}$ after every base change. This is exactly what is required to define the derived pullback and to compare two choices of replacement. Repeating this argument in the arrow category gives the Cartesian-square and pasting assertions. The terminal map $X \to e$ is in $\mathrm{Fib}_{\mathcal{W}}$, because its pullback operation is product with X and relative Theorem A gives product stability. Footnote 130 correctly flags that this last premise needs to be supplied in a general abstract category. No use of $\mathrm{Cof}_{\mathcal{W}}$ or of the reverse direction of a cocartesian-square axiom occurs in these arguments. In particular $\mathcal{W}_0$, which satisfies $(7F)$ but not $(6b)$, is a valid instance; the exact distinction is E.2. □

G.14. The two definitions of homotopy-cocartesian square agree. [Proved]

XII Proposition 20, manuscript pp. 262–264, is valid for every modern fundamental localizer: the Grothendieck-integral criterion for a homotopy-cocartesian square agrees with the criterion using a $\mathcal{W}$-cofibration replacement and ordinary pushout. Its invocation of the strong relative Theorem A axiom is legitimate under our conventions, and no reverse excision is required.

Proof. For a square $Q = (S, X_1, X_2, X_3)$, factor $S \to X_1$ functorially as a Dwyer map $S \to \widetilde{X}_1$ followed by $p_1 : \widetilde{X}_1 \to X_1$ in $\mathcal{W}_\infty$. This gives $\widetilde{Q} \to Q$ with the identity at the other vertices. The induced map of Grothendieck constructions

$$\mathcal{X}_{\widetilde{Q}} \longrightarrow \mathcal{X}_Q$$

is in $\mathcal{W}$, by relative Theorem A or B.2. Replace the other leg similarly if desired. The canonical comparison from the integral to the ordinary pushout along the Dwyer leg is in $\mathcal{W}_\infty$. The comparison from that pushout to X_3 belongs to $\mathcal{W}$ exactly when the comparison $\mathcal{X}_Q \to X_3$ does, by two out of three. This is the asserted equivalence.

The displayed arrow labelled λ at p. 263 points from $\mathcal{X}_Q$ to $\mathcal{X}_{\widetilde{Q}}$, opposite to the functor induced by the stated $\widetilde{Q} \to Q$. Reversing that diagram arrow repairs the typing; the argument uses a zigzag in $\mathcal{W}$, so its conclusion is unchanged. Footnote 134 explicitly identifies the stronger relative axiom. XVI footnotes 15–17, manuscript pp. 23–24, return to this comparison and correctly separate it from reverse excision. □

G.15. The opening assertion and the gap of XII §8 need different repairs. [Proved]

The opening sentence of XII §8, manuscript p. 265, reverses the implication proved in Proposition 6 at p. 78. The actual earlier result is W(7)⇒W(7 bis). The proposition stated immediately thereafter, deriving the forward and reverse cube clauses from W(7 bis) and the third clause with retract closure, is valid in the modern convention. Footnote 136 at p. 266 identifies a real missing comparison in its proof, which is supplied by A.8–A.12.

Proof. The two degenerate cubes of A.10 are the proof at p. 78 and establish its direction without additional premises. At p. 265 the proposed proposition has W(7 bis) as its hypothesis, so it is the converse being investigated. In the modern setting, the implication is A.12. The proof's first step replaces each f_i by a Dwyer cofibration followed by a weak equivalence. To preserve the homotopy-pushout conditions of the two faces, one needs a comparison map between their pushout models in $\mathcal{W}$. Termwise weak equivalences alone do not prove this for an arbitrary ordinary pushout. The functorial Dwyer cylinder and the relative Brown gluing lemma give precisely that comparison. This repairs the step without using the reverse conclusion that it is intended to prove. The historical weaker absolute-Theorem-A issue remains exactly as stated in A.16; it is not hidden by this modern repair. □

G.16. The cofibres lemma survives, but the reduction uses reverse detection. [Proved]

The lemma at XII, manuscript p. 268, is true for every modern fundamental localizer: for a map of homotopy-pushout squares, the square of its four pointed homotopy cofibres is again a homotopy-pushout square. The assertion at pp. 269–270 that the *whole reduction* of W(7) to aspherical vertices has used only the forward clause is false. The proof has already used cofibres to reflect equivalences at p. 267.

Proof. Let $f : X_\bullet \to Y_\bullet$ be a coherent map of the two squares. At each vertex let $Z_i = Y_i \amalg^h_{X_i} e$ with the specified cone point. The homotopy colimit of the span $Z_1 \leftarrow Z_0 \to Z_2$ is the iterated homotopy colimit of the corresponding two-dimensional diagram of spans. Interchanging these two homotopy colimits identifies it with

$$\left(Y_1 \amalg^h_{Y_0} Y_2\right) \amalg^h_{X_1 \amalg^h_{X_0} X_2} e.$$

The point diagram has homotopy colimit e, since its indexing span is contractible. The specified face comparisons to Y_3 and X_3 are in $\mathcal{W}$, so forward invariance gives a comparison from this displayed object to Z_3 in $\mathcal{W}$. The interchanges can be made in the bar construction, or in the Grothendieck construction with its natural comparison weak equivalences. This proves the lemma without reverse detection.

On the other hand, $f_i \in \mathcal{W} \iff Z_i \to e \in \mathcal{W}$ is the cofibres criterion of B.5. Its reverse direction fails for $\mathcal{W}_\infty$. The source invokes precisely that biconditional at p. 267 and again writes $\lambda \in \mathcal{W}$ "ou encore" $U \in \mathrm{Asph}_{\mathcal{W}}$ in its deployment argument at p. 269. The conclusion that the three *object* assertions are equivalent under suitable excision is separate from the claim that they detect the three *map* assertions. Thus we keep the true lemma and remove the unsupported claim about the weaker hypotheses of the entire reduction. □

G.17. The abstract cube clause and its first corollary are false as printed. [Refuted]

Theorem 3 ° at XII, manuscript p. 274, claims $f_0, f_3 \in \mathcal{W} \Rightarrow f_1, f_2 \in \mathcal{W}$ from enough cofibrations, a terminal object, and closure of aspherical objects under retracts. It is false. Corollary 1 on the same page, identifying this map assertion with its aspherical-object version, is also false without the missing reverse-detection hypothesis. Both failures already occur in $(\mathbf{Cat}, \mathcal{W}_\infty)$.

Proof. Consider the following strict map between sieve-cover squares:

vertex	0	1	2	3
$X_\bullet$	A	A	C_-A	C_-A
$Y_\bullet$	A	C_+A	C_-A	ΣA
$f_\bullet$	id_A	$A \hookrightarrow C_+A$	id_{C_-A}	$C_-A \hookrightarrow \Sigma A.$

(G.17.1)

All arrows in both squares and all four vertical maps are literal inclusions, so the cube commutes strictly. The two squares are homotopy pushouts; indeed they are MV squares. The maps f_0, f_2, f_3 are weak equivalences, while f_1 is not. The category **Cat** has a terminal object, enough $\mathcal{W}_\infty$-cofibrations by B.4, and retract closure of weakly contractible objects. Thus every hypothesis actually printed in part 3 ° is met. The program `ag_small_checks.py` checks both sieve covers, every category and inclusion, and all six commuting faces. The weak-equivalence decisions use the certified A and suspension proof.

The aspherical-object implication in Corollary 1 is nonetheless true for $\mathcal{W}_\infty$, by G.1. It therefore cannot be equivalent to the failed cube assertion. The printed page, checked against the attached edition PDF, contains no conservativity hypothesis in part 3 °; this is not an artefact of the AI transcription. Adding the reverse axiom repairs the map assertion, as A.12 proves. □

G.18. The literal Cartesian version of Cons 2 is false. [Refuted]

At XII, manuscript p. 273, the condition named Cons 2 says "carré cartésien". Read literally, its claim is false even for $\mathcal{W}_Z$. It therefore is not equivalent to the conservativity axiom stated immediately above it. Replacing "cartésien" by "cocartésien", retaining the two cofibration hypotheses, gives the valid intended implication.

Proof. The strict pullback square

$$\begin{array}{ccc} \varnothing & \longrightarrow & \{0\} \\ \downarrow & & \downarrow \\ \{1\} & \longrightarrow & [1] \end{array} \tag{G.18.1}$$

has three contractible nonempty vertices and empty fourth vertex. Its two initial arrows are $\mathcal{W}$-cofibrations for every modern fundamental localizer, because adjoining a disjoint fixed summand preserves $\mathcal{W}$. For a nonzero ring R, $\varnothing$ is not $\mathcal{W}_R$-aspherical, since $H_0(\varnothing; R) = 0 \neq R$. This verifies every stated hypothesis and refutes the conclusion. All four categories and the empty strict pullback are checked in `ag_small_checks.py`. The word "cartésien" is also present in the edition PDF, not just in its transcription.

For the corrected pushout clause, a pushout with a cofibration leg is a homotopy pushout. If Z_1, Z_2, Z_3 are aspherical, its edge $Z_1 \to Z_3$ is in $\mathcal{W}$; reverse cancellation gives $Z_0 \to Z_2 \in \mathcal{W}$, hence Z_0 is aspherical. Cons 1 is exactly detection by the homotopy cone after a cofibration replacement and is already equivalent, in this setting, to the reverse axiom by B.5. Thus no extra false pullback condition is needed. □

G.5 The appendix's gluing question

G.19. A sieve and its complementary cosieve always give the proposed homotopy pushout. [Proved]

Let U be a full sieve in a small category X, and let $V = X \setminus U$ be the full complementary cosieve. Let E be the full subcategory of the arrow category $\mathrm{Fl}(X)$ on arrows $a : u \to v$ with $u \in U, v \in V$, with source and target functors $s : E \to U$ and $t : E \to V$. The source-to-target natural transformation gives a specified comparison

$$\operatorname{hocolim}(BU \xleftarrow{Bs} BE \xrightarrow{Bt} BV) \longrightarrow BX,$$

and it is a weak homotopy equivalence. This proves the conjecture at XII appendix, manuscript pp. 17–18, for every modern fundamental localizer, without reverse excision. The extra reverse tests proposed in the same paragraph are false for $\mathcal{W}_\infty$.

Proof. We give a category-level proof to specify the coherent comparison. Let P be the fibred Grothendieck construction of the span, using the indexing category with arrows $1 \to 0 \leftarrow 2$. Its objects have three types: objects of U, objects of V, and arrows $a : u \to v$ in E. It retains the internal arrows of these three categories. An arrow from $u' \in U$ to $a \in E$ is an arrow $u' \to u$ in U, and an arrow from $v' \in V$ to $a \in E$ is an arrow $v' \to v$ in V; there are no other inter-fibre arrows. Composition uses s and t. Define $\varphi : P \to X$ by $u' \mapsto u'$, $v' \mapsto v'$, and $(u \xrightarrow{a} v) \mapsto v$. It takes the first kind of inter-fibre arrow $b : u' \to u$ to ab, the second to its given arrow in V, and an internal commutative square in E to its target component. These assignments are a functor precisely because the squares in the arrow category commute.

If $x \in U$, no object mapped into V has an arrow to x, so $P/x = U/x$ and has terminal object id_x. Suppose $x \in V$. Define an endofunctor R of P/x by

$$\begin{aligned}(u, h : u \to x) &\longmapsto (h : u \to x, \mathrm{id}_x),\\ (a : u \to v, h : v \to x) &\longmapsto (ha : u \to x, \mathrm{id}_x),\\ (v, h : v \to x) &\longmapsto (x, \mathrm{id}_x) \quad \text{in the } V \text{ fibre.}\end{aligned}$$

On an internal U arrow b, an internal E square with source component b, or an arrow b from the U fibre into the E fibre, it uses the square (b, id_x). It sends internal V arrows to the identity of its constant value. On an arrow from the V fibre into the E fibre it uses the inter-fibre arrow labelled id_x. The equations defining arrows over x make these assignments composable and functorial. There is a natural transformation $\mathrm{id}_{P/x} \Rightarrow R$: its components are the inter-fibre arrow labelled id_u on a U object, the square (id_u, h) on an E object, and $h : v \to x$ on a V object. There is also a natural transformation from the constant functor at the V object (x, id_x) to R, using the inter-fibre arrow id_x into each E object and the identity on its constant V value. Thus the nerve of P/x is contractible.

Theorem A gives $\varphi \in \mathcal{W}_\infty$. The fibred version of the Grothendieck-construction comparison identifies BP with the displayed homotopy pushout (apply the usual version to opposite categories if desired). Its comparison to BX is exactly $B\varphi$, using the specified source-to-target homotopy. This includes $U = \varnothing$, $V = \varnothing$, or $E = \varnothing$: the same comma-category proof applies and the span becomes the appropriate empty or disjoint union diagram.

Finally, take $X = \Sigma A$, $U = C_+A$, and $V = \{t_-\}$. Then $E \cong A$. Both inclusions $U \to X$ and $V \to X$ are weak equivalences, but neither $E \to U$ nor $E \to V$ is. Thus "the square is homotopy cocartesian" cannot be replaced by the reverse equivalences claimed in the question after an edge is assumed weak. These are precisely the false uses of the appendix's footnote 9. The positive comparison proves the stronger $\mathcal{W}_\infty$ claim conjectured in footnote 10 and corrects the companion's still-open gluing entry. □

G.20. The final conditional theorem is correct after the stated repairs. [Proved]

XII Theorem 1 ° at pp. 272–273, its equivalence of reverse cube cancellation with conservativity in part 2 °, and Corollary 2 at p. 275 are valid in **Cat** for modern fundamental localizers. Corollary 2 must retain its explicit conservativity hypothesis. The same corrected clauses hold in an arbitrary relative category under the precise pushout and factorization premises of G.26, which supplies the compatible finite homotopy-cofibre comparisons from those premises.

Proof. The forward cube theorem is B.2. Restricting reverse cube cancellation to the degenerate cubes of A.10 gives the single-square conservativity assertion. Its converse and the third cube

clause are B.5 and A.12, with the necessary retract premise explicit. Hence all three clauses hold together under Corollary 2's hypotheses. In **Cat** the functorial cylinder, the all-objects-cofibrant property, and Brown's relative gluing lemma supply the comparisons used in this argument. Footnotes 137–138 ask whether factorization alone supplies the abstract comparison calculus. G.26 proves that it supplies the required finite calculus, including the terminal-object comparison; no extra cofibrancy premise on that object is needed. The reference "XV §3" in footnote 137 is to be read as the Brown discussion in XVI §3. Neither the literal Cartesian Cons 2 nor the unconditional part 3° is used in the repaired proof. □

G.6 XVI §§3–7: the complete reverse-dependency register

G.21. The valid and invalid dependencies in XVI §§3–4. [Proved]

The implications in XVI Proposition 3.3, manuscript p. 22, summary 3.5, p. 25, and Corollary 3.7, pp. 26–27, are valid for modern fundamental localizers, with the qualifications on retracts specified in Part A. They are conditional statements about reversible excision; they do not assert that $\mathcal{W}_\infty$ has it. The cofibre calculus in Comment 3.4 and Proposition 3.6, pp. 22–26, and the digression 4.4.15, pp. 66–68, need only forward gluing. The arguments concerning fibration axioms throughout §4 have no reverse-cofibration premise.

Proof. At pp. 16′–19, the degenerate cubes give the three formal implications of A.10. At pp. 19–22, the application of Brown's lemma is legitimate once its four premises are checked; A.8 supplies the proof the source says it does not know, and A.11 supplies the Dwyer premises without a circular use of the desired cube theorem. The conditional reverse implications then give exactly A.12. The unresolved question at p. 23, whether forward single-square excision suffices for the needed cofibre calculus, is answered affirmatively in the modern convention because B.2 proves forward cube gluing directly from relative Theorem A. This does not purport to answer the distinct weaker historical axiom-system question in A.16.

The two questions at pp. 23–24 are the integral criterion and presheaf-monomorphism gluing. They are respectively G.14 and G.9. The references there to Théorème 4 and Corollaries 1–3 of XII need only their forward part, as G.8 proves. Proposition 3.6 is A.11; its Corollary 3.7 adds exactly the reverse condition $(6C)$. In 4.4.15 the text explicitly removes the unnecessarily inherited W(7 bis) assumption from these forward arguments. Modern strong saturation supplies the premise tentatively mentioned in footnote 35.

The eight fibration formulations and their formal arrows use the dual Brown lemma, path replacements, and at certain historically stated steps sequential-colimit closure or strong saturation; A.13–A.17 give their exact premises and modern resolution. None uses reverse excision. The source's $\mathcal{W}_0$ example at pp. 68–70 and footnote 36 is correct: it satisfies $(7F)$ but neither $(6b)$ nor $D_{\mathcal{W}_\infty}$. In particular the hoped-for inference from enough fibrations to the fibration axiom is false even with every additional forward, accessibility, and connectedness condition printed there. The exact factorization and classification are in Parts C and E. □

G.22. Theorem 5.4's list is false, but its accessibility assertion is true. [Refuted]

XVI Théorème 5.4, manuscript p. 79, is false when its list of axioms includes full Loc $(6) = (6a) + (6b)$. Its assertions about accessibility of $\mathcal{W}_\infty$ and its satisfaction of the other listed axioms remain true. The accessibility proof at pp. 80–83 does not need reverse cofibration excision.

Proof. The list includes $(6b)$, refuted by G.2. Its stronger unstated inclusion of $(6c)$ would fail as well by G.17. The remaining listed properties are established in Parts B, C and E; in

particular $\mathcal{W}_\infty$ is the weak-equivalence class of the combinatorial Thomason model category. The class of weak equivalences is accessible, or this may be checked by the countable-extension construction corresponding to the source's convention: begin with countably many objects and arrows in a weak equivalence $A \to B$, close under finite compositions, and successively add witnesses for surjectivity and injectivity of every relative homotopy group. Finite simplicial representatives and finite simplicial homotopies use only finitely many further arrows at each test. There are countably many tests at each countable stage, and countably many stages. Their union is still countable and the induced functor is a weak equivalence. This proves the countable-subcategory extension property stated in Corollary 5.3.2. If there are several components, include witnesses for equality and surjectivity on components first; if both categories are empty, no extension is needed. A weak equivalence with empty target already has empty source.

This argument can be made using the simplicial nerves followed by the accessible fibrant replacement Ex^∞: each finite test factors through a finite stage and a finite part of the categories. It is the same finite-witness mechanism behind the manuscript's path-space and iterated-loop argument. The source's Lemma 5.5 uses the fibration condition to transfer accessibility between maps and aspherical fibres; it does not use $(6b)$. Its unfinished extra hypothesis in part b is not silently filled in by full excision. The standalone theorem about $\mathcal{W}_\infty$ follows by the direct construction above or the model-category accessibility theorem, without that unfinished general lemma. The source's "$\aleph_0$-accessible" here means its countable-cardinality convention, not the modern claim that the arrow subcategory is finitely accessible. □

G.23. The reverse axiom changes the minimum in XVI §6. [Proved]

The fibration theorem XVI 6.1, manuscript pp. 83–85, and Corollary 6.3 have no reverse-cofibration dependency and remain valid; in the modern convention their nontrivial conclusion is $\mathcal{W} = \mathcal{W}_\infty$. In contrast, the full stability list proposed in 6.4(b), pp. 87–89, has total localizer as its least solution if it includes $(6b)$. There is no solution satisfying that list and the additionally asserted containment $\widetilde{\mathcal{W}}_\infty \subseteq \mathcal{W}_\infty$. If (6) is replaced throughout that list by $(6a)$, the least solution is $\mathcal{W}_\infty$.

Proof. The proof of 6.1 factors a map as a weak equivalence followed by a reference-perfect map, uses the fibration axiom to make its fibres aspherical, and then iterates based loop spaces. No homotopy pushout or cofibres criterion occurs. Its exact modern version, including the total-localizer case and no accessibility requirement, is E.5. Lemma 6.2's reduction to aspherical objects uses the same factorization and fibre criterion, and so is also independent of the cofibration axiom.

The list in 6.4(b) explicitly includes relative Theorem A. Therefore modern minimality applies to that list even though the earlier historical definition used absolute Theorem A. With $(6b)$ present, D.4 gives the total localizer; with only $(6a)$ it gives $\mathcal{W}_\infty$. Totality contradicts the separately stipulated containment in $\mathcal{W}_\infty$. The printed conjecture that this minimum is strictly smaller than $\mathcal{W}_\infty$ is therefore false under either proposed reading, for different reasons. Its simple-homotopy suggestion is examined with the correct weaker structures in Part D. The discussion in 6.4(c), pp. 89–91, that $(7F)$ alone does not identify the canonical path replacement with the needed fibration replacement is correct, and Part E gives its precise proper-localization meaning. The unresolved question there about $\mathcal{W}_n$ satisfying $(7F)$ is answered affirmatively in Part C. None of these conclusions about $(7F)$ requires reverse excision. □

G.24. Accessible localizers give derivators without reverse excision. [Known]

For every accessible modern fundamental localizer $\mathcal{W}$, the diagram-localization construc-

tion underlying $\mathrm{HOT}_{\mathcal{W}}$ is a strong derivator on all small indexing categories. No assumption of Loc $(6b)$, Loc (7), Loc $(7D)$, or Loc $(7F)$ is required. In particular the remark at XVI, manuscript pp. 119–120, that $(6b)$ plays no role in obtaining the derivator axioms is correct. This statement uses relative Theorem A and does not identify it with the weaker historical definition.

Proof. By Cisinski (2006, Theorem 5.2.15, p. 225), the category of small categories carries a cofibrantly generated left proper model structure whose weak equivalences are exactly $\mathcal{W}$ and whose cofibrations are the Thomason cofibrations. Its underlying category is locally presentable, so it is combinatorial. Thus Groth (2013, Proposition 1.36, with Theorem 1.34 and Lemmas 1.35–1.37) applies to this very relative category, not just to an abstract equivalent homotopy category. It proves that

$$J \longmapsto \mathrm{Ho}(\mathbf{Cat}^J) = \mathbf{Cat}^J[(\mathcal{W}^J)^{-1}]$$

is a strong derivator. With the manuscript's presheaf coefficient convention, put $J = I^{\mathrm{op}}$ and use the corresponding opposite convention on indexing 2-cells; the displayed values are exactly $\mathrm{HOT}_{\mathcal{W}}(I)$ used in this paper.

Here are the relevant comparisons in that application. Projective and injective model structures on $\mathbf{Cat}^J$ have the same objectwise weak equivalences. For $u : J \to K$, restriction is a right Quillen functor for the projective structures and a left Quillen functor for the injective structures, giving its derived left and right adjoints. The pointwise formulas are computed by homotopy colimits and homotopy limits over the actual comma categories. In the presheaf convention, the left formula at $k \in K$ uses $(k\backslash u)^{\mathrm{op}}$ and the right formula uses $(u/k)^{\mathrm{op}}$. The evaluation and discrete (op)fibration restrictions in these formulas preserve the requisite fibrations or cofibrations: their opposite adjoints are coproducts over the discrete fibres. Consequently the ordinary Kan isomorphism derives to the Beck–Chevalley isomorphism, exactly as in the cited proof. Isomorphisms are detected objectwise because weak equivalences in these diagram model structures are objectwise and saturated; disjoint unions of indexing categories give products of diagram categories. Finally, the model-category cylinder/path rectification for $[1]$ gives the full, essentially surjective partial underlying-diagram functor, which is the strongness assertion of Proposition 1.36.

This verifies the conclusion made in XVI 7.4–7.7, pp. 103–118, under the modern convention and its accessibility hypothesis. The source's tables at pp. 108 and 119 use the forward cofibration theory to construct the required homotopy operations; full Loc (6) is unnecessarily strong in the first table entry and is corrected to $(6a)$ in the discussion at p. 110. Footnotes 55, 57 and 58 correctly warn that recovery of $\mathcal{W}$ as the localizer of its derivator requires the relative axiom in that argument. We retain it from the outset. We do not assert a model presentation or a derivator existence theorem for arbitrary nonaccessible weak historical localizers. The positive modern theorem here already covers $\mathcal{W}_\infty$, all the accessible examples of Part C, and every accessible reverse-excision example, irrespective of whether it has enough intrinsic fibrations. □

G.25. Locations and interpretation of the reverse dependencies. [Proved]

The following register records the direct and inherited reverse-excision uses in the requested sections. A conditional assertion is recorded as valid only with its actual reverse premise; a redundant use in a proof is distinguished from a false conclusion. "Forward" in this table means the precise comparison and premise verified in the cited claim, not unrestricted preservation by ordinary pushout. The supplied XIIc-b excerpt continues through manuscript p. 275, so the register includes the three pages beyond the endpoint mentioned in the prompt.

Verification of the register. We matched the original transcription page markers to the arguments, checked the companion rows where available, and checked the suspect new printed formulas against the attached XII edition PDF. The executable lexical inventory `code/ag_source_inventory.py` locates explicit axiom names and the reverse/conservativity markers; its output is a navigation aid, and the mathematical dispositions below come from the proofs in this paper. The entries of XII §6 concerning truncation and the intervening purely fibration arguments do not introduce an additional reverse dependency: their separate corrections are in E.7 and G.13. □

Location	Statement or use	Disposition
XII 76–80	Object corollaries, W(7)⇒W(7 bis), Théorème 2 and dual-derivator explanation	G.1–G.2; false reverse for $\mathcal{W}_\infty$
XII 81–85	Proposition 6 cancellation; Proposition 7 and one-point complement	G.3–G.4; forward survives
XII 97–100	Total-asphericity gluing and retraction repair	G.5; false even with full W(7)
XII 104–107	Cone-based obstruction to cofibrancy	G.6; survives by suspension and homology
XII 108–114	Théorème 3, trivial cofibrations, boundary recognition	G.3, G.10; recognition is separately false
XII 114–115	Endpoint pushout-product desideratum	G.11, H.6; cannot replace the actual cylinder
XII 125–135	Proposition 13; integral and homotopy-square reverse comparisons	G.7; conditional reverse valid, false for $\mathcal{W}_\infty$
XII 146–157	Théorème 4 reverse smoothness; Corollaries 4–6; lifting-integral argument	G.4, G.8; finite counterexamples
XII 168–169	Théorème 5 converse implications	G.9; homotopy pushout and forward clauses survive
XII 227–234	Footnote 124 and the cofibre construction	G.3, G.11; remove reverse addenda
XII 235–239	Integral inclusion criterion and pasting Proposition 18	G.12; reverse use dispensable
XII 240–261	General fibre and Cartesian-square formalism	G.13; no reverse-cofibration use
XII 262–264	Proposition 20 comparison of definitions	G.14; forward only; correct λ direction
XII 265–267	Reversed introductory citation, Dwyer replacement, footnote 136, cone detection	G.15; repair comparison; retain reverse detection
XII 268–272	Cofibres square and reduction to aspherical vertices	G.16; lemma true, weaker-reduction claim false
XII 273–275	Cons 1–2, unconditional cube 3 °, Corollaries 1–2	G.17, G.18, G.20, G.26; two printed errors
XII appendix 15–16	One-point converse and cancellation after closed Dwyer attachment	G.6; conclusions repaired by forward argument
XII appendix 17–18	Open/closed gluing and its proposed reverse tests	G.19; gluing proved, reverse tests false
XVI 16′–22	Loc $(6a, b, c)$, Brown lemma, Proposition 3.3	A.8–A.12, G.21
XVI 23–27	Integral comparison, presheaf monos, Proposition 3.6, Corollary 3.7	G.9, G.14, G.21
XVI 27–70	Fibration diagram, 4.4.15 and $\mathcal{W}_0$ example, footnote 36	A.13–A.17, G.21; reverse not used
XVI 71–83	Accessibility and Théorème 5.4's full list	G.22; list false, accessibility true

Location	Statement or use	Disposition
XVI 83–97	Fibration theorem, proposed minimum, simple homotopy discussion	G.23, Part D; full list forces totality
XVI 98–123	Homotopy operations, derivator tables, statement that $(6b)$ is unused	G.24; modern accessible theorem, historical caveat explicit

G.26. The factorization-only hypotheses supply the required abstract finite calculus. [Proved]

Let $\mathcal{M}$ be a category, in a fixed universe large enough for the diagrams under consideration, with all pushouts. Let $\mathcal{W} \subseteq \mathrm{Fl}(\mathcal{M})$ contain the isomorphisms and satisfy two out of three. Assume that every arrow factors as an intrinsic $\mathcal{W}$-cofibration followed by a member of $\mathcal{W}$. These are the pushout-dual premises of XII §7 and the factorization premise at manuscript p. 272. They suffice for a choice-independent notion of homotopy-pushout square, with specified weak comparison maps between its models, invariance under objectwise $\mathcal{W}$-maps, and the compatible finite comparisons needed for the square-of-cofibres lemma at p. 268. If $\mathcal{M}$ has a terminal object e, that lemma holds for the actual cofibres of a map of homotopy-pushout squares whose four vertical maps are cofibrations.

Under these same hypotheses, theorem parts 1°–2° at XII pp. 272–274 are valid after correcting Cons 2 to say *cocartesian*. Corollary 2 is valid with its stated conservativity and object-retract assumptions. No global initial object, cofibrancy or fibrancy of the terminal object, pushout-product axiom, or globally functorial factorization is required. The finite absolute diagrams used here have a least vertex: spans, squares, and their products. We do not assert the existence of arbitrary finite homotopy colimits in $\mathcal{M}$.

Proof. **One-leg gluing.** Write $C = \mathrm{Cof}_{\mathcal{W}}$. Composition and cobase change preserve C, directly by associativity of the corresponding pushout functors, and isomorphisms belong to C. By A.9, the maps in $C \cap \mathcal{W}$ are preserved by arbitrary pushout. That claim also proves that a $\mathcal{W}$-map between two C-objects under a fixed base stays in $\mathcal{W}$ after any change of that base.

Consider a natural transformation of spans

$$(A \leftarrow S \rightarrow B) \longrightarrow (A' \leftarrow S' \rightarrow B')$$

whose three components $a : A \to A'$, $s : S \to S'$, $b : B \to B'$ are in $\mathcal{W}$. Assume that the corresponding legs $S \to A$ and $S' \to A'$ belong to C; no condition is imposed on the other legs. Put $A^* = A \amalg_S S'$. The map $A \to A^*$ belongs to $\mathcal{W}$, since it is the pushout of s along $S \to A$. Therefore the induced map $a^* : A^* \to A'$ lies in $\mathcal{W}$, and both $S' \to A^*$ and $S' \to A'$ belong to C. The canonical pushout comparison factors as

$$A \amalg_S B \longrightarrow A \amalg_S B' \cong A^* \amalg_{S'} B' \longrightarrow A' \amalg_{S'} B'. \tag{G.26.1}$$

The first map is in $\mathcal{W}$ by the definition of $S \to A \in C$, applied to b as a map under S. The last is in $\mathcal{W}$ by the relative assertion of A.9, applied to a^* under S'. Thus (G.26.1), which is the specified comparison induced by the original transformation, is in $\mathcal{W}$.

Finite replacements under a fixed object. For $T \in \mathcal{M}$, let $\mathcal{M}_T = T\backslash\mathcal{M}$. This category has initial object id_T and all pushouts, hence all finite colimits. Its full subcategory on the objects $T \to Z$ in C is closed under pushouts along C-maps and under the factorizations used below. Finite colimits used in this paragraph are taken in $\mathcal{M}_T$. The forgetful functor to $\mathcal{M}$ creates connected colimits: compatibility of an ordinary cocone over a connected diagram makes its

induced maps from T coincide. In particular it creates all the nonempty span and product colimits used subsequently. The empty colimit in $\mathcal{M}_T$ is T; no initial object of $\mathcal{M}$ is inferred.

Let J be a finite poset and $D : J \to \mathcal{M}_T$ a diagram. A diagram $Q : J \to \mathcal{M}_T$ is Reedy cofibrant here if, for each $j \in J$, its latching map

$$L_jQ = \operatorname{colim}_{i<j} Q_i \longrightarrow Q_j$$

belongs to C; an empty latching colimit means T. Choose a linear ordering extending the partial order. Having constructed Q_i and $q_i : Q_i \to D_i$ for $i < j$, the maps q_i and the structure maps of D induce a specified map $L_jQ \to D_j$. Factor it as

$$L_jQ \xrightarrow{c_j} Q_j \xrightarrow{q_j} D_j, \qquad c_j \in C, \quad q_j \in \mathcal{W}. \tag{G.26.2}$$

The factorization defines all incoming maps to Q_j and their composites. Thus $q : Q \to D$ is a natural transformation, objectwise in $\mathcal{W}$. At a least vertex with value T we may take both arrows to be identities. There are only finitely many choices.

Two elementary colimit facts give the needed control. If $K \subseteq J$ is downward closed and Q is Reedy cofibrant, then

$$\operatorname{colim}_K Q \longrightarrow \operatorname{colim}_J Q$$

belongs to C. Indeed, add the remaining vertices in a linear ordering; adding j is the pushout of $L_jQ \to Q_j$. In particular all $T \to Q_j$ and $T \to \operatorname{colim}_J Q$ are in C. Secondly, an objectwise $\mathcal{W}$-map $Q \to R$ of Reedy-cofibrant diagrams induces a $\mathcal{W}$-map on colimits. Induct on the number of vertices. When a last vertex j is added, both its latching diagram and the previously added downward-closed diagram have fewer vertices, so their comparison maps belong to $\mathcal{W}$ by induction. Apply (G.26.1) to the spans

$$Q_j \longleftarrow L_jQ \longrightarrow \operatorname{colim}_{i\neq j} Q_i, \qquad R_j \longleftarrow L_jR \longrightarrow \operatorname{colim}_{i\neq j} R_i.$$

Their left legs are the latching cofibrations. The empty case is id_T. This proves colimit invariance without a model-category or lifting assumption.

Common refinements and change of the initial vertex. The independence assertion uses actual comparison maps. Suppose $Q \to D$ and $Q' \to D$ are Reedy-cofibrant replacements in $\mathcal{M}_T$. Their pointwise coproduct $P = Q \amalg_T Q'$ is Reedy cofibrant, because latching colimits commute with this coproduct and a coproduct of two cofibrations is a composite of two pushouts of cofibrations. There is a natural map $P \to D$. Construct a relative replacement $P \to R \xrightarrow{r} D$ inductively by factoring

$$P_j \amalg_{L_jP} L_jR \longrightarrow D_j \quad \text{as} \quad P_j \amalg_{L_jP} L_jR \xrightarrow{d_j} R_j \xrightarrow{r_j} D_j, \qquad d_j \in C, \quad r_j \in \mathcal{W}. \tag{G.26.3}$$

The map $L_jR \to R_j$ is a composite of the pushout of $L_jP \to P_j$ and d_j, hence is in C. Thus R is Reedy cofibrant. The natural maps $Q \to R$ and $Q' \to R$ are objectwise in $\mathcal{W}$, since their composites with r are the given objectwise $\mathcal{W}$-maps to D. Their colimit maps are therefore in $\mathcal{W}$ by the preceding paragraph. These are commuting comparison diagrams over D, not an asserted abstract isomorphism after localization.

For a span $D = (D_1 \leftarrow D_0 \to D_2)$, a cofibrant span model means a span Q with both legs in C and an objectwise $\mathcal{W}$-map $q : Q \to D$. Its root need not equal D_0. Rectify that root by putting

$$Q_0^* = D_0, \qquad Q_i^* = Q_i \amalg_{Q_0} D_0 \quad (i = 1, 2). \tag{G.26.4}$$

The canonical maps $Q \to Q^*$ are objectwise in $\mathcal{W}$, since the two branch maps are pushouts of $q_0 \in \mathcal{W}$ along cofibrations. The induced $Q_i^* \to D_i$ lie in $\mathcal{W}$ by two out of three, and the two legs of Q^* are in C. Furthermore,

$$\operatorname{colim} Q \longrightarrow \operatorname{colim} Q^* \cong (\operatorname{colim} Q) \amalg_{Q_0} D_0$$

is in $\mathcal{W}$, because $Q_0 \to \operatorname{colim} Q$ is a composite of cofibrations. Now Q^* is a Reedy-cofibrant replacement under the fixed root D_0, so (G.26.3) compares it to any other such replacement. This proves that every cofibrant span model gives the same test for the membership in $\mathcal{W}$ of its specified comparison to a fourth vertex.

Consequently the definition of a homotopy-pushout square by cofibration replacement is independent of all choices and of an auxiliary object under which the calculation was performed. If one leg of the original span is already in C, (G.26.1), applied to a cofibrant span model and that original span, shows directly that the canonical comparison from the model to the ordinary pushout lies in $\mathcal{W}$. Thus an ordinary pushout with one cofibration leg is a homotopy pushout, even when the other leg is a map to a noncofibrant terminal object.

Product diagrams and the finite interchange. For finite posets J, K, let $Q : J \times K \to \mathcal{M}_T$ be Reedy cofibrant. Each fixed row and column is Reedy cofibrant. We check the point that a fixed row is not itself downward closed. For fixed $k \in K$, its latching map at j identifies with

$$\operatorname{colim}_{J_{<j} \times K_{\leq k}} Q \longrightarrow \operatorname{colim}_{J_{\leq j} \times K_{\leq k}} Q \cong Q(j,k). \tag{G.26.5}$$

The source is canonically $\operatorname{colim}_{i<j} Q(i,k)$, because k is terminal in $K_{\leq k}$. Both indexing subposets in (G.26.5) are downward closed in the product, and one is included in the other. Thus (G.26.5) belongs to C by the attachment argument. This also treats empty latching subposets. The column argument is identical with the factors interchanged.

The partial-colimit diagram $k \mapsto \operatorname{colim}_{j \in J} Q(j,k)$ is Reedy cofibrant as well: its latching map is

$$\operatorname{colim}_{J \times K_{<k}} Q \longrightarrow \operatorname{colim}_{J \times K_{\leq k}} Q \cong \operatorname{colim}_{j \in J} Q(j,k), \tag{G.26.6}$$

again the map from a downward-closed subposet. Ordinary colimits commute, giving the specified canonical isomorphisms

$$\operatorname{colim}_K \operatorname{colim}_J Q \cong \operatorname{colim}_{J \times K} Q \cong \operatorname{colim}_J \operatorname{colim}_K Q. \tag{G.26.7}$$

Equations (G.26.2)–(G.26.6) show that these are comparisons of valid compatible homotopy-colimit models whenever the relevant diagrams are formed under one fixed object. Different choices are compared by (G.26.3). No pushout-product of maps has entered the construction.

In particular, for a map of spans or squares choose a replacement of its whole diagram, indexed by the original poset times $[1]$, under the value of its least vertex. The two faces are cofibrant span models, including when their initial objects differ from those of the original faces; (G.26.4) justifies that change. If the given transformation is objectwise in $\mathcal{W}$, the replacement's corresponding vertical maps are in $\mathcal{W}$ by two out of three. Equations (G.26.1) and (G.26.7) then give the specified weak comparison of pushout models. The commutative square with the two fourth-vertex comparisons proves invariance of homotopy-pushout squares under objectwise $\mathcal{W}$-maps. More generally, for a map of two homotopy-pushout squares, $f_0, f_1, f_2 \in \mathcal{W}$ implies $f_3 \in \mathcal{W}$. This is the abstract theorem's part 1°.

The actual square of cofibres. Assume now that e is terminal. Let $I = [1] \times [1]$, with vertices $0, 1, 2, 3$ in the manuscript's order, and let J be the span poset with arrows $0 \to 1$, $0 \to 2$. Given a strict map $f : X \to Y$ of homotopy-pushout squares with each $f_i \in C$, form the actual diagram

$$D : I \times J \longrightarrow \mathcal{M}, \qquad D(i,0) = X_i, \quad D(i,1) = Y_i, \quad D(i,2) = e. \tag{G.26.8}$$

The two J-arrows are f_i and the unique $X_i \to e$; the I-arrows in the last column are identities. Its least vertex has value $T = X_0$. Choose a Reedy-cofibrant replacement $q : Q \to D$ in $\mathcal{M}_T$.

Let $I_0 = I \setminus \{3\}$, the initial three-vertex span. For $j = 0, 1, 2$ there are natural comparison maps

$$\lambda_j : \mathrm{colim}_{i \in I_0} Q(i, j) \longrightarrow Q(3, j). \tag{G.26.9}$$

For $j = 0, 1$, their composites with $q_{3,j}$ are the comparison maps of cofibrant span models to the fourth vertices of X and Y. They are in $\mathcal{W}$ by hypothesis and the independence argument. Since $q_{3,j} \in \mathcal{W}$, two out of three gives $\lambda_j \in \mathcal{W}$.

For $j = 2$, compare the span $Q(-, 2)|_{I_0}$ to the constant span $e \leftarrow e \rightarrow e$. Its three comparison maps lie in $\mathcal{W}$, its own legs are cofibrations, and the target legs are identities, hence cofibrations. Equation (G.26.1) proves

$$\mathrm{colim}_{i \in I_0} Q(i, 2) \longrightarrow e \in \mathcal{W}. \tag{G.26.10}$$

The composite of λ_2 with $q_{3,2} : Q(3, 2) \rightarrow e$ is this map, so $\lambda_2 \in \mathcal{W}$. This is the full justification that the terminal constant span has homotopy pushout e; it uses no cofibrancy of $T \rightarrow e$.

By (G.26.5)–(G.26.6), the two J-diagrams in (G.26.9) are Reedy cofibrant. Colimit invariance therefore sends their objectwise $\mathcal{W}$-map λ to a map in $\mathcal{W}$. Put

$$Z'_i = \mathrm{colim}_{j \in J} Q(i, j).$$

Using (G.26.7), the resulting map is exactly

$$Z'_1 \amalg_{Z'_0} Z'_2 \longrightarrow Z'_3 \in \mathcal{W}. \tag{G.26.11}$$

The square Z' is Reedy cofibrant by (G.26.6), so (G.26.11) says it is a homotopy-pushout square.

These Z'_i are replacement-row cofibres. The cofibres in the manuscript's lemma are the actual pushouts

$$Z_i = Y_i \amalg_{X_i} e,$$

with their specified maps $e \rightarrow Z_i$. The natural transformation q induces a natural comparison $Z' \rightarrow Z$. At each vertex its map is in $\mathcal{W}$ by (G.26.1), applied to the corresponding row of Q and (G.26.8): the required target leg is precisely $f_i \in C$. Thus the actual square Z is a homotopy pushout by the invariance already established. This supplies the exact square-of-cofibres lemma with actual comparison maps and its actual point data.

Conservativity and the final abstract theorem. For arbitrary f_i, first apply (G.26.2) to the whole $I \times [1]$ diagram of f, under X_0. Its vertical maps $g_i : \widetilde{X}_i \rightarrow \widetilde{Y}_i$ are in C by (G.26.5); its vertex comparisons to X, Y are in $\mathcal{W}$. Hence $g_i \in \mathcal{W}$ if and only if $f_i \in \mathcal{W}$, and its two faces are still homotopy pushouts. Its actual cone square

$$Z_i = \widetilde{Y}_i \amalg_{\widetilde{X}_i} e$$

is therefore a homotopy pushout by the preceding paragraph.

Suppose that homotopy-pushout squares are conservative in the source's sense. For a cofibration $g : A \rightarrow B$, its ordinary cone square is a homotopy pushout. Thus

$$g \in \mathcal{W} \iff e \rightarrow B \amalg_A e \in \mathcal{W} \iff (B \amalg_A e) \rightarrow e \in \mathcal{W}. \tag{G.26.12}$$

The first forward implication also follows directly from A.9; its reverse is conservativity. The second equivalence follows from two out of three, since the composite of the two displayed point maps is id_e.

If $f_1, f_2, f_3 \in \mathcal{W}$, then Z_1, Z_2, Z_3 are aspherical by (G.26.12). The edge $Z_1 \rightarrow Z_3$ is in $\mathcal{W}$, so conservativity gives $Z_0 \rightarrow Z_2 \in \mathcal{W}$. Hence Z_0 is aspherical and g_0, and therefore f_0, is in $\mathcal{W}$. Conversely this reverse cube clause implies conservativity by the degenerate cube in A.10; its

degenerate face is an ordinary pushout along an identity, hence is a homotopy pushout by (G.26.1). This proves part 2°.

Conservativity implies Cons 1 and the corrected cocartesian Cons 2 directly. Conversely assume those two conditions. In the preceding cone-square argument, factor the two initial legs of Z as cofibrations followed by $\mathcal{W}$-maps, keeping Z_0 fixed. The resulting ordinary pushout P maps to Z_3 by a member of $\mathcal{W}$. If Z_1, Z_2, Z_3 are aspherical, both replaced branches and P are aspherical. Corrected Cons 2 implies that Z_0 is aspherical, and Cons 1 applied to g_0 implies $g_0 \in \mathcal{W}$. This gives the reverse cube clause and hence conservativity. The literal Cartesian condition remains false, as G.18 proves.

Finally assume the source's literal premise that every retract *in* $\mathcal{M}$ of a $\mathcal{W}$-aspherical object is aspherical. In a homotopy-pushout square Z with Z_0, Z_3 aspherical, factor its two initial legs as

$$Z_0 \longrightarrow \widehat{Z}_i \longrightarrow Z_i \quad (i = 1, 2),$$

with first arrows in C and second arrows in $\mathcal{W}$. Let $P = \widehat{Z}_1 \amalg_{Z_0} \widehat{Z}_2$. Its comparison to Z_3 lies in $\mathcal{W}$, so P is aspherical. Push the two legs and this pushout along $Z_0 \to e \in \mathcal{W}$. All resulting maps

$$\widehat{Z}_i \longrightarrow Z_i^* = \widehat{Z}_i \amalg_{Z_0} e, \qquad P \longrightarrow P^* = Z_1^* \amalg_e Z_2^*$$

belong to $\mathcal{W}$, since the respective structural maps out of Z_0 are cofibrations. Thus P^* is aspherical. Each Z_i^* is an actual retract of P^*: use its identity and the composite $Z_{3-i}^* \to e \to Z_i^*$ on the other branch. These maps agree on e. Retract closure makes both Z_i^*, and then both Z_i, aspherical. Applying this object result to the actual cone square and using (G.26.12) proves the third cube clause under conservativity. Together with the forward and reverse clauses this is Corollary 2. It does not prove the unconditional part 3° or its Corollary 1, whose counterexample remains G.17. □

H The placement of the localizers and their derivators

H.1. The inclusions and the two poles. [Proved]

Every modern fundamental localizer contains $\mathcal{W}_\infty$. Among those satisfying Loc (5), exactly $\mathcal{W}_\infty$ satisfies Loc ($7D_{\mathcal{W}_\infty}$). The localizers satisfying Loc (6 b) have least member $\mathcal{W}_\mathbb{Z}$, but do not form the whole upper interval above $\mathcal{W}_\mathbb{Z}$. There are strict inclusions

$$\mathcal{W}_\infty \subsetneq \mathcal{W}_+ \subsetneq \mathcal{W}_\mathbb{Z} \subsetneq \mathcal{W}_0.$$

Neither $\mathcal{W}_+$ nor $\mathcal{W}_0$ satisfies Loc (6 b). For each finite $n \geq 1$, $\mathcal{W}_n$ and $\mathcal{W}_\mathbb{Z}$ are incomparable. No nontotal localizer satisfying Loc (6 b) contains $\mathcal{W}_1$.

Proof. Minimality is B.1; the fibration classification, with its connectedness qualification, is E.5. The minimal reverse-excision class is D.1. The plus-construction inclusions and its failure of reverse excision are F.9; the $\mathcal{W}_0$ counterexample is B.13. Strictness of $\mathcal{W}_\mathbb{Z} \subset \mathcal{W}_0$ follows, for example, from $B\mathbb{Z} \to e$, since its source is connected and has first homology $\mathbb{Z}$. For $n \geq 1$, the map $S^{n+1} \to *$ belongs to $\mathcal{W}_n$ and not to $\mathcal{W}_\mathbb{Z}$, while the nontrivial acyclic-group map $BG \to e$ of B.10 belongs to $\mathcal{W}_\mathbb{Z}$ and not to $\mathcal{W}_n$. The sphere map means any specified simplex-category model; all such comparisons are weak homotopy equivalences. The final assertion is D.3. Thus the phrase "the localizers above $\mathcal{W}_\mathbb{Z}$ with reversible cofibrations" must be read as a subclass subject to that extra axiom, not as an inclusion-only characterization. □

H.2. Total asphericity and ordinary final images. [Proved]

For a small category A, define $T_{\mathcal{W}}(A)$ by requiring that

$$B/(f_!1_A) \longrightarrow e \quad \text{belong to } \mathcal{W}$$

for every small B and every functor $f : A \to B$. Here $f_!$ is ordinary left Kan extension of presheaves along f^{op}, and B/H is the presheaf category of elements. Then $T_{\mathcal{W}}(A)$ holds if and only if every target B of an ordinary final functor $A \to B$ is $\mathcal{W}$-aspherical. Final means that every $b\backslash A$ is nonempty and connected; no homotopical finality is implicit.

Proof. For an arbitrary f, put $P_f(b) = \pi_0(b\backslash A)$. Precomposition by $\beta : b' \to b$ defines the presheaf action $P_f(b) \to P_f(b')$. The pointwise Kan formula is

$$(f_!1_A)(b) = \mathrm{colim}_{(b\backslash A)^{\mathrm{op}}}\{*\} = P_f(b).$$

There is a functor $q_f : A \to B/P_f$ given by $q_f(a) = (fa, [(a, \mathrm{id}_{fa})])$; on an arrow it uses its image under f. For an object (b, c) of B/P_f, its comma category under q_f is precisely the connected component c of $b\backslash A$. It is nonempty and connected. Thus q_f is ordinary final and the target in every Kan-extension test is a final image of A. Conversely, if f is final, $P_f = 1_B$, canonically, and $B/P_f \cong B$. This proves both directions, including empty categories. In particular, totality implies plain asphericity by the identity test. This verifies the exact final-image reformulation used in the attached TOTASPH return, with its presheaf variance specified. □

H.3. The entire interval is invisible to total asphericity. [Proved]

For every small category A,

$$T_{\mathcal{W}_\infty}(A) \quad \Longleftrightarrow \quad T_{\mathcal{W}_{\mathbb{Z}}}(A).$$

Consequently these conditions are equivalent to $T_{\mathcal{W}}(A)$ for every $\mathcal{W}_\infty \subseteq \mathcal{W} \subseteq \mathcal{W}_{\mathbb{Z}}$. Their intrinsic criterion is that A be nonempty and connected and that, for every covariant functor $H : A \to \mathbf{Set}$, every connected component of $\mathrm{El}_A(H)$ be integrally acyclic. In this criterion "integrally acyclic" may equivalently be replaced by "weakly contractible".

Proof. Denote the condition on all element categories by $\mathcal{P}_{\mathbb{Z}}(A)$, without imposing nonemptiness on an empty element category. Suppose first that $T_{\mathcal{W}_{\mathbb{Z}}}(A)$. The identity test makes BA nonempty, connected, and integrally acyclic. For a covariant H with $\mathrm{El}_A(H)$ nonempty and connected, adjoin an object x with $\mathrm{Hom}(x, a) = H(a)$, no arrows from A to x, and $\mathrm{End}(x) = \{\mathrm{id}\}$. The resulting category is the collage A_H of B.9. The inclusion $A \to A_H$ is ordinary final: over $a \in A$ its comma category is $a\backslash A$, with initial object, and over x it is $\mathrm{El}_A(H)$. Therefore BA_H is integrally acyclic by H.2. The mapping-cone description of B.9 and its reduced homology sequence show that $B\,\mathrm{El}_A(H)$ is integrally acyclic. For an arbitrary H, each connected component gives a subfunctor $H_E \subseteq H$, since every arrow preserves components. Applying the same argument to H_E proves $\mathcal{P}_{\mathbb{Z}}(A)$.

The property is inherited by each component E of each $\mathrm{El}_A(H)$. In fact, given $K : E \to \mathbf{Set}$, define

$$L(a) = \coprod_{h \in H_E(a)} K(a, h).$$

For $v : a \to a'$, send the summand $K(a, h)$ by $K(v)$ into the summand indexed by $H_E(v)h$. The correspondence $((a, h), k) \mapsto (a, (h, k))$ is an isomorphism $\mathrm{El}_E(K) \cong \mathrm{El}_A(L)$, on both objects and arrows. This proves the inheritance assertion with the exact comparison, not merely an identification of homotopy types.

Let such an E be connected and nonempty. The singleton functor makes BE acyclic. Fix $e_0 \in E$ and let ΠE be the groupoid obtained by formally inverting its arrows. The edge-path presentation identifies $G = \mathrm{Aut}_{\Pi E}(e_0)$ with $\pi_1(BE, e_0)$. For a subgroup $K \leq G$, put

$$H_K(e) = \mathrm{Hom}_{\Pi E}(e_0, e)/K,$$

where K acts by precomposition and arrows of E act by postcomposition. These actions are bijective. Thus $\mathrm{El}_E(H_K) \to E$ has unique arrow lifts with either prescribed source or prescribed target. Its nerve realizes to the covering associated with K: edge paths lift uniquely, its monodromy is the transitive coset action, and a loop lifts to a loop at the identity coset exactly when its class belongs to K. In particular the covering is connected and its fundamental group is K. These are also the assertions of Hatcher (2002, Theorem 1.38), with the subgroup selected by the displayed basepoint.

Take K to be the cyclic subgroup generated by an arbitrary element of G. The inherited property gives

$$0 = H_1(B\,\mathrm{El}_E(H_K); \mathbb{Z}) = K_{\mathsf{ab}} = K.$$

The first-homology identification is Hatcher (2002, Theorem 2A.1). Every element of G is therefore trivial. Since BE is a simply connected acyclic CW space, Hurewicz gives weak contractibility as in B.6. This proves that $\mathcal{P}_{\mathbb{Z}}$ and the corresponding weak-contractibility condition are equivalent.

Conversely, assume A is nonempty and connected and satisfies this last condition. Its singleton element category shows that BA is weakly contractible. For an ordinary final $f : A \to B$, every $b\backslash A$ is the element category of the covariant functor $a \mapsto \mathrm{Hom}_B(b, fa)$; it is nonempty and connected and hence weakly contractible. The final-variance Theorem A implies that the specified map Bf is a weak equivalence. Its variance follows either by applying Theorem A to opposites or by the relative-Theorem-A property for $\mathcal{W}_\infty$; reversing nerves does not change their realizations. Therefore BB is weakly contractible. Claim H.2 proves $T_{\mathcal{W}_\infty}(A)$. The reverse implication to $T_{\mathcal{W}_{\mathbb{Z}}}(A)$ follows from inclusion of the localizers. Monotonicity of every test in H.2 then proves the assertion for the entire interval. This independently verifies TOTASPH C.5. □

H.4. Above the one-type localizer, totality is plain asphericity. [Proved]

If $\mathcal{W}_1 \subseteq \mathcal{W}$, then $T_{\mathcal{W}}(A)$ is equivalent to $A \to e \in \mathcal{W}$ for every small A, including the empty category. No accessibility hypothesis is required.

Proof. Only sufficiency needs proof. Let $f : A \to B$ be ordinary final and suppose $A \to e \in \mathcal{W}$. If A is empty, finality forces B empty. Otherwise finality gives a bijection on components and a surjection on each fundamental group. Here is the edge-path verification of the latter statement. For each object b of B, choose an arrow $\beta_b : b \to f(a_b)$. For an arrow $v : b \to c$, the objects (a_b, β_b) and $(a_c, \beta_c v)$ in $b\backslash A$ are joined by a zigzag. Applying f to that zigzag identifies, in ΠB, the arrow $\beta_c v \beta_b^{-1}$ with the image of a path in ΠA. Multiplying these identities around any based loop cancels the β_b factors and lifts the loop. Connectedness of the same comma categories shows that choices of a component over b agree, and nonemptiness gives a component over every b; this proves the component bijection as well.

Choose one object a_i in each component of A. Let $\mathcal{S} \subset \Pi A$ and $\mathcal{T} \subset \Pi B$ be the full subgroupoids on a_i and $f(a_i)$ respectively. Their inclusions are equivalences, and $A \to \Pi A$, $B \to \Pi B$ are $\mathcal{W}_1$-equivalences by their edge-path presentations. The restricted functor $\psi : \mathcal{S} \to \mathcal{T}$ is bijective on objects and surjective on each endomorphism group. In particular it is a $\mathcal{W}$-cofibration by C.11. The ordinary pushout of $\mathcal{T} \leftarrow \mathcal{S} \to e$ is e: a compatible functor to any

category must send all objects to the selected object and all arrows to its identity, since every arrow lifts under ψ. Cobase change along ψ therefore sends the $\mathcal{W}$-map $\mathcal{S} \to e$ to $\mathcal{T} \to e$. The specified groupoid comparisons give $B \to e \in \mathcal{W}$. Claim H.2 proves totality. For the total localizer the empty case is included automatically; for every other localizer, $\varnothing \to e$ is excluded by the argument in D.3. This verifies TOTASPH D.19. □

H.5. The precise boundary for all sieves to be cofibrations. [Proved]

For a modern fundamental localizer $\mathcal{W}$, the following are equivalent: $\mathcal{W}_1 \subseteq \mathcal{W}$; a category of homotopy type S^2 is $\mathcal{W}$-aspherical; every object-injective functor belongs to $\mathrm{Cof}_{\mathcal{W}}$; and every full sieve belongs to $\mathrm{Cof}_{\mathcal{W}}$.

Proof. The equivalence of the first two conditions is Cisinski (2006, Proposition 9.2.17). Their implication to all object injections is C.11; a full sieve is object-injective. We prove the remaining implication by a complete finite example. Let Q have objects a, b, z, x and nonidentity arrows

$$r : a \to z, \quad v : b \to z, \quad \lambda, \mu : z \rightrightarrows x, \quad \alpha : a \to x, \quad \beta : b \to x,$$

with the only nonidentity composites $\lambda r = \mu r = \alpha$ and $\lambda v = \mu v = \beta$. Its full subcategory $U = \{a, b, z\}$ is a sieve with terminal object z. The two full subcategories on $\{a, z, x\}$ and $\{b, z, x\}$ have initial objects and intersect in the parallel-arrow category $z \rightrightarrows x$, whose nerve is a circle. Their nerves cover NQ, so $BQ \simeq S^2$. Equivalently, this is the opposite of two terminal cones on that circle; reversing the nerve gives the same realization.

Collapsing U to e forces r, v to be identities and hence forces $\lambda = \mu$. The pushout is the arrow category $[1]$. If $U \hookrightarrow Q$ is a $\mathcal{W}$-cofibration, applying its pushout functor to $U \to e \in \mathcal{W}_\infty$ gives $Q \to [1] \in \mathcal{W}$, so BQ is $\mathcal{W}$-aspherical. The program `code/sieve_pushouts.py` constructs the categorical pushout by the defining coend relations, checks all compositions, and verifies the counts $(4, 10)$ for Q, $(3, 5)$ for U, and $(2, 3)$ for its pushout. Its normalized integral chain calculations give Betti numbers $(1, 0, 1)$ for Q, with every nonzero Smith factor equal to 1. The cone decomposition supplies the homotopy-type assertion, beyond what the homology calculation alone proves. □

H.6. The pushout-product substitution for the manuscript's cylinders. [Refuted]

Let $n \geq 2$ and let $\Phi_n = (0 \to 1 \leftarrow 2 \to \cdots \leftarrow 2n)$. If S^2 is not $\mathcal{W}$-aspherical, then the pushout-product of $\{0\} \hookrightarrow [1]$ and $\{0, 2n\} \hookrightarrow \Phi_n$ does not belong to $\mathrm{Cof}_{\mathcal{W}}$. In particular this fails for $\mathcal{W}_\infty$ and for every nontotal localizer satisfying Loc (6 b). Moreover the only class of full-sieve $\mathcal{W}$-cofibrations stable under taking this pushout-product is a class consisting of isomorphisms. Products with the endpoint inclusion, as used in B.4, have no such defect.

Proof. Write $P = [1] \times \Phi_n$, with lower and upper objects $l_k = (0, k)$ and $u_k = (1, k)$. Let $A \subset P$ contain every l_k and only the two upper endpoints u_0, u_{2n}. It is a full sieve. Its inclusion $\alpha : A \hookrightarrow P$ is precisely the pushout-product in the statement. The nerve of A is a tree, the lower alternating interval with two attached endpoint leaves. The category P is a product of two contractible categories. Consequently $\alpha \in \mathcal{W}_\infty$.

Put $C = Q^{\mathrm{op}}$ for the four-object category of H.5, retaining labels so its arrows are $\lambda, \mu : x \rightrightarrows z$, $r : z \to a$, $v : z \to b$, and their equal composites. Define $p : A \to C$ by $p(l_0) = z$, $p(u_0) = b$, and $p(w) = a$ at every other object. The arrows $l_0 \to l_1$ and $l_0 \to u_0$ go to r and v; all remaining nonidentity arrows go to identities. Form the ordinary pushout $Q_0 = P \amalg_A C$. We identify its crossing arrows to establish its homotopy type without replacing the categorical pushout by a homotopy pushout.

Its objects are the four objects of C and the u_k, $1 \le k \le 2n-1$. It retains C and the full subcategory of these upper objects, has no arrows from an upper object to C, and has crossing sets

$$\mathrm{Hom}_{Q_0}(c, u_k) = \int^{w \in A} \mathrm{Hom}_C(c, pw) \times \mathrm{Hom}_P(w, u_k).$$

Composition is induced by that of C and P. For $k \ge 2$, every predecessor of u_k in A maps to a, and its predecessor category is nonempty and connected. The crossing presheaf is therefore $C(-, a)$. For $k = 1$ its predecessor category has the four objects l_0, l_1, l_2, u_0 and arrows $l_0 \to l_1$, $l_2 \to l_1$, $l_0 \to u_0$; its image is the fork $\Psi = (z \to a, z \to b) \subset C$. This functor to Ψ is final: its three comma categories are respectively that connected tree, the arrow $l_2 \to l_1$, and the singleton u_0. The fork inclusion $\Psi \to C$ is final as well. At x, its comma category has two objects over z and one over each of a, b, with four crossing arrows, hence is a connected square; at the other objects its comma category has an initial object. Thus the coend for $k = 1$ is the singleton presheaf on C. The transition from u_2 to u_1 is the unique map $C(-, a) \to 1$; all transitions between the other representable weights are identities.

The full subcategory $R_0 = C \cup \{u_1\} \subset Q_0$ is therefore the terminal cone on C. Define $\rho : Q_0 \to R_0$ to fix this subcategory and to send every u_k, $k \ge 2$, to a. On a crossing arrow represented by $h : c \to a$, use h; on an upper arrow use the corresponding identity of a or its unique arrow to u_1. The coend description proves that these assignments respect composition. If $j : R_0 \hookrightarrow Q_0$, then $\rho j = \mathrm{id}$ and there is a natural transformation $j\rho \Rightarrow \mathrm{id}_{Q_0}$, with components at u_k, $k \ge 2$, the arrows represented by id_a, and identity components elsewhere. Naturality on crossing arrows is exactly composition with id_a; on $u_2 \to u_1$ it is the unique-map transition above. Thus BQ_0 is contractible.

We use one general consequence of enough intrinsic cofibrations, already supplied by B.4. Every $i \in \mathrm{Cof}_{\mathcal{W}} \cap \mathcal{W}$ is couniversally in $\mathcal{W}$. Indeed factor an arbitrary $f : A \to D$ as $A \xrightarrow{k} D' \xrightarrow{w} D$ with $k \in \mathrm{Cof}_{\mathcal{W}}$ and $w \in \mathcal{W}$. Cobase change of i along k is in $\mathcal{W}$, since k preserves $\mathcal{W}$. Cobase change of w along i is in $\mathcal{W}$, since i preserves $\mathcal{W}$. The commuting square with left edge w and these two maps makes the cobase change along f belong to $\mathcal{W}$ by two out of three. If α were a $\mathcal{W}$-cofibration, its weak equivalence would therefore give $C \to Q_0 \in \mathcal{W}$. Since BQ_0 is contractible, this would make $BC \simeq S^2$ aspherical, a contradiction.

There is also a literal weak-equivalence input exhibiting failure for $\mathcal{W}_\infty$. Replace C by $D = C^{\triangleright}$ with terminal object s, and use the composite $A \to C \to D$, which is a weak equivalence between contractible categories. The pushout $Q = P \amalg_A D$ retracts by the same formulas onto the full subcategory on C, s, u_1. The two added objects are terminal cones on C, with no arrows between them. Thus $BQ \simeq \Sigma BC \simeq S^3$, whereas BP is contractible. Cobase change along α has sent the weak equivalence $A \to D$ to the non-weak-equivalence $P \to Q$.

All objects, arrows, and compositions for arbitrary n have been specified by a finite poset, a four-object category, and explicit coend relations. The program checks the complete cases $n = 2, 3$, including every possible composite representative, all associativity equations, the pushout maps, the retraction, and its natural transformation. For $n = 2$, Q_0 has 7 objects and 25 arrows and Q has 8 objects and 30 arrows; their Betti numbers are respectively $(1, 0, 0, 0, 0)$ and $(1, 0, 0, 1, 0)$. For $n = 3$ the respective counts are $(9, 35)$ and $(10, 40)$, with the same homology patterns. The uniform argument above proves the assertion for every $n \ge 2$; these checks are certificates of finite instances, not an induction inferred from a sample.

Finally let $i : U \hookrightarrow X$ be a full sieve. If there is an arrow from $u \in U$ to $x \notin U$, selecting that arrow and using the characteristic functor $X \to [1]$ exhibit $\{0\} \hookrightarrow [1]$ as a retract of i in the arrow category. The characteristic functor sends U to 0 and its complement to 1; the sieve condition excludes arrows in the forbidden direction. Pushout-product with the fixed endpoint map preserves this retract diagram. Intrinsic cofibrations are closed under arrow

retracts because the induced pushout comparison is an arrow retract of the corresponding preserved $\mathcal{W}$-map, and $\mathcal{W}$ is retract closed by B.1. Thus i cannot belong to a class as in the statement. If there are no crossing arrows and i is a nonisomorphism, then $X = U \amalg V$ with V nonempty; its pushout-product has, for $v \in V$, the crossing arrow $(v, 0) \to (v, 1)$. Applying the first case to this new sieve contradicts closure under one further pushout-product. Hence every member of the class is an isomorphism. Claim D.3 supplies the assertion about nontotal reversible localizers. This verifies, in the precise range $n \geq 2$, the pushout-product objections in TOTASPH F.23–F.24 and F.31. □

H.7. Stability of the unpointed derivator. [Proved]

Whenever the unpointed prederivator

$$\mathrm{HOT}_{\mathcal{W}}(I) = \mathbf{Cat}^{I^{\mathrm{op}}}[(\mathcal{W}^{I^{\mathrm{op}}})^{-1}]$$

has the derivator structure under discussion, it is stable in Grothendieck's sense if and only if $\mathcal{W} = \mathrm{Fl}(\mathbf{Cat})$. In particular none of the homology localizers for nonzero theories gives a stable unpointed derivator, although its equivalences admit the stable detection of Part F.

Proof. Stability includes pointedness: initial and terminal objects agree. The images of $\varnothing$ and e remain initial and terminal in $\mathbf{Cat}[\mathcal{W}^{-1}]$. To check this directly, the natural transformations from the constant $\varnothing$ functor to the identity and from the identity to the constant e functor descend through localization: all three functors preserve $\mathcal{W}$, and naturality for inverted arrows follows from naturality for the original arrows. At $\varnothing$ and e their respective components are identities. Naturality then says that every map from $\varnothing$, or to e, equals the displayed component, proving uniqueness as well as existence. If the derivator is pointed, its canonical map $\varnothing \to e$ is therefore invertible. Strong saturation B.1 puts this specified functor in $\mathcal{W}$. The product and two-out-of-three argument of D.3 makes $\mathcal{W}$ total.

Conversely, for the total class every morphism of every diagram category is inverted. The constant terminal-category diagram is terminal before localization and gives a contraction after groupoid completion: each object has its unique natural transformation to that terminal diagram. Thus every displayed value is equivalent to the terminal category. The resulting terminal derivator has its unique object both initial and terminal, and every square is both homotopy cartesian and homotopy cocartesian. It is stable. This concerns the unpointed $\mathrm{HOT}_{\mathcal{W}}$ requested here; neither a stable spectral test nor passage to a pointed or stabilized construction changes this conclusion about that specific derivator. □

H.8. The stronger separation from enough intrinsic fibrations. [Proved]

Every nontotal modern fundamental localizer satisfying Loc (6 b) satisfies Loc (5) and fails Loc (7 F). Equivalently, Loc (6 b) and Loc (7 F) together force the total localizer, without any accessibility assumption. In particular every nontotal reversible localizer lies between $\mathcal{W}_{\mathbb{Z}}$ and $\mathcal{W}_0$, with a strict upper inclusion.

Proof. Let $f : X \to Y$ be a map inverted by such a localizer, interpreted in spaces through the natural simplex-category models. Its pointed homotopy cofiber C_f is $\mathcal{W}$-aspherical by B.5. If f misses a component of Y, the cone component and an omitted component are distinct. Select one point in each and collapse all components into these two labels. This exhibits S^0 as a homotopy retract of C_f, contradicting D.3 and retract closure.

Suppose instead that $\pi_0 f$ is onto but not injective. Then C_f is connected, and the ordinary homology exact sequence gives a surjection

$$H_1(C_f; \mathbb{Z}) \twoheadrightarrow \ker\big(H_0(X; \mathbb{Z}) \to H_0(Y; \mathbb{Z})\big).$$

The kernel is a nonzero free abelian group: over each component of Y it is generated freely by differences from one selected component of X above it. This also describes the kernel for infinite component sets. It has a quotient $\mathbb{Z}$. The resulting surjection $\pi_1 C_f \twoheadrightarrow \mathbb{Z}$ is represented by a pointed map $h : C_f \to K(\mathbb{Z}, 1) = S^1$. Choose a loop $s : S^1 \to C_f$ whose image is 1. The composite hs has degree one and is pointed homotopic to the identity. Thus S^1 is a homotopy retract of C_f, again excluded by D.3. Homotopy retractions can be tested in $\mathbf{Cat}[\mathcal{W}_\infty^{-1}]$ and then in $\mathbf{Cat}[\mathcal{W}^{-1}]$; strong saturation B.1 transfers the resulting invertibility of $S^i \to *$ back to membership in $\mathcal{W}$. We have proved that every $f \in \mathcal{W}$ is a π_0-bijection, which is Loc (5).

Every $\mathcal{W}$-aspherical space A is therefore connected and nonempty. No homomorphism $\pi_1 A \to \mathbb{Z}$ is nonzero: otherwise its image, rescaled to $\mathbb{Z}$, and a chosen loop provide the same S^1 homotopy retraction. The free-subgroup argument in C.4 now applies verbatim to show that $Y = BF_2$ is null for every such A; the constant-map map $Y \to \mathrm{Map}(A, Y)$ is an equivalence.

This last assertion needs no localization at a proper class of objects. Define directly

$$V_Y = \{u : A \to B : \mathrm{Map}(Bu, Y) \text{ is an equivalence of spaces}\}.$$

It is a modern fundamental localizer. Weak saturation and the terminal axiom follow from the mapping-space functor. For relative Theorem A, use the natural comma-diagram comparisons $\mathrm{hocolim}_c B(A/c) \simeq BA$ and $\mathrm{hocolim}_c B(B/c) \simeq BB$ in the proof of F.2. Mapping into Y turns those specified homotopy colimits into homotopy limits, so an equivalence for each comma map implies the required equivalence for u. Thus $\mathcal{W}^{\mathrm{asph}} \subseteq V_Y$.

If Loc (7 F) held, the accessibility-free direction of E.2 would give $\mathcal{W} = \mathcal{W}^{\mathrm{asph}} \subseteq V_Y$. But the endomorphism $B\alpha : BF_2 \to BF_2$ of F.8 belongs to $\mathcal{W}_{\mathbb{Z}} \subseteq \mathcal{W}$ and does not belong to V_Y. Indeed, if precomposition by $B\alpha$ were an equivalence of $\mathrm{Map}(Y, Y)$, surjectivity on components would supply $r : Y \to Y$ with $rB\alpha \simeq \mathrm{id}_Y$. Injectivity on components, applied to $(B\alpha r)B\alpha \simeq B\alpha = \mathrm{id}_Y B\alpha$, would then give $B\alpha r \simeq \mathrm{id}_Y$. This would make $B\alpha$ a homotopy equivalence, contradicting its explicitly nonsurjective fundamental-group homomorphism. Hence (7 F) fails. The total class satisfies both axioms directly, and the strict upper inclusion follows from the failure of reverse excision for $\mathcal{W}_0$ in B.13. □

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Wolcott, F. L. (2015). Variations of the telescope conjecture and Bousfield lattices for localized categories of spectra. *Pacific Journal of Mathematics*, 276(2):483–509. verified: publisher PDF opened; Propositions 6.2 and 6.4 and the caveat following Proposition 6.4 inspected.